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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 317 — Past Questions & Materials
by Renance

Course: MEE 317

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 9

Levels: 200, 300

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Paper 1 of 9 — MEE 317 Basic Safety Engineering

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
FIRST SEMESTER EXAMINATION 2024/2025 SESSION
COURSE CODE AND TITLE: MEE 317 BASIC SAFETY ENGINEERING
TIME ALLOTTED: TWO HOURS (2HRS)
INSTRUCTION: ANSWER ALL QUESTION

QUESTION 1

- (a) (i) What is hazard, and how is it different from risk? (ii) Differentiate between an incident and an accident.
- (b) Describe three different types of hazards in an engineering environment.
- (c) Mention three common causes of accidents in an engineering environment.
- (d) Define the following: (i) Safety, (ii) First Aid, (iii) Personal Protective Equipment, (iv) Toolbox talk (v) Emergency Preparedness.
- (e) Mention the basic steps of first aid

[25 Marks]

QUESTION 2

- (a) What do you understand by the term Hazard Identification and Risk Assessment?
- (b) Give a detailed risk assessment for a building construction site.
- (c) What are the most common work place hazard, and how can they be mitigated?

[15 Marks]

QUESTION 3

- (a) What is the rationale behind the good Samaritan law? (At least mention three applicable incidents).
- (b) Is work place safety important? Give justifications to your answer.
- (c) Explain in details how you would handle a fire extinguisher when faced with a fire outbreak in an office environment.

[15 Marks]

QUESTION 4

- (a) What is the purpose of Audit and Inspection in safety?
- (b) In order to ensure safety of lives, certain safety equipment or devices must be installed or placed in a vehicle, mention any five (5) of such equipment or devices and state its function.

[15Marks]

Paper 2 of 9 — MEE 317 - Basic Safety Engineering

Level 300 • First Semester • PastQuestion

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Paper 3 of 9 — MEE 317 - 2024/2025 - First Semester - BASIC SAFETY ENGINEERING

Level 200 • First Semester • PastQuestion

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Paper 4 of 9 — MEE 317 BASIC SAFETY ENGINEERING 2024/2025

Level 300 • First Semester • PastQuestion

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Paper 5 of 9 — MEE 317 Basic safety engineering 2024/2025.

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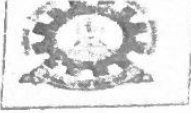
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Paper 6 of 9 — MEE 317 Basic safety engineering 2025

Level 300 • First Semester • PastQuestion

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Paper 7 of 9 — MEE 317 - Industrial Safety Analysis

Level 300 • First Semester • Handout

INDUSTRIAL SAFETY ANALYSIS

GENERAL

Safety analysis is a necessity in today's litigious society. Companies that do not analyze may not prevent accidents which are preventable and may find their ability to defend themselves in a post accident lawsuit compromised. The standard to which the courts hold management is prudence, foresee ability, and reasonableness. Prudence requires that (1) production systems, products, and facilities be analyzed for safety, and (2) proper preventive measures follow the analysis. A prudent manager analyzes. Attempt must be made to foresee the foreseeable. This attempt must be reasonable. The key is reason. Analysis is simply the application of reason. There are two types of reason inductive and deductive. If a company can demonstrate that they made a reasonable attempt to foresee the foreseeable by applying system safety analysis inductive and deductive techniques and responding to the analyses with proper preventive actions, the company has a strong legal position. They can show they were reasonable and prudent in trying to foresee and avoid accidents, provided that they have carefully documented their efforts.

The philosophy developed by system safety engineers is that accident prevention can and must begin as soon as the idea for a new system or operation is conceived. Methods of analyses have been developed to this end which can be undertaken early. The analyses can indicate tentatively any hazards that might be present in any proposed operation, the types and degrees of accidents that might result from hazards, and the measures which can be taken to avoid or minimize accidents or their consequences. As more and more details become available after the initial analysis on the characteristics of the proposed system or operation, they can be expanded further and further to determine more intimately where there might be potential causes of accidents, their effects and safeguards needed. Almost 90 techniques have been developed, used, or proposed. This chapter includes a few important ones to indicate some of the methodologies now in wide use. This may make occupational safety engineers aware of some types of detailed analyses they might want to undertake.

Some of these techniques start at the component level and reason upward to find what "undesirable end events" might occur from that components fault or failure. These "bottom-up" techniques are based on inductive reasoning. Some techniques start with the undesired event which is to be avoided and reason down to identify which components of the system could contribute to the undesired event through their fault or failure. In systems where there is a critical potential for catastrophe, the prudent manager must make every effort to foresee and avoid the catastrophe. Everything is not foreseeable, but the manager must make a thorough effort to foresee when there is a potential for a catastrophe. This means the manager will use both inductive and deductive system safety analysis, if feasible. Preferably, the inductive analysis will be conducted blind to the deductive analysis, and vice versa. The results of the two analysis will then be compared. Hopefully, the undesirable end events identified in the inductive analysis will match those the deductive analysis started with; the component faults and failures identified in

the deductive analysis will match those components the inductive analysis started with. Where the inductive and deductive techniques do not match, the deficient analysis is corrected. This double-blind cross-check is a strong indicator of a reasonable effort to foresee the foreseeable, and when accompanied by prudent corrective actions, places management in a very defensible position.

PRELIMINARY HAZARDS ANALYSIS

The first method generally used in the systematic process of determining hazard causes, effects, and deterrents is the preliminary hazards analysis. As its name implies, this is an initial study from which analysis efforts can be expanded further. It is fairly broad in scope, investigates what hazards might be present, whether they can be eliminated entirely, or if not, the best way to control them. If the hazard cannot be eliminated, the analyst determines whether there are standards or methods by which the hazard could, should, or must be controlled. A review is made of the functions to be performed and whether the environments in which they must be performed will have any adverse effects on personnel, equipment, facilities, or operations. According to Greek mythology, Daedalus wanted to escape from Crete where he and his son Icarus were being held captive by the king because of his skill as an artificer. Daedalus made wings of feathers, flax, and beeswax with which he and Icarus could fly to Greece. Before they flew off, Daedalus told Icarus:

My boy, take care

To wing your course

Along the middle air: If low, the surges

If high, the sun The melting wax consumes.

Daedalus's instruction to his son can be taken to be an elementary preliminary hazard analysis. However, Daedalus, not being an experienced safety engineer, did not go far modern Safety Analysis

Safety engineers are smarter than designers. We safety engineers can figure out potential ways of getting killed which Daedalus didn't even consider, and we can even indicate safety measures by which he could have prevented Icarus's accident (he flew too close to the sun, the wax melted off his wings and he fell to his death). Modern designers, on the other hand are not nearly as smart as Dardalus, they have never been able to figure out how to make flyable wings out of feathers, flax, and beeswax.] Each product or operation have a limited number of hazards, which can be determined as soon as a few facts are known. For example, the proposed product is to use electrical power. The hazards, which can potentially be present when electrical power is in me, could include any or all of the following electrical shock, burns from hot equipment, fire due to arcing, sparking, or very hot surfaces; inadvertent starts of equipment, failure of equipment to operate at a critical time so that an accident will occur, radiation effects, and electrical explosions. Numerous conditions can lead to each of these, but these are the basic hazards. If the electrical product uses small batteries such as for transistorized equipment, most of the hazards mentioned will not present. The analysis can therefore be limited to those which could possibly exist, no

matter how improbable Progress of the design can be monitored thereafter to determine whether the hazard can be eliminated; and if it cannot be, how best it could be controlled, and if it is controlled, whether or not the control appears to be adequate and how the adequacy of the control can be verified. At this point, very rough estimates might be made of the probability of an accident due to that hazard, and of the severity of its probable effects.

A practice has been initiated in which the preliminary hazards analysis list, which is prepared in tabular form, is broken down into individual items. Each item is recorded (Fig. 15-2) and then tracked through development or modification of a system, product, or operation to ensure that adequate consideration is given to its hazard elimination control. (To assist in the tracking, visualizations, such as in the fault-tree method described later, can be used.) When all necessary actions are completed to eliminate or acceptably control the hazard, the item is signed off, since no further action is required. (An item should never be signed off with a comment that an action will be taken, is scheduled to take place, or a similar entry Signoff is considered accomplished only after the proposed action has been completed and shown to be adequate.) As design or planning progresses, studies are made of the hardware and facilities through reviews of assemblies of their major components, their proposed interrelationships and interfaces with each other and with personnel, environments that could affect them, and the effects they could generate on personnel, other components and assemblies, and the environment.

Some information does not lend itself well to an analysis presented solely in a tabular form. A narrative format can then be added to include addition data. For example an analysis presented in columnar form might indicate that a hazard was a accidental fire. To have a fire requires a fuel, oxidizer, ignitable mixture, and source of ignition. Detailed information on each item could then be listed separately, with details on the many facts, oxidizers, and sources of modern Safety Analysis.

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FAILURE MODES AND EFFECTS ANALYSIS

One of the methods used to accomplish safety analyses has been derived from reliability engineering Failure Modes and Effects Analysis (FMEA). FMEA is an inductive technique. It is

used to determine how long a piece of complex equipment will operate satisfactorily and what the effects of any failure of individual components might be. The method is intended for analyses of proposed equipment and systems. An occasion might arise in which it would be highly desirable for an occupational safety engineer to determine whether or not the manufacturer of industrial equipment and system to be purchased and installed had had analysis made, and to understand how failure might occur, modes and frequencies of failures, and the necessity for proper and timely maintenance and replacements.

In this method of analysis, the constituent major assemblies of the product to be analyzed are listed. Each assembly broken down into subassemblies and their components. Each component is studied to determine how it could malfunction and cause downstream effects. Effect might result on other components, and then higher-level subassemblies, assemblies, and the entire product or system. Failure rates for each item are determined and listed. The calculations are used to determine how long a piece of hardware is expected to operate between failures, and the overall probability that it will operate for a specific length of time. It is the best and principal means of determining where components and designs must be improved to increase the operational life of a product. It is best used to analyze how often and when parts must be replaced if a failure, possibly affecting safety, must be avoided.

Until the use of Boolean mathematics, described in the next paragraph, FMEA calculations were often erroneous. Also, because many component failures would have to effect on safety, that aspect of an FMEA does not involve accident possibilities. Also failure modes and effects analysis is limited to determination of all causes and effects, hazardous or not. Further, the FMEA does very little to analyze problem which could arise from operator errors (unless the system analysis includes the human as a component), or hazardous characteristics of equipment created by bad design of adverse environments (unless the scope of the analysis includes these elements). The FMEA is excellent for determining optimum points for improving and controlling product quality. Another method, fault-tree analysis has been found more effective for safety purposes.

FAULT-TREE ANALYSIS (FTA)

In 1959 the Air Force became concerned with the potentially catastrophic events that could occur with the Minuteman missile then being developed by Boeing. The Air Force contracted with Bell Laboratories to develop a method of analysis by which probabilities of occurrence of events with which they were concerned could be computed. The Air Force wanted to know the possibilities and probabilities that a missile could be launched or a warhead activated inadvertently, and the chance of either of these being done intentionally by an unauthorized person in an act of sabotage. Two years later Bell Laboratories completed the project. The new method, fault-tree analysis involved Boolean logic in ways similar to those being used increasingly in electronics industries. It is a deductive technique. Safety engineers of the Boeing Company adopted the proposed method and became its foremost proponents. They added to the symbols that Bell Laboratories had proposed. They organized the method so it could be computerized to permit calculating the probabilities of the problem with which the analyst was concerned. Although still used to determine probabilities of shapes in complex systems or

operations, fault tree analysis is being used far more frequently in logically analyze the possibilities of potential accidents due to hazards, and the quantitative safety level,

At first the fact that Boolean logic (and arithmetic), with which few engineers were familiar at the time was its basis gave many people the impression the method could be used only by mathematicians (Some of the first Boeing employees involved in making those early safety analyses were mathematicians) It turned out that Boolean logic was fairly simple to understand. One of the fundamental principles is that any statement, condition, act, situation, or process could be described as being in either one of two states Something could be true or false, on or off, up or down-but not both, it would go or not go; or it could be fully open or not fully open (closed or partially open). There were no middle intermediate positions.

FAULT-TREE SYMBOLS

Primary Event Symbols

Basic Event, such as a component failure Circle.

-Conditioning Event, a condition which must be satisfied before the event above the gate can occur- Ellipse

- Undeveloped Event, an event not fully developed because of a lack of information or significance - Diamond

Normal Event, expected to occur during system operation-House-shape

Intermediate and Top Event Symbol

-Output Event, those that should be developed or analyzed further to determine how they can occur (the top output event is called the end event; others are called intermediate events) - Rectangle for gate): Any event leading into it from underneath will cause the event leading out of it at the top to occur.

Limitations of Fault-Tree Analyses One of the very few limitations of fault-tree analysis is that the tree indicates only the events included which will contribute to the occurrence of the top event.

SAFETY ANALYSIS METHODS MANDATED FOR PROCESS SAFETY MANAGEMENT

Safety analysis techniques are now mandated by regulations for some industries. This program requirement is examined below, and the techniques it suggests that are not already covered in this chapter are described and explained.

In the mid-1990s, Industries with processes that involved "highly hazardous chemicals" (HHC) were placed under requirements by OSHA (29 CFR 1910.119) to conduct process hazard analysis and have a process safety management program. The program is triggered by above-threshold quantities of any of 136 different, listed chemicals. The purpose of this standard is to prevent or minimize the consequences of a catastrophic release of toxic, reactive, flammable, or explosive highly hazardous chemicals from their use, storage, manufacturing, handling or on-site movement by industry.

The importance of this standard is that it requires safety analysis and names certain analytic

techniques to use or their equivalent. The employer shall one or more of the following methodologies that are appropriate to determine and evaluate the hazards of the process being analyzed... What-if, Checklist, What-if/Checklist. Hazard and Operability Study (HAZOP), Failure Mode and Effects Analysis (FMEA), Fault Tree Analysis, or an appropriate equivalent methodology" [29 CFR 1925.04(012))

What-if. In "What-if?" analysis, an analysis team reviews the covered process from beginning to end. At each step in the process they think of questions that begin with "What-if... They address procedural, hardware and software errors and produce a tabular listing of these questions and their answers. This constitutes a set of potential accident scenarios, their consequences, and remedial recommendations Report form can be narrative, but use of a matrix table often provides clarity

Checklist. The "Checklist" method applies a list of specific items, usually from industry standards, consensus codes, industry guidelines, etc. The list is prepared by an experienced safety engineer, familiar with the covered process and the Limitations of Fault-Tree Analyses

Process Safety Management program

Process Safety Information

Requires compilation of written process safety information, including hazard information on HHCS and information on technology and equipment at the covered process

Employee Involvement

Requires a written plan of action for involving employees in the conduct and development of process hazard analyses and process safety management and for providing employees access to the information required by the standard.

Process Hazard Analysis

Specifies that process hazard analysis be conducted and updated and revalidated at least every five years. Must be retained for the life of the process

Operating procedure

Requires these be in writing and include clear instructions for safely conducting activities of the covered process Steps for each operating phase, operating limits, safety and health considerations and safety systems and their functions must be readily accessible to involved workers. They must reflect current operating practice, and must implement safe work practices for special circumstances such as lockout/tag out and confined-space entry.

Training

Operators of the covered process must be trained in the overview of the process and in the operating procedures. Training must emphasize specific safety and health hazards, emergency operations, and safe work practices. Refresher training is required at least every three years.

Contractors

Contract employers are required to train their employees to safely perform their jobs at the process site. Must document that employees received and understood training and sure that contract employees know about the potential process hazards and the worksite employer's

emergency action plan. Must assure that contract employees follow safety rules of the facility
Must advise the worksite employer of hazards that contract work itself poses or hazards identified by contract employees.

Prestart up Safety Review

A safety review is required for new facilities and significantly modified work sites to confirm that the construction and equipment of a process are to design specs. Must assure that adequate safety, operating, maintenance, and emergency procedures are in place and that process operator training has been completed For new facilities process hazard analysis must be performed and its outstanding section items resolved and safety implemented before startup. Modified facilities must meet management of change requirements

Mechanical integrity

Written procedures must be established and maintained for the ongoing integrity of process equipment, particularly those components which are part of a covered process.

Hot Work

Hot work permits must be issued for hot work operations conducted in or near a covered process

Management of Change

Written procedures must be established and implemented to manage changes to facilities that affect a covered process Employees must be informed of changes

Incident Investigation

Require s employers to investigate as soon as possible, but no later than 48 hours after incidents which did result or could reasonably have resulted in catastrophic releases of covered chemicals. An investigation team which includes at least one person knowledgeable in the process involved and others with knowledge and experience in incident investigation and analysis must develop a written report on the incident. Reports must be retained for five years

Emergency Planning and Response Compliance Audits

An emergency action plan must be developed and implemented. It must include procedures for handling small releases

Employers must certify that they have evaluated compliance with process safety

Requirements at fast every three years. Prompt response to audit findings and documentation that deficiencies are corrected are required. The two most recent audit reports must be retained.

Trade Secret

Requires that certain information be available to employees Employers may enter into confidentiality agreement with employees to prevent disclosure of trade secrets questionnaire be applied to the covered process or system. Those parts of the design which are not adequate should be revealed when the checklist is applied. Then corrective measures are identified. What-it/Checklist. The What-if/Checklist Analysis uses the list of What-d questions and supplements it with a checklist. Therefore, it is a combination of the two methods, as the name suggests.

Hazard and Operability Study (HAZOP), HAZOP addresses each element in a process. The

elements must first be identified. The expected way each element operates is determined. Deviations from the intended or expected operations and parameters are identified using guide words. The consequences from the deviating element are estimated. Causes of the deviations, currently planned (or used) hazard controls, and inadequate controls are identified. The study is conducted by a team made up of five or six persons usually including a team leader familiar with HAZOP safety and health expert, a manager/supervisor, a technical person, a senior process operator, a maintenance person, a chemist, etc. The quality of the results will depend upon the quality of the team selected. The team will start by collecting process information from piping and instrumentation diagrams (P&ID), written operating instructions, material safety data sheets, plant models, Safety Analysis Reports (SARs), etc. They then review each element step by step through each line or process. At each element they apply key words a guide words to tease out the potential deviations from normal expectations of a design parameter.

The hazard and operability consequences, if any, for each deviation are determined and documented. The actions or changes in plant methods which will prevent the deviation or reduce the consequences are identified. If the cost of the changes can be justified, the team then must agree on the change and who will be responsible for the action. The team then must follow up to make sure that the action has been taken. If the cost is not considered to be justifiable, the team must agree to accept the risk. One of more reports of the evaluation are made, usually including tabular representation of the results. Team decision making is at the heart of this methodology. This technique as well as the others in this chapter are supported by software. The reader is advised to "surf the net" for the latest. Arthur D. Little Company is one source.

HAZOP GUIDE WORDS

No/Not Complete negation of design intention Quantitative increase or decrease

No part of the design intention happens e.g. <No flow=, No pressure= etc

More, less Quantitative increase or decrease

Refers to quantities, properties and duration of design parameters, such as flow rate, temperature, etc

As well as Something else occurs in addition to design intention

Design intention is achieved, plus additional things occur.

Part of Only part of design is achieved

Partial addition or removal of material occurs. Activities are incomplete.

Reverse The opposite of the design intention

Reverse flow of material, electrical current, voltage polarity, Backward installation (e.g. check valve), opposite chemical reaction (e.g. decomposition).

Other than Complete substitution

Original design intention is not accomplished, a different activity occurs-e.g. leak, Rupture

Job Safety Analysis (JSA). The purpose of this technique is to uncover and correct hazards which are intrinsic to or inherent in the workplace. It should be done with a team that includes the

worker, supervisor, safety engineer, and management. Its success depends upon the rigor this team exercises during five steps: (1) Select a job, (2) Break the job down into steps (3) Identify the hazards and determine the necessary controls of the hazards, (4) Apply the controls to the hazards, and (5) Evaluate the controls

Eliminating, controlling, and minimizing hazards are becoming more and more necessary for work plants to comply with the standards of regulatory agencies. Formerly, chemical plants were built in which hazardous environments existed and where operators were safeguarded, theoretically, by use of protective equipment, such as gas masks, whose need was established after operations had begun. Some government agencies, such as OSHA, have adopted policies of "deterrence" and will not countenance the existence of hazardous conditions. Intrinsically safe environments are being required in which operators can work without the need for protective equipment, which is often inadequate, and which workers find burdensome and inconvenient and will not wear. To ensure that regulatory agencies do not require extensive and expensive retrofits of corrective equipment, it is highly advisable that analyses be made early to recognize potential problems and their solutions. The billions required to revise and correct designs and procedures at Three Mile Island and other nuclear power plants are an example of need for proper and adequate analysis and planning for safety purposes.

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PEDESTRIAN SAFETY (PERSONAL SAFETY TIPS)

The Nigeria Police once wrote:

"You have the primary responsibility to prevent crime against yourself, loved ones and your properties. Thinking that crime can only happen to someone else and doing nothing to prepare yourself, or take precautionary preventive measures, makes you a perfect target for criminals. The ultimate goal of crime prevention is to reduce the risk of being a victim. In order to accomplish this effectively, it is important to remove opportunities for a criminal to take advantage of you or your property.

You are attempting to prevent either victimization or criminalization by presenting an unattractive target to the criminal. This effort. removing opportunities is often referred to as

target-hardening. Target -hardening can be as complex as installing a high tech alarm system, or taking a self defense course, to something as simple as locking your door. Making yourself less of a target, is simpler than you might think it further gave the following information on safety at home, ATM Security tips

Make Sure They See You: When someone does create a . conflict that threatens your safety, it means he either did not see your car or was counting on you to avoid the conflict. Be certain that the other person sees your car and shows clearly by his actions that he will "stay put. Always tap your horn or flick your lights if in doubt. Do it early so you could stop if necessary.

Never insist on taking the right-of-way, even when you are sure it is yours. The right-of-way should never be taken - it must be given. This is a rule of common courtesy, which also requires that you yield the right-of-way to others when it cases a traffic situation to do so. Courtesy in yielding the right-of-way is an important factor in modern driving Courtesy smooths the roadway, and it saves lives .Never drive without a Valid Driver's License in your immediate possession.

AFINAL RULE

No driver will give perfect attention to his driving, but habits based on observance of correct seeing in every day driving will prevent you from becoming involved in many critical situations.

AFINAL REMINDER

Keep Your Eyes Moving Build a strong habit of forcing your eyes to move at least every two seconds in driving. Check the rear view mirror each five seconds if any traffic is ahead or behind. Check the rear also before changing lanes and before any slowdown that could surprise a driver behind. The eye moving habit is restful for your eyes, keeps your mind straying too far from your driving and literally forces you to adjust speed to traffic. weather and visibility conditions.

SAFETY AT HOME

7Make sure there are good locks on exterior doors and windows.

7 Lock up when you go out, are asleep, or alone, even if it is only for a few minutes. It takes a thief ten seconds or less to enter an open room and steal property.

7Get to know your neighbors and their daily routines,

8Keep a list of emergency numbers especially of the Police Station close to you.

7Make sure your home is fenced and if you can afford it.

7Trim flowers that hide doorways or windows.

7Set timers on lights and radios to make your home appear lived-in, when you are away.

7 Have a neighbor regularly check on your home when you are away, and report any suspicious activities. Remember that one good turn deserves another

8Mark your valuables with a number or identification unique to you, with an engraving pen and keep a list of them in a safe place .

Replace solid walls in front yard with open fencing to eliminate hiding places and make climbing more difficult

- Install a wide-angle peephole in your front door. so you can look out to an on-coming visitor without being seen yourself

Do not give out information about your neighbors.

Don't make it easy on burglars

8 Keep valuables away from windows

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ATM SECURITY TIPS

7 Be alert and conscious of your surroundings when using ATM.

Don't use an ATM machine located in a dark or obscure location.

8 Never give your card or Personal Identification Number (PIN) to anyone, for any reason

Be very conscious of persons hanging around the ATM machine Do not write your PIN on the card or anything that is kept with the card

Do not insert your card until asked to do so by the display screen

Never use an ATM with a blank screen, and if the ATM is obscured from view or properly lit, leave immediately and

find another ATM

7 Stand close to the ATM and use your body and hand as shield to make sure nobody sees you keying in your PIN Never accept help from strangers when using an ATM. You should be wary of strangers asking for help. The elderly, or physically challenged persons should take along with them trusted persons, to assist them..

7 Criminals work in teams, for example, one to distract you while the other steals your card or money. If your card is retained or swallowed by the ATM, it is advisable to contact your bank immediately Sign up for SMS alert, so that you are alerted whenever any withdrawal is made in your account

VEHICLE SECURITY TIPS

Most car crime is preventable. Police are making a concerted effort to tackle car crime, and target offenders: but motorists need to help themselves avoid becoming victims, by removing the opportunity for crime in the first place

Don't leave anything on display: it is a prime invitation to passersby. Even an old coat or plastic bag can tempt a thief. If items have to be left in the car put them out of sight at the start of journey.

Get a music system that is removable, or has a front that can be taken off, and remove ipads and mp3 players.

Driving documents and personal correspondence could help a thief to sell your vehicle or provide a cover story, if stopped by the police. Do not keep these in your vehicle Some car thieves break into houses to steal car keys.

Always keep the keys in a safe place. Do not leave them in the hallway or kitchen, as these are the obvious places.

A-Z PERSONAL SAFETY TIPS

Personal Safety is the act of ensuring that one is free from harm or danger, and or the ability to take steps towards preventing such harm

In other words, it is a list of actions that are designed to keep situations from occurring or escalating, and to make sure that physical, emotional and verbal abuse is avoided, stopped at the first moment, if it at all begins to happen. It borders on ideas such as when to stop a conversation, when to leave the room, and or call for help.

The following tips can help secure you in your daily living, while hopefully inspiring more awareness, when you are out and about

- Always be alert and aware of your surroundings, and of the people around you
- Make casual eye contact with people when walking
- Do not be easily distracted or take your eyes off someone you feel looks suspicious in a possibly dangerous situations.

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- Always carry ID Card and only the much cash need
- Whenever possible buddy up to walk out for dark parking
- Informants trusted family or friends if you are traveling your itinery dates and location if necessary
- Stay away from the bush or trees when walking of running Always be prepared to run away from an attacker and scream of confront where possible.
- Turn and go in the opposite direction of the car if a stranger approaches you, offering a ride. If possible rite down the license plate number and description of the vehicle.
- .. Do not respond to passing comments or conversation from strangers on the street or elsewhere.
- Walk confidently at a steady pace, and have your keys ready in your hand
- Avoid being on a cell phone with an ipod in your other ear unaware of your surroundings. This makes you an easy target.
- Avoid dangerous places at night time. Visit them during the time hours.
- Stop and look around if you feel unsafe, entering an area Trust your instincts
- Avoid isolated bus stops otherwise, continually look all around. Be Alart
- Avoid suspicious passenger cars/buses, one chance or all male etc.
- Don't stay in the same spot at bus stops for long. It makes you an easy target.

If you are carrying money, don't open your purse or wallet while boarding a vehicle Have your fare ready.

- Keep flashy jewelry or other valuables of easy prey out of sight
- If you must be in an isolated vehicle, sit close to the driver
- Check your purse or wallet, if someone is jostling, crowding or pushing you
- Never leave your purse, backpack or briefcase in plain view Lock it up when you leave your desk or office
- Keep the office door locked if you work alone or before and after normal business hours
- Try to find another worker or a security guard to walk out with you, if you work late
- Do not get in the elevator with another person if you do not feel comfortable with that person, take the next one. If you have to get in, stand next to the control panel, so that if you are attacked you can press the alarm and the control buttons as hard as possible
- Be alert for pick pockets on crowded elevators, and other busy places
- Be aware of escape routes for emergencies and post phone numbers of police, security agencies and fire departments in your Cell phones, so that you do not have to fumble pushing many numbers.

Young women should avoid provocative clothing, and carelessly immoral gesture.

TRAVELLING, SAFETY TIPS

7 Notify your Resident Security Officer (RSC) or Personal Security Officer (PSO) your departure and return dates but don't otherwise publicize your travel or vacation plans.

7 Check plane, tram and bus times before you travel,

8 Sit near other people, aisles or doors Learn the location of emergency alarms and exits

7 Stay awake and alert when using public transportation

7 Conceal your passport, money, or credit cards when traveling

8 Keep other valuables out of sight, and luggage where possible close at hand if carrying a handbag, keep it in front of you, closed with the fastening towards your body. Keep wallet in your front pant's pocket and not back

7 Let go if your bag is snatched. Do not run after, especially in unfamiliar surroundings

7 Do some research on the area you are visiting, especially at first time. Talk to your security officer or host regarding travel advisories and warnings.

7 When traveling, dress casually dress down where appropriate. Be aware of local customs.

7 Do not wear excess jewelry Reduce wallet and purse contents, particularly cards denoting affiliations, memberships and accounts, etc

8 At airports proceed through security checks, and go to the boarding area, as quick as possible. These areas are usually the most secure in the airport. Be very careful anytime you make a telephone conversation.

LETTER AND PARCEL BOMBS SAFETY TIPS

Letter and parcel bombs generally are "victim activated" incaring that a victim or intended target must activate the device by opening it. They do not normally contain timing devices.

Bombs can range from the size of a cigarette packet to a large parcel. Letter and package bombs have been disguised as letters, books, candy and figurines. Delivery methods have included mail systems, personal delivery, or placement at the recipient's site. A letter or parcel bomb might have some of the following indication.

7 Suspicious origin especially if the postmark or name of sender is unusual, unknown, or no further address is given.

7 Excessive inadequate postage:

7 Off-balance or lopsided letter or package.

8 Unusual weight for the size of the letter or package. Letters also may be unusually thick.

Softness or spring of contents when checking do not bend excessively.

Protruding wires or components, unusual grease or oil stains on the envelope.

7 Strange smell, particularly almond or other suspicious odours.

7 Handwriting of sender is not familiar, or indicates foreign style not normally received by recipient. Common words or names are misspelled.

7 Rub-on or black lettering.

7 Restrictive markings such as confidential, "personal", or an horrific title appended to the name of the addressee.

7 Wrapping that could be a provision for an arming safety wire.

8 Rattling inside the envelope or package, possibly loose components of a device.

7 Visual distractions for example currency or pornography etc.

WARNING

If you identify a letter or package as suspicious, don't let anyone near it. Notify your local Bomb Disposal Unit or the nearest Police Station immediately. Leave the letter or package in open area such as a court yard, where it is easily accessible to Bomb Squad Personnel. Never submerge it in water or burn.

CAR JACKING SAFETY TIPS

8 When in your car, always keep the doors locked. Anytime you drive through areas containing stoplights, stop-signs or anything that significantly reduces vehicular speed, keep your windows up, especially in narrow ways and isolated places.

7 Leave ample maneuvering space between your vehicle and the one in front of you.. If you are approached by suspicious persons while you are stopped do not roll down windscreen, drive away quickly.

If you are being followed or harassed by another driver, try to find the nearest police station, hotel or other public facility. Once you find a place of safety, don't worry about using a legal parking space, park as close as you can, and get inside fast, especially if the threat is on your life.

7 If another driver tries to force you to pull over, or to cut you off, keep driving and try to get away. Try to note the license plate number of the car, and a description of the driver. If this effort places you in danger, don't do it. The information is not as important as your safety.

If you are being trailed, never lead the person back to your home or stop and get out. Drive to the nearest police station or public facility

7 If you are traveling alone and a car "bumps" into you, don't stop to exchange accident information. Go to the nearest service station, or other public place to call the police if necessary.

8 When you park, look for a spot that offers good lighting and is close to a location where there are a lot of people. Lock valuables in the trunk and lock all doors

7 Be wary of unsolicited assistance.

7 Be extra careful while shopping and loading your car trunk.

SURVEILLANCE

The purpose of surveillance is to identify a potential target, based on the security precautions that an individual takes, and the most suitable time, location, and method of attack. Surveillance may last for days or weeks. Naturally the surveillance of a person who has set routines, and who takes few precautions will take less time

Detecting surveillance requires a fairly constant state of alerts and therefore, must become a habit. A good sense of what normal, and what is unusual in your surroundings could be more important, than any other type of security precautions you may take. Above all, do not hesitate to report any unusual event. There are three forms of surveillance foot, vehicular, and stationary

People who have well-established routines, permit surveillance to use methods that are much more difficult to detect: If for example, you leave the office at the same time each day, and travel by the most direct route to your home, or if you live in a remote area with few or no alternative routes to your home surveillants have no need to follow you all the way to your residence

Therefore, it is advisable to:

7 Vary your routes and times of travel 7 Be familiar with your route and have alternate routes

7 Check regularly for surveillance. Stationary surveillance is most commonly used by terrorists' organizations. Most attacks take place near the victims residence, because that part of the route is least easily varied. People are generally most vulnerable in the mornings when departing for work. because these times are more predictable than evening arrivals

Many surveillance teams use vans with windscreen in the sides or back, that permit observation from the interior of the van. Often, the van will have the name of a business or Utility Company to provide some pretext for being in the area. Where it is not possible to watch the residence unobserved, surveillants must come up with a plausible reason for being in the area. Women and children are often used to give an appearance of innocence. Try to check the street in front of your home from a window before you go out each day. If you suspect surveillance, report to the Police giving details of the vehicle including color and make.

7 Be alert to people disguised as public utility crews, road workers, vendors, etc. who might enter or station themselves near your home or office.

7 Always check your vehicle inside before entering

PERSONAL SAFETY TIPS

. If you notice anything unusual, do not enter the vehicle.

7 Household staff and family members should be reminded to look for suspicious activities around your residence

7 Tell your household staff and family members to note descriptions and license numbers of suspicious vehicles Advise them to be alert for details.

MENTAL DISORDERS AND SAFETY

When dealing with people who show signs of mental disorders, ensuring personal safety should be paramount. Sometimes one may have to be willing to do certain things that seem uncomfortable, break some taboos, be firm, get out of dangerous situations, call for help or do whatever is necessary at that particular moment, to remove oneself from potential violence. abuse, and or threats.

A certain man narrated how he disguised as a madman in order to escape danger along a lonely rural road. A lady equally told the story of how she regained her freedom by begging a rapist gang in tears, not to rape her and be infected with HIV which she suffered from.

Conversation can occasionally lead to danger. It is easy to spend a great deal of time and energy in conversations that leave you feeling hurt and accomplish little or nothing. Some conversations may be urgent in nature or cannot be put off to another time. Note that in such circumstances, less is more, and your words will have more impact if you don't repeat the same things over and again. But just say what you have to say once. And then stop talking. Remember, a word is a enough for the wise.

TIPS ON WHEN TO STOP A CONVERSATION TO AVOID GETTING HURT

8When your personal boundary is violated.

8When you are verbally insulted, mocked or invalidated

7When someone engages in thought-policing or tells you how you must think or feel.

7When you are engaged in a circular conversation.

7When your words start falling on deaf ears

7When the conversation is one way

7When you are angry and at a risk of taking the emotional elevator down to a less mature level.

7When you are tired and not able to engage in further constructive communication.

7When your attention is needed for a more responsible engagement.

WHEN TO LEAVE A PLACE TO AVOID BEING HURT

8When you feel unsafe or isolated.

8When you fear violence or abuse.

7When a person refuses to end the conversation or leave you alone

7When you are alone with an abusive or potentially dangerous person.

Do not hesitate to call for help or the Police when you are faced with situations such as:

7Someone threatens harm to him or herself.

7Someone threatens to hurt you.

7Someone threatens to hurt the children

7Someone performs any act of violence towards other people or property.

By applying these basic safety tips, and your own common sense, you can help protect yourself as much as possible. Above all, implicit trust in the Almighty Creator and godliness is good step towards a safe and Secure life.

The object of Defensive

TRAFFIC SAFETY(SAFETY DRIVING TECHNIQUE)

DEFENSIVE DRIVING CODE

The object of Defensive Driving is to drive without having a preventive accident.

Defensive Driving requires the knowledge and strict adherence to all traffic rules and regulation applicable to the area in which vehicle is being operated.

Defensive Driving requires a constant alertness for the illegal acts and driving errors of other drivers and the willingness to make timely adjustments in one`s own driving. So that these illegal acts and errors will not involve one in an accident.

Defensive Driving requires knowledge of an d intelligent adjustments in one`s own driving to the special hazards presented by abnormal, unusual or changing conditions in the mechanical functioning of one`s vehicle, type of road surface, weather; degree of light, kind of traffic and one`s physical condition and state of mind.

Defensive Driving requires a thorough knowledge of the rules of right - of way and the willingness to yield the right-of-way to the other whenever necessary an accident.

In Defensive Driving requires an attitude of confidence that one can drive without ever having a preventable accident. Drivers are required by law to comply with road/traffic rules set forth in the Highway code.

THE DRIVER`S SIGNALS

An important rule of the road is that every driver must give signals to tell other drivers on the road, what he intends to do in guiding the vehicle he is driving. They must be given by hand or by mechanical/electrical devices which are maintained in good operating conditions. These signals are intended to alert other drivers or pedestrians in advance before you change speed, direction or stop in any direction. Give the signal for 100meters continuously before reaching the turning point. You are not the required to continue the signal actually making the turn. Be sure your electrical signal does not continue to flash after you have completed your turn.

When you plan a series of driving changes- such as stop for a stop sign, followed by a right turn you should always signal first for the action you intend to take. In the situation mentioned, you would signal first for a stop and come to complete stop; in this position you would then give the signal for your right turn. Common sense should guide you in giving and reading driver`s signals.

At night when the arm to use electrical lamp signals. In bright sunlight it is often difficult to see the trafficator signals and it is better to use the hand and arm signal as well.

Give your own signals clearly and do what you have indicated by means of the signals. Read other driver's signal with the same care you give when making your own. Be alert to other drivers who may neglect to use signals.

STARTING, STANDING PARKING RULES

In starting from a proper parking place at the curb, walk around your vehicle and look for children, tool, materials, etc before you start up. Look for cars approaching from the rear before pulling away from the curb – slowly. Remember, you do not have the right-of-way.

In starting from a shop or a drive way be sure to watch for approaching persons and vehicles. Move continuously to join the traffic at a suitable speed.

BACKING

If your starting position requires backing, inspect to see what is behind your vehicle before you begin to move it. Then keep the situation well in view through your mirrors and windows. Look in the direction your vehicle is moving or using rear view camera on your dashboard which displays images from the vehicles on the center console screen, helping driver to park and maneuver more easily.

STOPPING AND PARKING

Always give a clear signal for slowing or stopping, Park your vehicle properly. Never leave your vehicle until you have stop the engine, set the hand brake and remove the keys. Do not park in restricted or prohibited areas. These areas are designated by signs or by curbs that are painted yellow. When you have to stop on a high way, be sure to park with all four wheels off the pavement, if possible. Turn on your dome light and parking light at night. Do not stop, park or leave your vehicle in any of the following places.

1. Within an intersection
2. By fire hydrants
3. In front of a driveway
4. On a crosswalk or sidewalk.
5. Beside or opposite any street excavation or obstruction when such stopping would further obstruct traffic.
6. On the roadway side of any car parked at the curb or edge of the road.
7. Beside curb space marked as bus stop; either by sign or curb painting.
8. Beside curbs painted yellow, except to discharge passengers or cargo.
9. On hill, get curve on the highway. If you must park on a hill, get off the road, set the parking brake and transmission in 8park=, low or reverse gear, as appropriate.
Block the the gear wheels of the heavily loaded trucks to prevents motion of the vehicle, if brake should fail.
10. On the vehicle on the wrong side of a street or road, headed in the wrong direction

OTHER BASIC RULED OF THE BASIC OF THE DRIVE TO THE RIGHT

Always drive on the right hand side of the roadway except when passing cars going in the same direction. When turning left, when the right half of the roadway is closed or obstructed or on a one-way road.

SLOW VEHICLES

Any vehicle moving slower than normal flow of traffic should be driven as close to the right edge of the roadway as is safe.

DON`T WEAVE

Stay in one lane but if you have to change lanes look to be sure you can do so safely, then give the proper signal before changing.

FOLLOWING

Keep a safe distance from the vehicle ahead; for each 15 kph allow one car length. At speed over 50kph , increase your safe position to two car lengths for 15kph.

RIGHT OF WAY

The car approaching on your left has the right-way. He will probably insist on it so be prepared to stop. The man on your right does not have the right-of -way, but he may take it so again,be prepared to stop. Having the right-of-way will not save your life.

HOW TO MAKE TURNS

RULES FOR RIGHT AND LEFT TURNS

Keep your vehicle in the proper line of traffic for all types of turns. Keep to the far right side of the road for turns to the right.

Keep to the left side of your traffic lane as close to the center line as possible for a turn to the left.Signal for 100 meters before turning. You may need to signal for a longer distance if you must change lanes to make the turns properly.

Do not move to the right or the left on a roadway, unless you know such a movement can be made safely , and not until after you the proper signal. For safety`s sake, avoid turning <wide= from one street to another. You may enter any lane that is safely available after first making sure that there are no conditions of danger present.

SPEED

Although the Nigerian Highway Code clearly defines speed limits, you should be guided by what is known as the <Basic Speed Rule=

It states, No person shall drive a vehicle upon a high way at speed greater than is reasonable or prudent having due regards for the traffic theon, and the surface and width of the highway, and in no event at which endangers the safety of persons or property=

This precept does not set an exact limit in miles or kilometers per hour at which you may drive. It considers all conditions that may affect your safety. These include the number and speed of other cars on the road; the present of pedestrians, the surface of other road whether smooth or rough, wet or dry, wide or narrow, how far ahead you can see in existing light conditions; and such other things as rain , fog, wind, dust and sand. For example, if you fail to notice that the car ahead of you has stopped and you strike it in the rear, you are, you are in violation of the Basic speed Rule=

WHEN YOU SHOULD SLOW DOWN

Regardless of speed limit signs you should always look for either cars or animals on the road and slow down if necessary under these conditions;

1. When you come to or across an intersection or railroad crossing.
2. When you come to or go around a curve.
3. When you come to the top or crest of a hill, keep well to the right
4. When driving upon any narrow or winding roadway.
5. When you approach any special conditions such as these:
 - a. Pedestrian,
 - b. A bicycle rider
 - c. Slow-moving vehicle
 - d. School zones and playgrounds
 - e. School guide safely patrol
 - f. Bad roads
 - g. Animals
 - h. When entering a town or village
 - i. When approaching the scene of an accident

You should slow down when there are children walking or playing in or near the street.

Children often do not watch for cars so you must watch for them. You should always slow down after dark or at times when you cannot see clearly, as in dust, fog or rain. Even 40kilommeters an hour on the open road may be too fast if it violates the precept of the basic speed rules.

PASSING AND CLEARANCE

When you overtake another vehicle and wish to pass it, pass to the left. There are also certain conditions under which you may pass to the, when such passing can be done safely. They are:

1. When the driver of the vehicle is making or about to make a left turn. In this case you may pass to the right with caution, because the driver ahead may turn right.
2. Upon a road where the pavement is wide enough and clearly marked for two or more lanes of traffic in your direction of travel. You may pass on either the left or the right in this case but only if it is done safely.

Before moving to the left or right on a highway, give a signal tell the vehicle ahead of you and the one behind, that you are changing position, don`t make the change until you have looked in the rear view mirror and to the side you wish to move toward to be the lane is clear.

PASSING ON THE LEFT- AND RESTRICTION

- 1 .On the highway where there is a broken center line,

Passing is permitted. On the positions of the highway where there is a solid center line, or two lines with a solid line closest to your side of the road, passing is prohibited.

The solid line indicates some obstruction to vision(hill curve, intersection or railroad crossing) that makes passing at this point hazardous.

2. Make sure there is time to pass before changing lanes to make the pass. Make sure there is time to

pass before changing lanes to make the pass. Make sure you have time to get back on the right hand

side of the road to avoid getting dangerously close to any car coming from the opposite direction. Allow 100 meters to make you pass.

3. In passing at a safe distance to the left , do not return to the right hand lane until you are sure that

you are safely out of the way of the vehicle behind you in the right lane. A safe rule is to wait until you

can see the car you have passed your rear view mirror, but do not turn sharply in front of it do not <cut

- in=

4.You must not pass or attempt to pass on hills, curve, hump of bridge or within 100meters of an intersection. Do not pass at any time when the road ahead is not clearly visible(87)

5.Except in a location where another vehicle may pass you on the right, when another vehicle comes

up behind you and sounds its horn, you must move over to your right and let it pass. You must never

speed up when another vehicle tries to pass you.

6. If you meet or overtake any animal or livestock, you must reduce your speed or stop as may be necessary or may be requested by any person in charge o the animal or livestock.

DRIVING AT NIGHT

Driving speed should be reduced for night driving; particularly so at dawn and dawn dusk. Too many

drivers try to drive just as fast at night as they do during the day. We are not as alert, nor do we see as

well when we are tired after a day`s work. Wide awake driving is necessary at all times and especially

at night , since we can see objects only a limited distance ahead. Driving at night with a sand pitted windshield is a safety hazard.

Don`t overdrive your headlights. You should hold to a speed that allows you to come to a complete stop when necessary in the distance you can clearly see ahead. After-dark driving requires different skills and extra care for safe travel. These are the things you should do:

1. When you meet a car at night, you should lower headlight beams(dim your lights)when within 500

meters of the approaching vehicle. Even if the other driver does not lower his beams., will be safer to keep your beams lowered and drive with due care.

2. Lower your headlight beams when following another vehicle within 200 meters

3. Lower your headlight beams when you are driving on well lighted streets.

4. Use your lower headlight beams when driving in fog or haze , and reduce your speed. Driving with

your headlight beams on in a fog is like shining your light on a mirror light is reflected back to your own eyes and blinds you.

5. Avoid looking directly into the light

6. Keep your headlights properly adjusted so that the lower beams are not aimed upward into an approaching driver's eyes.
7. Ensure that your tail lights, stop lights and turn signals are working
8. Keep your headlights and windshield clean
9. Slow down when facing the glare from approaching headlights
10. Be sure that you can stop, when it is necessary, within the distance you can see clearly ahead and watch carefully for pedestrians and animals along the road side.

CAUTION-CAR COONTROL

Keep your car under control and your brakes in good condition so that you can slow down or stop short of any possible danger or emergency. To do this, always allow yourself a SAFE DISTANCE between your car and another vehicle or person, so that you can make smooth stop, jerky stop often cause dangerous skidding.

At high speeds, even on dry roads a car may go a long way before it will slide much further. That is why you must be very careful on slippery roads.

To avoid skidding on wet or slippery surfaces, drive at a slow speed. Allow more distance between your car and others. Apply your brakes or gas pedal slowly and easily so that you can start or stop smoothly. Try to keep your wheels out of ruts. No matter what causes a skid, the method of control is always the same. If your vehicle goes into a skid.

1. Turn your front wheels in the direction the rear end is skidding. That tends to straighten you out. Turning your wheels in the wrong direction will throw you into a dangerous spin.

2. Don't over steer. That could throw the rear end into a reverse skid in the other direction and you'll be worse off than you were before.

3. Keep your foot on the accelerator to keep your rear wheels rolling, if there's nothing in front of you, feed gas gently to maintain traction.

4. Keep your foot off the brake. Jamming of the brake will make the skid worse. When you have the car under control, pump the brakes gently to stop.

If you turn off the pavement onto the sand to avoid an oncoming vehicle, do not attempt to return to

the pavement too quickly. By over correcting you're steering the front wheels tend to move sideways

until they <dig-in= the sand. This increases the chances of your vehicle overturning. Reduce your speed gradually and hold the steering firmly with both hands.

Turn slowly and carefully __ it's better to get stuck than to overturn your vehicle. Do not return to the

roadway until you are sure traffic is clear and you can do so safely. A complete stop is recommended.

SIGNS AND SIGNALS

For your guidance, a display of the traffic signs approved for use in Nigeria is included.

REMEMBER __ the stop means S-T-O-P, a complete stop. The yield sign means that you must yield right-of-way to vehicles on the street that you are about to cross or enter.

Nigerian traffic expect a straight ahead signal by hand when a driver is proceeding straight through at

an intersection under his control. This signal will assist the policeman in directing a smoother flow of safer traffic.

SAFETY EQUIPMENT OF YOUR VEHICLE

The Traffic Department and Division have many requests for the standards for efficiency of brakes, light and horns. We are not aware of any local laws giving a clean cut definition of all these standard

in some other areas.

BRAKES

Every passenger motor vehicle must have a service brake, which is usually a foot brake, capable of stopping it in 25 meters from a speed of 35 kilos per hour. An emergency brake (parking brake) is also required.

This is usually a hand brake, but in some newer vehicle models sometime it is applied by foot pressure separately from the service brake. For some late model cars, vigorous of the foot brake and

the hand brake is necessary to prevent the car from rolling back. The emergency brake must be able

to hold the vehicle on any grade upon which it is driven.

HEADLIGHTS

Your car must have at least two, headlights, one on each side of the front of the vehicle. The headlights must be strong enough to light the road so that you can see a person a person or vehicle 100 meters ahead on you, low beam, and 350 meters ahead on the high beam. They must not blind an

oncoming driver.

TAIL LIGHTS

Every vehicle must be EQUIPPED with at least one tail- light showing a red light for easy visibility a distance of 500meter to the rear. It is normally required that headlights(not parking light) and tail-lights be lighted when a vehicle is driven between 30 minutes after sunset and 30 minutes before sunrise, and at other times when needed.

STOPLIGHT

Every vehicle must be equipped with at least one rear stop light, which may be part of the tail-light or

a separate system. The stop light should display a red light when the service brake pedal is applied. The red light must be strong enough to be seen 300meters to the car day or night.

TIRES

No tire should be used which has worn to the point that fabric is exposed or has sidewall cuts or breaks which have damaged the fabric. All regular tires must have a tread configuration on the part making contact with the road surface. Specific exception to this rule may be in cases of heavy trucks

and trailers

Some people think the chief function of tires is to provide a soft ride. They do far more than that, they

are an integral part of the braking system and the transmission of power from the motor to the road.

Under-inflation not only affects the safe performance of these systems, but can weaken the tires rapidly to the point when they fail and often result in dangerous blow outs .If you should have a blowout, positive smooth braking effort at intervals much more efficient and safer than quick, hard brake applications, Hold as straight a line as possible with your vehicle by means of a tight, firm grip

on the steering wheel; otherwise, the car will turn toward the side the blown out tire is on.

OTHER EQUIPMENT

A horn which can be heard 200 meters away should be on every vehicle.

Muffler systems should be gas-tight and should not release exhaust gases to the side of the vehicle between 2 feet and 11 feet above the ground.

Never close a fuel tank spout with such materials as a plug of wood, rags or other combustible items.

An inside rear view mirror should be attached so that you can see the road to the rear car for at least

200 meters. Left and right hand out side rear view mirrors will improve your visibility.

AUTOMOTIVE SEAT BELTS

It is the responsibility of the organization to which the vehicle is assigned to see that appropriate seat

belt installations are made if the vehicle is not already so equipped. It is the responsibility of each driver to use the seat belt. It is an additional responsibility of the driver to insist that his passengers also use seat belt as provided.

Owners of privates vehicles are urged to install and use seat belts. we all know that seat belts do not

prevent accidents. We do not prevent accidents. we do know that the use of seat belt does minimize

the opportunity for, and severity of,injuriesthat might arise out of an accident.

WHEN ON THE SCENCE OF AN ACCIDENT

1. When approaching the scene of an incident, slow down. Do not park your in a manner that will congest traffic.

2. Use available means to alert other traffic.

3. Assist the injured.

4. Dial emergency and briefly describe the accident, give the exact location and stay on the phone to

answer necessary questions.

5.

WHEN YOU ARE INVOLVED IN AN ACCIDENT

1. Assist injured persons.

2. Use available means to alert other traffic.
3. Do not accept responsibility or liability for the accident to anyone.
4. Do not make payment of money to anyone claiming that you are responsible or liable
5. Do not move the vehicle until permission is received from competent authority. (FRSC law permits vehicle to transport injured person to a medical facility if an ambulance is not readily available). The vehicle must then be return to the accident venue immediately.

SAFER DRIVING SKILLS

It takes three kinds of skills to drive successfully:

1. Mechanical operation of the vehicle.
2. Correct application of traffic laws.
3. How to see and what to look for.

Most of this guide deals with the first and second of these skills. The third skill, art of seeing and knowing what to look for, is as important as the first two but is usually overlooked or at least not given sufficient attention.

By understanding and practicing from the start the seeing methods used by the experts, most drivers can form seeing habits which are the safest approach to proper driving. These methods emphasize allowance timing and spacing.

The secret is selective seeing. In a moving automobile, the driver has no time to see every detail along the road. This selective seeing is based on the following five seeing steps:

1. Aim High in Steering: Correct steering is simple ignore the fact that you sit left of the center in the car. Depend chiefly on your lower fringe vision for occasional glance well ahead at the center of your traffic lane. Seeing is usually of two kinds, sharp center eyesight which is a narrow three degree cone of vision and fringe vision which is the upper, lower and side parts your part of eye sight, your car then travels on center in its lane, and to the sides for traffic conditions.
2. Get the Big Picture: keep your central vision scanning a wide, deep roadway scene- foreground, background, side ground- for a block ahead in town; up to half a kilo away from towns.
3. Keep your eyes moving: Building a strong habit of forcing your eyes to move at least every two seconds in the driving. Check the rear view mirror each five second if any traffic is ahead or behind. Check the rear also before changing lanes and before slow down that could surprise a driver behind. The eye moving habit is restful for your eyes, keeps your mind straying too far from you driving and literally forces you to adjust speed to traffic. Weather and visibility conditions.
4. Make Sure They See You: When some one does create a conflict that threatens your safety, it means he either did not see your car or was counting on you to avoid the conflict. Be certain that the other person sees your car and shows clearly by his action that he will <stay put=.

Always tap your horn or flick your lights if in doubt. Do it early so you could stop if necessary.

5. Leave yourself an out: Recognize a basic rule of self preservation; protect yourself from

being trapped by error of other people. Adjust your pace in traffic to keep a space cushion around your car. Whenever possible poise your foot lightly above the brake when in doubt. Allow extra space when the road is slippery, when large vehicles block your view or when other cars drive too close behind you. With the habit of leaving yourself an out, you avoid impulsive moves and almost never need a hurried move. It's the smart and easy way to drive.

A FINAL RULE

Never insist on taking the right-of-way even when you are sure it is yours. The right-of-way should never be taken it must be given. This is a rule of common courtesy, which also requires that you yield the right-of-way to others when it eases a traffic situation to do so. Courtesy in yielding the right-of-way is an important factor in modern driving. Courtesy the roadway, and it saves lives.

OCCUPATIONAL HEALTH SAFETY PRACTICE INDUSTRIES

Safety is 25% Common sense, 80% Compliance, and the rest is goodliness – Safety spud.

Industrial safety can be defined as the ability to manage the risks inherent to operations, or related to the environment. Industrial safety is not a dislike of risk, rather it is a commitment to clearly identify them in relation to production operations, assess them in terms of quality, and quantity and manage them. It is the policies and protections put in place, to ensure plant and factory worker protection from hazards that could cause injury.

INDUSTRIAL SAFETY TIPS

Total, a globally reputable company in her golden rules of occupational safety: (Thoroughness and vigilance) Summarized it in these words:

<Everyone must understand the basic rules of prevention and comply with them.=

These golden rules reinforce and complete our procedures for identifying dangers and controlling risks. The essential rules were established using records of accidents, observed in our different business. They must be followed to the letter.

Everyone should intervene as soon as a deviation is observed, and stop work if risk is not controlled. The twelve themes covered by these are:

High Risk Situations

Do not start-up or shut-down equipment or installations without using the appropriate written operating procedure. Report any situation where the risk level is temporally increased. Identify and analyze induced risks implement compensatory measures.

Traffic: Machine/Vehicle/Cyclist/ Pedestrians

Do not exceed the speed limits. Traffic rules apply inside and outside sites.

Body Mechanics and Tools

Do not carry out work if you do not have the right tools the job and the environment: Adopting an inappropriate posture when handling objects or using tools can cause physical harm.

Protective Equipment

Do not access installations and perform work without wearing, general or task-specific personal Protective Equipment (PPE). Collective protection is preferred. Individual protection

completes preventive measures already taken.

Work on Powered Systems

Do not perform work without checking that the power and product source supply has been rendered inoperative. Some works require the energy to be safely discharged or the equipment to be purged and vented. In Such cases, a lock-out system is set up to Isolate the energy or product according to a specific method.

Confined Spaces

Do not enter a confined space until isolation has been verified and the atmosphere checked. Do not enter a confined space(container, tanks, well, etc) without supervision, and only after the atmosphere and isolation have been checked.

Excavation Work

Do not perform excavation Work without valid work permit comprising a map of all underground hazards.

Manual or mechanical excavation work including dredging rivers and sea beds, can only start after a risk analysis has been done performed, to identify all hazards in the zone, and related precautions have to be taken.

Working at Height

Do not work at heights without a safety harness when there is no collective protective equipment. Work at height is performed on fixed or mobile platforms with a guardrail, designed for the task at hand. Work on rooftops(buildings, reservoirs) is performed only after the roof solidity has been checked and appropriate has been set. Ladders are a means access only, their use must remain exceptional.

Change Management

Do not make any technical or organization changes without prior authorization. Any change in technique or organization must be the subject of a risk analysis.

Simultaneous Operations or Co-activities

Do not perform simultaneous or co-activities without a prior visit.

All work or operations inside a unit in operation can risk levels.

Interferences due to simultaneous operations or joint-activities must be identified and controlled.

PREVENTING ELECTRICAL HAZARDS IN THE WORK PLACE

Good electrical safety habits can help protect you and your co-workers from injury. And you can play a role in recognizing and preventing workplace hazards. Keep in mind the following:

1. Electrocution is one of the top fire causes of workplace deaths.
2. Workers younger than 25 have the highest rate of death from electrical shock.
- 3.Many workplace electrocutions involve touching a power line with long or tall equipment.

STAYING ALERT

Many electrical injuries could be prevented if people where alert to hazards. Stay aware by keeping focused on your job and do not let emotions like anger and frustration get in the way.

CHECK CORDS AND CONNECTION

Before you start work, check electric cords for wears. If you are outside or in a wet location, be

sure tools and extension cords are suitable for outdoor use and circuits are equipped with Ground Fault Circuit Interrupter (GFCI). The ground f-fault circuit interrupters (GCFI) is a fast- acting circuit- breaker designed to shut-off electric power in the event of a ground-fault, within as little as 1/40 of a second. It works by comparing the amount of current going to, and returning from equipment along the circuit conductors. It is also called Residual Current Device (RCD) and shuts off an unintended path, possibly through water or through a person. It is used to reduce the risk of electrical shocks. Also keep cords free of oil, heat and corrosion chemicals. Never yank. Kink or bend cords, and store them loosely coiled in a dry place.

POWER TOOLS

Never carry a tool by its cord. Be sure a tool is switched <off= before plugging or unplugging. This protects you and the next person who uses it. Watch out for energized areas when reaching into equipment. Shields, barriers, insulation and GFCI protects you, so do not modify them just to get a job done faster. Learn and follow your company lockout/tagout procedures. If in doubt ask a qualified electrical worker for help.

Dirt and dampness increase the risk of shock. Keep your tools, work areas and storage space clear and dry. When cleaning electrical equipment, be sure it is unplugged and follow the manufacturer`s cleaning instructions.

USE SAFETY GEAR

Personal Protective Equipment(PPE) is your first line defense against shock and electrical burns. Keep boots, gloves and other gear in good condition, even a pinhole let electricity through. Wear non- conductive protection on your head, face, hand, and feet. Use insulated tools or handling equipment, such as no-conductive ropes and protective shields.

PREVENTING ELECTRICAL FIRE

Most electrical fires can be traced to overheated circuits or overloaded equipment. When abused, insulation may melt or burn, exposing live wires. Electrical fires can also occur when equipment is driven beyond capacity; when accumulated oil and dirt overheat a motor or when sparks ignite scraps, dirt, dust or flammable liquids. Observe the following rules;

- Use and maintain wiring, tools and equipment correctly. Keep everything oil and dust-free.
- Uncoil an extension cord fully before use. Find the amperage marked on it. Is it adequate ?
- Don not use equipment that gives off mild shock, unusual heat or odd smells. If in doubt, have it checked and repaired or replaced.
- Sweep up scraps and sawdust, and store flammable liquids in approved containers.
- Do not use electrical equipment when flammable gases, vapours, liquids, dust or fibers are present.

A FIRE SAFETY CHECKLIST

Visualize your plan of response in a fire, so you can move quickly if one occurs. Take into account;

- The nearest multi-purpose fire extinguisher and how to use it.
- The nearest emergency exit or fire escape.
- Your company procedures for notifying fire fighters and other emergency personal.
- Try to extinguish a fire yourself, only if you are trained and it is small or non-threatening. When

in doubt, get out and take others with you.

INDUSTRIAL ACCIDENTS

An industrial accident is sudden and unanticipated event that can result in injuries, fatalities property damage and lost production time. Mishaps in the in the work place are fairly frequent, despite federal laws to protect workers safety. Ignoring safety procedures and using poor judgment are among factors that put workers at greater risk, resulting in serious job-related injury or illness. Industrial accident, work accident, work place accident, occupational accident or accident at work is a <discrete occurrence in the course of work is a <leading to physical or mental occupational injury.

According to the International Labor Organization (ILO). More than 337 million diseases in more than 337 million accidents happen on the job each year, resulting together with occupational disease in more than 2.3 million death annually. The phase <in the course of work= can include work-related accidents happening off the company`s premises, and can include accidents caused by third parties according to Eurostats. The definition of work accidents include occurring 8while engaged in an economic activity, or at work, or carrying on the business of the employer=. According to ILO, the phrase <Physical or mental harm= means any injury, disease or death.

Occupational accidents differ from occupational disease, as accidents are unexpected and unplanned occurrences e.g(mine collapse,etc).while occupational disease, contracted as a result of an exposure over a period of time, to risk factors arising from work activity= e.g(miners lung)

Incidents that fall within the definition of occupational accidents include cases of acute poisoning, attacks by humans and animals, insect,slips and falls, on payments or staircases, traffic collisions, and accidents on board means of transportation in the course of work, accidents in airports stations and so on.

There is no consequences as to whether commuting accidents, that is, =accidents on the way to work and while returning home after work= should be considered to be work accident. The Europeans statistics on accidents at work(ESAW) methodology exclude them, The international Labour Organization (ILO) includes them in its conventions concerning health and Safety at work, although it list them as a separate category of accidents and some countries for example Greece do not distinguish them from other work accidents.

A fatal accident at work is defined as an accident which leads to death of a victim. The time within which the death may occur, varies among countries. In Netherland, an accident is registered as fatal if the victim dies during same day that the accident happened. In Germany, if death came within 30 days, while Belgium, France and Greece set no time limit. Where the accident involve multiple fatalities, they are often referred to as industrial, they are often referred to as industrial disasters.

COMMON CAUSES OF ACCIDENT

About 80 out of every 100 accidents are the fault of the person involved in the incident. Unsafe acts cause four times time as many accidents and injuries conditions.

Accidents occur for many reasons. In most industries, people tend to look for things to blame= when an accident happened=; because it is easier than looking for= root causes=such as the underlisted:

Taking short-cuts: Everyday we, make decisions we hope will make the job faster and more efficient. But do time-severs ever your own safety or that of other crew members? Short -cuts that reduce your safety on the job are not short-cuts, but an increased chance for injury.

Over-confidence: Confidence is a good thing; over-confident is too much of a good thing. <it will never happen to me= is an attitude that can lead to improper procedures, tool or methods in your work. Any of these can lead to an injury.

Incomplete Instructions: To do the job safely and right the first time, you need complete information. Don`t be shy about asking for explanations about work procedures and safety precautions.

Poor House Keeping: When clients, managers or safety professionals walk through your work site, house keeping s an accurate indication of every ones attitude quality, production and safety. Poor house keeping creates hazards of all types. A well- maintained area sets a standard for others to follow. Good house keeping involves both pride and safety.

Ignoring Safety Procedures: Purposely failing to observe safety procedures can endanger you and your coworkers. You are being paid to follow the company safety policies: not to make your own rules. Being casual about safety can lead to a casualty.

Mental Distractions: Having a bad day at home and worrying about it at work is hazardous combination. Dropping your <mental= guard can pull your focus away from work procedures. You can also be distracted when answering phones not relating to your work., or when a friend comes by to talk while you are busy with work. Do not become a statistic because you took your eyes or focus off the machines <just a minute=.

Failure to plan: There is a lot of talk today about job hazard Analysis. JHA`S are an effective way to figure out the smartest ways to work safely and effectively .Being hasty in starting a task. Or in not thinking through the process can put you in a harm`s way. Instead plan your work, and work your plan.

Remember. Mark Twain said

<It is better to careful many times than to die once=

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- Use and maintain wiring, tools and equipment correctly. Keep everything oil and dust-free.
- Uncoil an extension cord fully before use. Find the amperage marked on it. Is it adequate?
- Don not use equipment that gives off mild shock, unusual heat or odd smells. If in doubt, have it checked and repaired or replaced.
- Sweep up scraps and sawdust, and store flammable liquids in approved containers.
- Do not use electrical equipment when flammable gases, vapours, liquids, dust or fibers are present.

A FIRE SAFETY CHECKLIST

Visualize your plan of response in a fire, so you can move quickly if one occurs. Take into account;

- The nearest multi-purpose fire extinguisher and how to use it.
- The nearest emergency exit or fire escape.
- Your company procedures for notifying fire fighters and other emergency personal.
- Try to extinguish a fire yourself, only if you are trained and it is small or non-threatening. When in doubt, get out and take others with you.

INDUSTRIAL ACCIDENTS

An industrial accident is sudden and unanticipated event that can result in injuries, fatalities property damage and lost production time. Mishaps in the in the work place are fairly frequent, despite federal laws to protect workers safety. Ignoring safety procedures and using poor judgment are among factors that put workers at greater risk, resulting in serious job-related injury or illness. Industrial accident, work accident, work place accident, occupational accident or accident at work is a <discrete occurrence in the course of work is a <leading to physical or mental occupational injury.

According to the International Labor Organization (ILO). More than 337 million diseases in more than 337 million accidents happen on the job each year, resulting together with occupational disease in more than 2.3 million death annually. The phrase <in the course of work= can include work-related accidents happening off the company`s premises, and can include accidents caused by third parties according to Euro stats. The definition of work accidents include occurring 8while engaged in an economic activity, or at work, or carrying on the business of the employer=. According to ILO, the phrase <Physical or mental harm= means any injury, disease or death.

Occupational accidents differ from occupational disease, as accidents are unexpected and unplanned occurrences e.g(mine collapse,etc).while occupational disease, contracted as a result of an exposure over a period of time, to risk factors arising from work activity=

e.g(miners lung)

Incidents that fall within the definition of occupational accidents include cases of acute poisoning, attacks by humans and animals, insect, slips and falls, on pavements or staircases, traffic collisions, and accidents on board means of transportation in the course of work, accidents in airports stations and so on.

There is no consequences as to whether commuting accidents, that is, <accidents on the way to work and while returning home after work= should be considered to be work accident. The Europeans statistics on accidents at work(ESAW) methodology exclude them, The international Labour Organization (ILO) includes them in its conventions concerning health and Safety at work, although it list them as a separate category of accidents and some countries for example Greece do not distinguish them from other work accidents.

A fatal accident at work is defined as an accident which leads to death of a victim. The time within which the death may occur, varies among countries. In Netherland, an accident is registered as fatal if the victim dies during same day that the accident happened. In Germany, if death came within 30 days, while Belgium, France and Greece set no time limit. Where the accidents involve multiple fatalities, they are often referred to as industrial, they are often referred to as industrial disasters.

COMMON CAUSES OF ACCIDENT

About 80 out of every 100 accidents are the fault of the person involved in the incident.

Unsafe acts cause four times time as many accidents and injuries conditions.

Accidents occur for many reasons. In most industries, people tend to look for things to blame= when an accident happened=; because it is easier than looking for= root causes= such as the under listed:

Taking short-cuts: Every day we, make decisions we hope will make the job faster and more efficient. But do time-severs ever your own safety or that of other crew members? Short -cuts that reduce your safety on the job are not short-cuts, but an increased chance for injury.

Over-confidence: Confidence is a good thing; over-confident is too much of a good thing. <it will never happen to me= is an attitude that can lead to improper procedures, tool or methods in your work. Any of these can lead to an injury.

Incomplete Instructions: To do the job safely and right the first time, you need complete information. Don`t be shy about asking for explanations about work procedures and safety precautions.

Poor House Keeping: When clients, managers or safety professionals walk through your work site, housekeeping an accurate indication of every ones attitude quality, production and safety. Poor housekeeping creates hazards of all types. A well- maintained area sets a standard for others to follow. Good housekeeping involves both pride and safety.

Ignoring Safety Procedures: Purposely failing to observe safety procedures can endanger you and your coworkers. You are being paid to follow the company safety policies: not to make your own rules. Being casual about safety can lead to a casualty.

Mental Distractions: Having a bad day at home and worrying about it at work is hazardous combination. Dropping your <mental= guard can pull your focus away from work procedures.

You can also be distracted when answering phones not relating to your work., or when a friend comes by to talk while you are busy with work. Do not become a statistic because you took your eyes or focus off the machines <just a minute=.

Failure to plan: There is a lot of talk today about job hazard Analysis. JHA`S are an effective way to figure out the smartest ways to work safely and effectively .Being hasty in starting a task. Or in not thinking through the process can put you in a harm`s way. Instead plan your work, and work your plan.

Remember. Mark Twain said

<It is better to be careful many times than to die once=

Disaster

Management

INTRODUCTION

Disaster, means an event, which brings sudden great fortune bringing disruption to normal life and causing colossal loss of life and property and disruption of humans activities. Disasters are situations of great calamity like earthquake, floods etc.

Disaster can also be defined as catastrophe, mishap, calamity or grave occurrence in any area arising from natural or man made causes, or by accident or negligence which results in substantial loss of life or human suffering or damage to, and destruction of, property, or damage to degradation of environment, and is of such a nature or magnitude as to be beyond the coping capacity of the community of the affected areas.

Natural disasters are global phenomena, which may occur anywhere at any time. These adversely affect the lives of the people and cause considerable damage to property, environment and infrastructure worldwide. Although the basic cause of most of the natural disasters is also dependent on man made changes caused in the environment. However, hazards and disasters are untimely related and are synonymous to each other. Virtually hazard is a threat whereas disaster is an event.

□ India is one of the most disaster-prone countries of the world. The following data clearly show that India is a disaster-prone country:

- 65 per cent of the land area is vulnerable to earthquakes
- 12 per cent of the land area is vulnerable to cyclones
- 70 per cent of the land under cultivation is prone to drought.

Crisis is an event of acute danger, which can cause sudden disruption of work.

The event may be caused due to natural cause, human error, equipment failure or sabotage by anti-social elements. The basic idea in crises(emergency situation) is to avoid wastage of crucial time. Terrorist threats and attacks, bomb explosion, strike, accidents, fire accidents, etc. are the examples of situations.

Disaster management Is a comprehensive term which includes all aspects of preventive and protective measure, preparedness and relief operations for reducing the impact of disaster humans.

Disaster management is a continuous and integrated process of planning, organizing, coordination and implementing measure, which are necessary or expedient for:

- (i) Prevention of danger or threat of any disaster
- (ii) Mitigation or reduction of risk of and disaster or its severity or consequences.
- (iii) Capacity building
- (iv) Preparedness to deal with any disaster
- (v) Prompt response to any threatening disaster situation or disaster
- (vi) Assessing the severity or magnitude of the effect of any disaster
- (vii) Evacuation, rescue and relief
- (viii) Rehabilitation and reconstruction

Poverty and lagging development amplify the adverse effects of natural disasters. Therefore developing countries (particularly their most densely populated regions) are more vulnerable to natural disasters because of their limited capacity to prevent and absorb these effects. Poor communities get forced to live in crowded, makeshift houses and in areas vulnerable to bad weather or seismic activity having low quality of infrastructure.

Mitigation efforts lessen the impact disasters have on people and property. Mitigation measures include keeping homes away from flood plains, designing structures to withstand earthquakes. The vulnerability to natural hazards can be reduced by altering, averting, avoiding and adapting to the hazard.

Beside the natural disasters, there are many man-made or technological oriented physical events, which result in a significant disruption in the normal working. The event which causes the sudden disruption of work, is known as crisis. The crisis management plan shall be able to respond quickly and effectively to such unexpected events and situation. The management includes the preventive measures need to be adopted to avoid any crisis and the ways and means to tackle a crisis, if it occurs in spite of preventive measures. Thus, crisis management helps in avoiding or minimizing wastage of crucial time and cost.

OBJECTIVES

The following are the major objectives of disaster/crisis

management

1. To improve state of preparedness to meet any contingency,
2. To reduce response time in organizing the assistance,
3. To identify major resources, manpower and the equipment needed to make the disaster management plan operational.
4. Making optimum use of the combined resources,
5. To identify mitigating measures and take timely action,
6. To avoid panic situations and rumors at the time of contingency.

The objective of preparing disaster management plan is to enable various departments/agencies to plan for, quickly respond to and recover from unexpected events and situations. This plan shall provide necessary guidelines for assistance to various sections for ensuring of people, protection of property and restoration of normal life including water and power supply, communication, etc.

CLASSIFICATION OF DISASTERS

The classification of disaster can be based on the origination agent, or the impact, or whether the disaster is preceded by a warning phase or not. The disasters may be broadly grouped into major or minor types depending upon their potential to cause damage to human life and property. The disasters are generally classification as major or minor.

NATURAL DISASTER

These can be further sub-classified as under as under:

1. Meteorological Disasters. These include:
 - (i)Cyclones ,(ii)Hurricanes,(iii)Floods,(iv)Droughts,(v)Hailstorm, cloudburst
- 2.Topographical Disasters. These include:
 - (i)Landslides,(ii)Avalanches(i.e, fall of mass of snow, rock and ice down a mountain)
- 3.Tectonic Disasters. These include:
 - (i)Earthquakes, (ii) Volcanoes
- 4.Infestic Disasters. These include:
 - (i)Epidemics,(ii)Pest attack, Locust invasion of crops
- 5.Chemical, Industrial and Nuclear Disasters
- 6.Accident related disasters. These include:
 - (i)Fires,(ii) Mine Flooding.(iii)Major building collapse(iv)Air, rail, road
 - (v)stampede

The accidents may be either natural or man-made (intentionally like bomb blast) or due to mistake or carelessness, etc

Different natural calamities can be distinguished from each other in terms of their nature and extent of their impact.

(a) Calamities like earthquakes hailstorms, avalanches and landslides occurs quite suddenly but they are restricted in their impact in terms of time and space.

(b) Floods and Cyclones occur with some element of warning yet their occurrence is confined in duration.

(c) Drought spans over a much longer time-frame and its adverse impact on the economic activities and life of an area is of a more lasting nature.

Natural disasters may be broadly grouped into major and minor types depending upon their potential to cause damage. For example:

(i) Major disasters. Earthquake, flood, drought, cyclone.

(2) Minor disasters. Landslide, avalanche, tornadoes, hailstorm, heat wave, cold wave.

MAN-MADE DISASTERS

Man-made disasters are the results of follies and unwise acts of human beings. Following man-made factors increase vulnerability:

1. Density of population
2. Population living below poverty line(PBL)
3. Low percentage of literacy
4. Pattern of occupation and situation in employment.

About 70% people of our country reside in villages and the mode of their employment revolve around agriculture. The agriculture sector is always vulnerable to drought, floods and man-made problems. such as spills from oil and gas industrial activities By and large farmers lack knowledge to face disasters. Natural disasters with man-made factors of vulnerability make the overall situation highly volatile.

As compared to the developed countries, the third world countries suffer more relative loss to lives, livestock, and property. The third world countries have weak socio-economic infra-structure and do not have adequate prevention, preparedness and mitigation against disasters.

Other man-made disasters are:

(i) Communal riots, (ii) Ethnic conflict, (iii) Refugee situation, (iv) Terrorism, (v) stampede, (vi) Food poisoning, (vii) Alcohol/Liquor tragedies.

NATIONAL POLICY ON DISASTER MANAGENT

As discussed above disasters can strike our area at any time with or without any fore warning. Therefore disaster management policy is proactive and is comprehensive and includes the measures of preparedness, mitigation and prevention . In view of the changing dimension and probabilities of the natural disasters a new approach of strategic management is being followed in the national policy.

Paper 8 of 9 — MEE 317 - HAZARD/RISK ASSESSMENT AND POSSIBLE SOLUTIONS TO AVOIDING ACCIDENTS ON FUPRE ROA

Level 300 • First Semester • Handout

HAZARD/RISK ASSESSMENT AND POSSIBLE
SOLUTIONS TO AVOIDING ACCIDENTS ON FUPRE
ROAD(NEW ROAD PROJECT)

By
GROUP 2

Course code: MEE 317

Course Title: BASIC SAFETY ENGINEERING

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ABSTRACT

The FUPRE New road project is a very significant road project to the university as it is the only access into and out of the school upon completion it is therefore necessary to ensure the safety of the road both for the workers on the site and for the subsequent users of the road. This study is aimed at assessing the risks that are posed by the current condition of the road and to seek for means to mitigate the harm which such risks could cause. A systematic analysis carried out on this road project identified several hazards on the new road site such as : The presence of an unprotected and uncovered drainage system , the

presence of grass and refuse in the drainage system, the presence of wood stumps and plants roots on the site e.t.c. The likelihood of these hazards were studied as well as their potential impact and possible solutions to mitigate the risk the hazards may pose were suggested.

INTRODUCTION

The FUPRE New Road Project is a route to the university connecting the FUPRE main gate to other parts of the school and is as well the only access route into and out of the university. It is thereby very necessary to ensure that road is free from all possible causes of harm to man and vehicles upon the completion of the road project.

Hazard/Risk assessment is a systematic process that identifies potential hazards, evaluates the risks the hazards pose and mitigates those risks in an environment. An assessment of the FUPRE road project was done by visual inspection and observation to identify the different hazard that are on-site, adequate measures taken can also aid in mitigating the risk which the identified hazards could pose.

THEORY

A hazard is a situation or process that can cause harm, such as injury, property damage, or environmental degradation. It simply refers to the potential for a harm to occur, while a risk simply means the likelihood of a harm to occur due to the presence of a hazard. The presence of hazards in the environment is an inevitable factor since anything in one's environment could be a hazard when not in its right place or when not handled appropriately, it is however necessary to take precautions to prevent imminent danger in an environment. In hazard/risk assessment several actions are carried out to control or eliminate those risks. In this study focus was made on the FUPRE New road project assessing the risks with respect to the health and safety risk that the hazard may pose. The steps taken in the assessment of risk on FUPRE New road project are as follows;

1. Identification of hazards on the FUPRE New road site, a consideration was made to what could cause harm in the work area.
2. Determination of who or what could be harmed on the site , (in most cases who could be harmed were the workers on the site, the students and the staff accessing the road)
3. Evaluation of the risk, an analysis was made as regarding

the risk and appropriate solutions were suggested

4. Review and update, a review was carried out to check the assessment and make update where necessary.

DISCUSSION OF RESULTS

The primary aim of this activity was to evaluate hazards and risks on the New FUPRE road site and prefer adequate solutions in each case in order to mitigate such risk , during this risk assessment and analysis, a visual inspection of the new FUPRE road site was carried out to observe and identify potential hazards on the site, some of the hazards identified during the study include;

¥ Undulating or uneven land area

¥ The presence of an unprotected and uncovered drainage system

¥ The presence of grass and refuse in the drainage system

¥ Presence of wood stumps and plants roots on the site

¥ Presence of a soak away pit close to the road site

¥ Looseness of soil particles due to continuous planting over the previous years

¥ Absence of caution signs to indicate that it is a construction area

1. Undulating and uneven land area: The uneven land area of the FUPRE New road site is a factor which could lead to the loss of traction between the vehicles being used on site and the ground this could lead to the overturning of the vehicles as well as the instability of the vehicles while being used. A solution to this could be to resurface the road and creating an evenly surfaced for the road to be constructed.

2. Presence of unprotected drainage system : Although drainage networks are needed by the FUPRE New route site to facilitate to run off of rain water it is necessary that it should be done to avoid future harm in the environment, an unprotected drainage system poses the risk of workers tripping into the drainage and also of vehicles skidding into the open drainage system. In order to prevent this hazard, guardrails and barriers can be provided over potential drainage entries and with mesh installed at the various drainage openings.

3. Grass ,plants and refuse in the drainage system: The presence of vegetation and debris in the drainage could

pose the risk of erosion and flooding due to the difficulty for it to allow water to pass through it, this could subsequently also damage the drainage system. A solution to this hazard is to ensure a proper periodic clearing of the canal to remove debris and grass from it.

4. Presence of wood

stumps and plant roots: The presence of wood stump on the new FUPRE road site could lead to the vehicle being used to swerve or lose control and poses a possibility of damages to the tyre of such vehicles wood stumps could also cause the workers on site to trip and also get injured. In order to mitigate this danger, it is necessary to remove such wood stumps and plants roots from the site to ensure a safer work area.

5. Presence of a soak away pit close to the road site: Soak away pits being close to a road site stands the risk of an unplanned collapse due to the vibrations of the surrounding area caused by passing vehicles. As a solution to this hazard soak away pits should only be constructed with reinforcements at a reasonable distance away from the road. Putting up warning signs in the soak away region could also warn drivers of their speed in the soak away area, regular checks can also be done to monitor and prevent the collapse of the soak away pit.

6. Looseness of soil particles due to previous continuous cultivation : The cultivation of the FUPRE new road site over the previous year has led to the particles of the soil being loose and could pose serious dangers to the vehicles being used at the site for construction, dangers such as the sinking of the tyres of the vehicles, and can also lead to erosion of the soil in the site. To mitigate these hazard it is necessary to stabilize the soil with compacting methods to ensure that the soil particles that tightly held together

7. Absence of caution signs for drivers and workers on the site: The absence of caution signs on the new road site can mislead the drivers and workers on the road site making them unaware of the impending dangers they may face in several areas while working on the site. To mitigate this hazard caution signs should be placed at different locations on the site to indicate the different conditions of the site.

8. Bushy vegetation close to the road: the presence of a bushy

vegetation close to the road site could be a hide out for predators that can be of harm to the workers on-site , to mitigate this hazard a regular clearing of the area around the road can should be done to expose possible hideouts for such predators and rodents.

CONCLUSION

The hazard assessment of the FUPRE New Road project emphasizes the need for a systematic approach to evaluating risks on the FUPRE road which could compromise the safety and functionality of the road on it's completion. The hazards identified on the road could pose significant challenges to the road users if not addressed on time. Addressing such hazards is essential to ensure the road meets the needs of its users while minimizing risks to safety both of site workers the road users and the road itself. The proposed solutions, provide a comprehensive strategy to mitigate the identified hazards, adopting a proactive monitoring system will enhance the long-term sustainability and reliability of the road. By prioritizing hazard assessment and evaluation, the FUPRE New Road project will serve the road users better on its completion and will be free from potential hazards. This assessment however shows the importance of safety in engineering, especially on the FUPRE New road project to ensure it's safety both for workers , staff and students who will in time use the road after it's completion.

INDUSTRIAL SAFETY ANALYSIS

GENERAL

Safety analysis is a necessity in today's litigious society. Companies that do not analyze may not prevent accidents which are preventable and may find their ability to defend themselves in a post accident lawsuit compromised. The standard to which the courts hold management is prudence, foresee ability, and reasonableness. Prudence requires that (1) production systems, products, and facilities be analyzed for safety, and (2) proper preventive measures follow the analysis. A prudent manager analyzes. Attempt must be made to foresee the foreseeable. This attempt must be reasonable. The key is reason. Analysis is simply the application of reason. There are two types of reason inductive and deductive. If a company can demonstrate that they made a reasonable attempt to foresee the foreseeable by applying system safety analysis inductive and deductive techniques and responding to the analyses with proper preventive actions, the company has a strong legal position. They can show they were reasonable and prudent in trying to foresee and avoid accidents, provided that they have carefully documented their efforts. The philosophy developed by system safety engineers is that accident prevention can and must

begin as soon as the idea for a new system or operation is conceived. Methods of analyses have been developed to this end which can be undertaken early. The analyses can indicate tentatively any hazards that might be present in any proposed operation, the types and degrees of accidents that might result from hazards, and the measures which can be taken to avoid or minimize accidents or their consequences. As more and more details become available after the initial analysis on the characteristics of the proposed system or operation, they can be expanded further and further to determine more intimately where there might be potential causes of accidents, their effects and safeguards needed. Almost 90 techniques have been developed, used, or proposed. This chapter includes a few important ones to indicate some of the methodologies now in wide use. This may make occupational safety engineers aware of some types of detailed analyses they might want to undertake.

Some of these techniques start at the component level and reason upward to find what "undesirable end events" might occur from that components fault or failure. These "bottom-up" techniques are based on inductive reasoning. Some techniques start with the undesired event which is to be avoided and reason down to identify which components of the system could contribute to the undesired event through their fault or failure. In systems where there is a critical potential for catastrophe, the prudent manager must make every effort to foresee and avoid the catastrophe. Everything is not foreseeable, but the manager must make a thorough effort to foresee when there is a potential for a catastrophe. This means the manager will use both inductive and deductive system safety analysis, if feasible. Preferably, the inductive analysis will be conducted blind to the deductive analysis, and vice versa. The results of the two analysis will then be compared. Hopefully, the undesirable end events identified in the inductive analysis will match those the deductive analysis started with; the component faults and failures identified in the deductive analysis will match those components the inductive analysis started with. Where the inductive and deductive techniques do not match, the deficient analysis is corrected. This double-blind cross-check is a strong indicator of a reasonable effort to foresee the foreseeable, and when accompanied by prudent corrective actions, places management in a very defensible position.

PRELIMINARY HAZARDS ANALYSIS

The first method generally used in the systematic process of determining hazard causes, effects, and deterrents is the preliminary hazards analysis. As its name implies. this is an initial study from which analysis efforts can be expanded further. It is fairly broad in scope, investigates what hazards might be present, whether they can be eliminated entirely, or if not, the best way to control them. If the hazard cannot be eliminated, the analyst determines whether there are standards or methods by which the hazard could, should, or must be controlled. A review is made of the functions to be performed and whether the environments in which they must be performed will have any adverse effects on personnel, equipment, facilities, or operations. According to Greek mythology, Daedalus wanted to escape from Crete where he and his son Icarus were being held captive by the king because of his skill as an artificer. Daedalus made wings of feathers, flax, and beeswax with which he and Icarus could fly to Greece. Before they flew off, Daedalus told Icarus:

My boy, take care

To wing your course

Along the middle air: If low, the surges

If high, the sun The melting wax consumes.

Daedalus's instruction to his son can be taken to be an elementary preliminary hazard analysis. However, Daedalus, not being an experienced safety engineer, did not go far modern Safety Analysis

Safety engineers are smarter than designers. We safety engineers can figure out potential ways of getting killed which Daedalus didn't even consider, and we can even indicate safety measures by which he could have prevented Icarus's accident (he flew too close to the sun, the wax melted off his wings and he fell to his death). Modern designers, on the other hand are not nearly as smart as Dardalus, they have never been able to figure out how to make flyable wings out of feathers, flax, and beeswax.] Each product or operation have a limited number of hazards, which can be determined as soon as a few facts are known. For example, the proposed product is to use electrical power. The hazards, which can potentially be present when electrical power is in me, could include any or all of the following electrical shock, burns from hot equipment, fire due to arcing, sparking, or very hot surfaces; inadvertent starts of equipment, failure of equipment to operate at a critical time so that an accident will occur, radiation effects, and electrical explosions. Numerous conditions can lead to each of these, but these are the basic hazards. If the electrical product uses small batteries such as for transistorized equipment, most of the hazards mentioned will not present. The analysis can therefore be limited to those which could possibly exist, no matter how improbable Progress of the design can be monitored thereafter to determine whether the hazard can be eliminated; and if it cannot be, how best it could be controlled, and if it is controlled, whether or not the control appears to be adequate and how the adequacy of the control can be verified. At this point, very rough estimates might be made of the probability of an accident due to that hazard, and of the severity of its probable effects.

A practice has been initiated in which the preliminary hazards analysis list, which is prepared in tabular form, is broken down into individual items. Each item is recorded (Fig. 15-2) and then tracked through development or modification of a system, product, or operation to ensure that adequate consideration is given to its hazard elimination control. (To assist in the tracking, visualizations, such as in the fault-tree method described later, can be used.) When all necessary actions are completed to eliminate or acceptably control the hazard, the item is signed off, since no further action is required. (An item should never be signed off with a comment that an action will be taken, is scheduled to take place, or a similar entry Signoff is considered accomplished only after the proposed action has been completed and shown to be adequate.) As design or planning progresses, studies are made of the hardware and facilities through reviews of assemblies of their major components, their proposed interrelationships and interfaces with each other and with personnel, environments that could affect them, and the effects they could generate on personnel, other components and assemblies, and the environment.

Some information does not lend itself well to an analysis presented solely in a tabular form. A narrative format can then be added to include additional data. For example an analysis presented in columnar form might indicate that a hazard was a accidental fire. To have a fire requires a fuel, oxidizer, ignitable mixture, and source of ignition. Detailed information on each item could then be listed separately, with details on the many facts, oxidizers, and sources of modern Safety Analysis.

A practice has been initiated in which the preliminary hazards analysis list, which is prepared in tabular form, is broken down into individual items. Each item is recorded and then tracked through development or modification of a system, product, or operation to ensure that adequate consideration is given to its hazard elimination control. (To assist in the tracking, visualizations, such as in the fault-tree method described later, can be used.) When all necessary actions are completed to eliminate or acceptably control the hazard, the item is signed off, since no further action is required. (An item should never be signed off with a comment that an action will be taken, is scheduled to take place, or a similar entry Signoff is considered accomplished only after the proposed action has been completed and shown to be adequate.) As design or planning progresses, studies are made of the hardware and facilities through reviews of assemblies of their major components, their proposed interrelationships and interfaces with each other and with personnel, environments that could affect them, and the effects they could generate on personnel, other components and assemblies, and the environment.

FAILURE MODES AND EFFECTS ANALYSIS

One of the methods used to accomplish safety analyses has been derived from reliability engineering Failure Modes and Effects Analysis (FMEA). FMEA is an inductive technique. It is used to determine how long a piece of complex equipment will operate satisfactorily and what the effects of any failure of individual components might be. The method is intended for analyses of proposed equipment and systems. An occasion might arise in which it would be highly desirable for an occupational safety engineer to determine whether or not the manufacturer of industrial equipment and system to be purchased and installed had had analysis made, and to understand how failure might occur, modes and frequencies of failures, and the necessity for proper and timely maintenance and replacements

In this method of analysis, the constituent major assemblies of the product to be analyzed are listed. Each assembly broken down into subassemblies and their components. Each component is studied to determine how it could malfunction and cause downstream effects. Effect might result on other components, and then higher-level subassemblies, assemblies, and the entire product or system. Failure rates for each item are determined and listed. The calculations are used to determine how long a piece of hardware is expected to operate between failures, and the overall probability that it will operate for a specific length of time. It is the best and principal means of determining where components and designs must be improved to increase the operational life of a product. It is best used to analyze how often and when parts must be replaced if a failure, possibly affecting safety, must be avoided

Until the use of Boolean mathematics, described in the next paragraph, FMEA calculations were

often erroneous. Also, because many component failures would have to effect on safety, that aspect of an FMEA does not involve accident possibilities. Also failure modes and effects analysis is limited to determination of all causes and effects, hazardous or not. Further, the FMEA does very little to analyze problem which could arise from operator errors (unless the system analysis includes the human as a component), or hazardous characteristics of equipment created by had design of adverse environments (unless the scope of the analysis includes these elements). The FMEA is excellent for determining optimum points for improving and controlling product quality. Another method, fault-tree analysis has been found more effective for safety purposes

FAULT-TREE ANALYSIS (FTA)

In 1959 the Air Force became concerned with the potentially catastrophic events that could occur with the Minuteman missile then being developed by Boeing. The Air Force contracted with Bell Laboratories to develop a method of analysis by which probabilities of occurrence of events with which they were concerned could be computed. The Air Force wanted to know the possibilities and probabilities that a missile could be launched or a warhead activated inadvertently, and the chance of either of these being done intentionally by an unauthorized person in an act of sabotage. Two years later Bell Laboratories completed the project. The new method, fault-tree analysis involved Boolean logic in ways similar to those being used increasingly in Analysis electronics industries. It is a deductive technique Safety engineers of the Boeing Company adopted the proposed method and became its foremost proponents. They added to the symbols that Bell Laboratories had proposed They organized the method so it could be computerized to permit calculating the probabilities of the problem with which the analyst was concerned Although still used to determine probabilities of shapes in complex systems or operations, fault tree analysis is being used far more frequently in logically analyze the possibilities of potential accidents due to hazards, and the quantitative safety level,

At first the fact that Boolean logic (and arithmetic), with which few engineers were familiar at the time was its basis gave many people the impression the method could be used only by mathematicians (Some of the first Boeing employees involved in making those early safety analyses were mathematicians) It turned out that Boolean logic was fairly simple to understand. One of the fundamental principles is that any statement, condition, act, situation, or process could be described as being in either one of two states Something could be true or false, on or off, up or down-but not both, it would go or not go; or it could be fully open or not fully open (closed or partially open). There were no middle intermediate positions.

FAULT-TREE SYMBOLS

Primary Event Symbols

Basic Event, such as a component failure Circle.

-Conditioning Event, a condition which must be satisfied before the event above the gate can occur- Ellipse

- Undeveloped Event, an event not fully developed because of a lack of information or significance - Diamond

Normal Event, expected to occur during system operation-House-shape

Intermediate and Top Event Symbol

-Output Event, those that should be developed or analyzed further to determine how they can occur (the top output event is called the end event; others are called intermediate events) –

Rectangle for gate): Any event leading into it from underneath will cause the event leading out of it at the top to occur.

Limitations of Fault-Tree Analyses One of the very few limitations of fault-tree analysis is that the tree indicates only the events included which will contribute to the occurrence of the top event.

SAFETY ANALYSIS METHODS MANDATED FOR PROCESS SAFETY MANAGEMENT

Safety analysis techniques are now mandated by regulations for some industries. This program requirement is examined below, and the techniques it suggests that are not already covered in this chapter are described and explained.

In the mid-1990s, Industries with processes that involved "highly hazardous chemicals" (HHC) were placed under requirements by OSHA (29 CFR 1910.119) to conduct process hazard analysis and have a process safety management program. The program is triggered by above-threshold quantities of any of 136 different, listed chemicals. The purpose of this standard is to prevent or minimize the consequences of a catastrophic release of toxic, reactive, flammable, or explosive highly hazardous chemicals from their use, storage, manufacturing, handling or on-site movement by industry.

The importance of this standard is that it requires safety analysis and names certain analytic techniques to use or their equivalent. The employer shall use one or more of the following methodologies that are appropriate to determine and evaluate the hazards of the process being analyzed... What-if, Checklist, What-if/Checklist. Hazard and Operability Study (HAZOP), Failure Mode and Effects Analysis (FMEA), Fault Tree Analysis, or an appropriate equivalent methodology" [29 CFR 1925.04(012)]

What-if. In "What-if?" analysis, an analysis team reviews the covered process from beginning to end. At each step in the process they think of questions that begin with "What-if... They address procedural, hardware and software errors and produce a tabular listing of these questions and their answers. This constitutes a set of potential accident scenarios, their consequences, and remedial recommendations Report form can be narrative, but use of a matrix table often provides clarity

Checklist. The "Checklist" method applies a list of specific items, usually from industry standards, consensus codes, industry guidelines, etc. The list is prepared by an experienced safety engineer, familiar with the covered process and the Limitations of Fault-Tree Analyses

Process Safety Management program

Process Safety Information

Requires compilation of written process safety information, including hazard information on

HHCS and information on technology and equipment at the covered process

Employee Involvement

Requires a written plan of action for involving employees in the conduct and development of process hazard analyses and process safety management and for providing employees access to the information required by the standard.

Process Hazard Analysis

Specifies that process hazard analysis be conducted and updated and revalidated at least every five years. Must be retained for the life of the process

Operating procedure

Requires these be in writing and include clear instructions for safely conducting activities of the covered process Steps for each operating phase, operating limits, safety and health considerations and safety systems and their functions must be readily accessible to involved workers. They must reflect current operating practice, and must implement safe work practices for special circumstances such as lockout/tag out and confined-space entry.

Training

Operators of the covered process must be trained in the overview of the process and in the operating procedures. Training must emphasize specific safety and health hazards, emergency operations, and safe work practices. Refresher training is required at least every three years.

Contractors

Contract employers are required to train their employees to safely perform their jobs at the process site. Must document that employees received and understood training and sure that contract employees know about the potential process hazards and the worksite employer's emergency action plan. Must assure that contract employees follow safety rules of the facility Must advise the worksite employer of hazards that contract work itself poses or hazards identified by contract employees.

Prestart up Safety Review

A safety review is required for new facilities and significantly modified work sites to confirm that the construction and equipment of a process are to design specs. Must assure that adequate safety, operating, maintenance, and emergency procedures are in place and that process operator training has been completed For new facilities process hazard analysis must be performed and its outstanding section items resolved and safety implemented before startup. Modified facilities must meet management of change requirements

Mechanical integrity

Written procedures must be established and maintained for the ongoing integrity of process equipment, particularly those components which are part of a covered process.

Hot Work

Hot work permits must be issued for hot work operations conducted in or near a covered process

Management of Change

Written procedures must be established and implemented to manage changes to facilities that affect a covered process Employees must be informed of changes

Incident Investigation

Require s employers to investigate as soon as possible, but no later than 48 hours after incidents which did result or could reasonably have resulted in catastrophic releases of covered chemicals. An investigation team which includes at least one person knowledgeable in the process involved and others with knowledge and experience in incident investigation and analysis must develop a written report on the incident. Reports must be retained for five years

Emergency Planning and Response Compliance Audits

An emergency action plan must be developed and implemented. It must include procedures for handling small releases

Employers must certify that they have evaluated compliance with process safety

Requirements at fast every three years. Prompt response to audit findings and documentation that deficiencies are corrected are required. The two most recent audit reports must be retained.

Trade Secret

Requires that certain information be available to employees Employers may enter into confidentiality agreement with employees to prevent disclosure of trade secrets questionnaire be applied to the covered process or system. Those parts of the design which are not adequate should be revealed when the checklist is applied. Then corrective measures are identified. What-it/Checklist. The What-if/Checklist Analysis uses the list of What-d questions and supplements it with a checklist. Therefore, it is a combination of the two methods, as the name suggests.

Hazard and Operability Study (HAZOP), HAZOP addresses each element in a process. The elements must first be identified. The expected way each element operates is determined. Deviations from the intended or expected operations and parameters are identified using guide words. The consequences from the deviating element are estimated Causes of the deviations, currently planned (or used) hazard controls, and inadequate controls are identified. The study is conducted by a team made up of five or six persons usually including a team leader familiar with HAZOP safety and health expert, a manager/supervisor, a technical person, a senior process operator, a maintenance person, a chemist, etc. The quality of the results will depend upon the quality of the team selected. The team will start by collecting process information from piping and instrumentation diagrams (P&ID), written operating instructions, material safety data sheets plant models, Safety Analysis Reports (SARs), etc. They then review each element step by step through each line or process. At each element they apply key words a guide words to tease out the potential deviations from normal expectations of a design parameter.

The hazard and operability consequences, if any, for each deviation are determined and documented. The actions or changes in plant methods which will prevent the deviation or reduce the consequences are identified. If the cost of the changes can justified. the team then must agree on the change and who will be responsible for the action. The team then must follow up to make sure that the action has been taken If the cost is not considered to be justifiable, the team must

agree to accept the risk One of more reports of the evaluation are made, usually including tabular representation of the results Team decision making is at the heart of this methodology This technique as well as the others in this chapter are supported by software. The reader is advised to "surf the net" for the latest. Arthur D. Little Company is one source.

HAZOP GUIDE WORDS

No/Not Complete negation of design intention Quantitative increase or decrease

No part of the design intention happens e.g <No flow=, No pressure= etc

More, less Quantitative increase or decrease

Refers to quantities properties and duration of design parameters, such as flow rate, temperature ,etc

As well as Something else occurs in addition to design intention

Design intention are achieved, plus additional things occur.

Part of Only part of design is achieved

Partial addition or removal of material occurs. Activities are incomplete.

Reverse The opposite of the design intion

Reverse flow of material, electrical current.voltage polarity, Backward installation (e.g check valve), opposite chemical reaction (eg.decomposition).

Other than Complete substitution

Original design intention is not accomplished, a different activity occurs-eg.leak.Rupture

Job Safety Analysis (JSA). The purpose of this technique is to uncover and correct hazards which or intrinsic to or inherent in the workplace. It should be done with a team that includes the worker, supervisor, safety engineer, and management. Its success depends upon the rigor this team exercises during five steps: (1) Select a job, (2) Break the job down into steps (3) Identify the hazards and determine the necessary controls of the hazards, (4) Apply the controls to the hazards, and (5)Evaluate the controls

Eliminating, controlling, and minimizing hazards are becoming more and more necessary for work plants to comply with the standards of regulatory agencies Formerly, chemical plants were built in which hazardous environments existed and where operators were safeguarded, theoretically, by use of protective equipment, such as gas masks, whose need was established after operations had begun. Some govern ment agencies, such as OSHA, have adopted policies of "deterrence" and will not countenance the existence of hazardous conditions. Intrinsically safe environments are being required in which operators can work without the need for protective equipment, which is often inadequate, and which workers find burdensome and inconvenient and will not wear. To ensure that regulatory agencies do not require extensive and expensive retrofits of corrective equipment, it is highly advisable that analyses be made early to recognize potential problems and their solutions. The billions required to revise and correct designs and procedures at Three Mile Island and other nuclear power plants are a example of need for proper and adequate analysis and planning for safety purposes.

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PEDESTRIAN SAFETY (PERNONAL SAFETY TIPS)

The Nigeria Police once wrote:

"You have the primary responsibility to prevent crime against yourself, loved ones and your
properties. Thinking that crime can only happen to someone else and doing nothing to prepare
yourself, or take precautionary preventive measures, makes you a perfect target for criminals.
The ultimate goal of crime prevention is to reduce the risk of being a victim. In order to
accomplish this effectively, it is important to remove opportunities for a criminal to take
advantage of you or your property.

You are attempting to prevent either victimization or criminalization by presenting an
unattractive target to the criminal. This effort. removing opportunities is often referred to as
target-hardening. Target -hardening can be as complex as installing a high tech alarm system, or
taking a self defense course, to something as simple as locking your door. Making yourself less
of a target, is simpler than you might think it further gave the following information on safety at
home, ATM Security tips

Make Sure They See You: When someone does create a . conflict that threatens your safety, it
means he either did not see your car or was counting on you to avoid the conflict. Be certain that
the other person sees your car and shows clearly by his actions that he will "stay put. Always tap
your hom or flick your lights if in doubt. Do it early so you could stop if necessary.

Never insist on taking the right-of-way, even when you are sure it is yours. The right-of-way
should never be taken - it must be given. This is a rule of common courtesy, which also requires
that you yield the right-of-way to others when it cases a traffic situation to do so. Courtesy in
yielding the right-of-way is an important factor in modern driving Courtesy smooths the roadway,
and it saves lives .Never drive without a Valid Driver's License in your immediate possession.

AFINAL RULE

No driver will give perfect attention to his driving, but habits based on observance of correct
seeing in every day driving will prevent you from becoming involved in many critical situations.

AFINAL REMINDER

Keep Your Eyes Moving Build a strong habit of forcing your eyes to move at least every two seconds in driving. Check the rear view mirror each five seconds if any traffic is ahead or behind. Check the rear also before changing lanes and before any slowdown that could surprise a driver behind. The eye moving habit is restful for your eyes, keeps your mind straying too far from your driving and literally forces you to adjust speed to traffic. weather and visibility conditions.

SAFETY AT HOME

7Make sure there are good locks on exterior doors and windows.

7 Lock up when you go out, are asleep, or alone, even if it is only for a few minutes. It takes a thief ten seconds or less to enter an open room and steal property.

7Get to know your neighbors and their daily routines,

8Keep a list of emergency numbers especially of the Police Station close to you.

7Make sure your home is fenced and if you can afford it.

7Trim flowers that hide doorways or windows.

7Set timers on lights and radios to make your home appear

lived-in, when you are away.

7 Have a neighbor regularly check on your home when you are away, and report any suspicious activities. Remember that one good turn deserves another

8Mark your valuables with a number or identification unique to you, with an engraving pen and keep a list of them in a safe place .

Replace solid walls in front yard with open fencing to eliminate hiding places and make climbing more difficult

- Install a wide-angle peephole in your front door. so you am look out to an on-coming visitor without being seen yourself

Do not give out information about your neighbors.

Don9t make it easy on burglars

8Keep valuables away from windows

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ATM SECURITY TIPS

7Be alert and conscious of your surroundings when using ATM.

Don't use an ATM machine located in a dark or obscure location.

8 Never give your card or Personal Identification Number (PIN) to anyone, for any reason

Be very conscious of persons hanging around the ATM machine Do not write your PIN on the card or anything that is kept with the card

Do not insert your card until asked to do so by the display screen

Never use an ATM with a blank screen, and if the ATM is obscured from view or properly lit, leave immediately and

find another ATM

7 Stand close to the ATM and use your body and hand as shield to make sure nobody sees you keying in your PIN Never accept help from strangers when using an ATM. You should be wary of strangers asking for help. The elderly, or physically challenged persons should take along with them trusted persons, to assist them..

7 Criminals work in teams, for example, one to distract you while the other steals your card or money. If your card is retained or swallowed by the ATM, it is advisable to contact your bank immediately Sign up for SMS alert, so that you are alerted whenever any withdrawal is made in your account

VEHICLE SECURITY TIPS

Most car crime is preventable. Police are making a concerted effort to tackle car crime, and target offenders: but motorists need to help themselves avoid becoming victims, by removing the opportunity for crime in the first place

Don't leave anything on display: it is a prime invitation to passersby. Even an old coat or plastic bag can tempt a thief. If items have to be left in the car put them out of sight at the start of journey.

Get a music system that is removable, or has a front that can be taken off, and remove ipads and mp3 players.

Driving documents and personal correspondence could help a thief to sell your vehicle or provide a cover story, if stopped by the police. Do not keep these in your vehicle Some car thieves break into houses to steal car keys.

Always keep the keys in a safe place. Do not leave them in the hallway or kitchen, as these are the obvious places.

A-Z PERSONAL SAFETY TIPS

Personal Safety is the act of ensuring that one is free from harm or danger, and or the ability to take steps towards preventing such harm

In other words, it is a list of actions that are designed to keep situations from occurring or escalating, and to make sure that physical, emotional and verbal abuse is avoided, stopped at the first moment, if it at all begins to happen. It borders on ideas such as when to stop a conversation, when to leave the room, and or call for help.

The following tips can help secure you in your daily living, while hopefully inspiring more awareness, when you are out and about

- Always be alert and aware of your surroundings, and of the people around you
- Make casual eye contact with people when walking
- Do not be easily distracted or take your eyes off someone you feel looks suspicious in a possibly dangerous situations.

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- Always carry ID Card and only the much cash need
- Whenever possible buddy up to walk out for dark parking
- Informants trusted family or friends if you are traveling your itinery dates and location if necessary
- Stay away from the bush or trees when walking of running Always be prepared to run away from an attacker and scream of confront where possible.
- . Turn and go in the opposite direction of the car if a stranger approaches you, offering a ride. If possible rite down the license plate number and description of the vehicle.
- .. Do not respond to passing comments or conversation from strangers on the street or elsewhere.
- .Walk confidently at a steady pace, and have your keys ready in your hand
- .Avoid being on a cell phone with an ipod in your other ear unaware of your surroundings. This makes you an easy target.
- Avoid dangerous places at night time. Visit them during the time hours.
- Stop and look around if you feel unsafe, entering an area Trust your instincts
- Avoid isolated bus stops otherwise, continually look all around. Be Alart
- . Avoid suspicious passenger cars/buses, one chance or all male etc.
- . Don't stay in the same spot at bus stops for long. It makes you an easy target.

If you are carrying money, don't open your purse or wallet while boarding a vehicle Have your fare ready.

- Keep flashy jewelry or other valuables of easy prey out of sight
- If you must be in an isolated vehicle, sit close to the driver
- . Check your purse or wallet, if someone is jostling. crowding or pushing you
- Never leave your purse, backpack or briefcase in plain view Lock it up when you leave your desk or office
- Keep the office door locked if you work alone or before and after normal business hours
- Try to find another worker or a security guard to walk out with you, if you work late
- . Do not get in the elevator with another person if you do not feel comfortable with that person, take the next one. If you have to get in, stand next to the control panel, so that if you are attacked you can press the alarm and the control buttons as hard as possible
- Be alert for pick pockets on crowded elevators, and other busy places
- Be aware of escape routes for emergencies and post phone numbers of police, security agencies and fire departments in your Cell phones, so that you do not have to fumble pushing many numbers.

Young women should avoid provocative clothing, and carelessly immoral gesture.

TRAVELLING, SAFETY TIPS

- 7 Notify your Resident Security Officer (RSC) or Personal Security Officer (PSO) your departure and return dates but don't otherwise publicize your travel or vacation plans.
- 7 Check plane, tram and bus times before you travel,
- 8 Sit near other people, aisles or doors Learn the location of emergency alarms and exits
- 7 Stay awake and alert when using public transportation
- 7 Conceal your passport, money, or credit cards when traveling
- 8 Keep other valuables out of sight, and luggage where possible close at hand if carrying a handbag, keep it in front of you, closed with the fastening towards your body. Keep wallet in your front pant's pocket and not back
- 7 Let go if your bag is snatched. Do not run after, especially in unfamiliar surroundings
- 7 Do some research on the area you are visiting, especially at first time. Talk to your security officer or host regarding travel advisories and warnings.
- 7 When traveling, dress casually dress down where appropriate. Be aware of local customs.
- 7 Do not wear excess jewelry Reduce wallet and purse contents, particularly cards denoting affiliations, memberships and accounts, etc
- 8 At airports proceed through security checks, and go to the boarding area, as quick as possible. These areas are usually the most secure in the airport. Be very careful anytime you make a telephone conversation.

LETTER AND PARCEL BOMBS SAFETY TIPS

Letter and parcel bombs generally are "victim activated" meaning that a victim or intended target must activate the device by opening it. They do not normally contain timing devices.

Bombs can range from the size of a cigarette packet to a large parcel Letter and package bombs have been disguised as letters, books, candy and figurines. Delivery methods have included mail systems, personal delivery, or placement at the recipient's site. A letter or parcel bomb might have some of the following indication.

- 7 Suspicious origin especially if the postmark or name of sender is unusual, unknown, or no further address is given.
- 7 Excessive inadequate postage:
- 7 Off-balance or lopsided letter or package.
- 8 Unusual weight for the size of the letter or package. Letters also may be unusually thick
- Stiffness or spring of contents when checking do not bend excessively.
- Protruding wires or components, unusual grease or oil stains on the envelope
- 7 Strange smell, particularly almond or other suspicious odours
- 7 Handwriting of sender is not familiar, or indicates foreign style not normally received by recipient Common words or names are misspelled
- 7 Rub-on or black lettering
- 7 Restrictive markings such as confidential "personal" or an horrific title appended to the name

of the addressee

7Wrapping that could be a provision for an arming safety wire.

8Rattling inside the envelope or package, possibly loose components of a device.

7Visual distractions for example currency or pornography etc

WARNING

If you identify a letter or package as suspicious, don't let anyone near it. Notify your local Bomb Disposal Unit or the nearest Police Station immediately. Leave the letter or package in open area such as a court yard, where it is easily accessible to Bomb Squad Personnel. Never submerge it in water or burn

CAR JACKING SAFETY TIPS

8When in your car, always keep the doors locked. Anytime you drive through areas containing stoplights, stop-signs or anything that significantly reduces vehicular speed, keep your windows up, especially in narrow ways and isolated places

7Leave ample maneuvering space between your vehicle and the one in front of you.. If you are approached by suspicious persons while you are stopped do not roll down windscreen, drive away quickly

If you are being followed or harassed by another driver, try to find the nearest police station, hotel or other public facility. Once you find a place of safety, don't worry about using a legal parking space, park as close as you can, and get inside fast, especially if the threat is on your life.

7 If another driver tries to force you to pull over, or to cut you off, keep driving and try to get away. Try to note the license plate number of the car, and a description of the driver. If this effort places you in danger, don't do it. The information is not as important as your safety

If you are being trailed, never lead the person back to your home or stop and get out. Drive to the nearest police station or public facility

7 If you are traveling alone and a car "bumps" into you, don't stop to exchange accident information. Go to the nearest service station, or other public place to call the police if necessary.

8When you park, look for a spot that offers good lighting and is close to a location where there are a lot of people Lock valuables in the trunk and lock all doors

7Be wary of unsolicited assistance.

7Be extra careful while shopping and loading your car trunk.

SURVEILLANCE

The purpose of surveillance is to identify a potential target, based on the security precautions that an individual takes, and the most suitable time, location, and method of attack. Surveillance may last for days or weeks. Naturally the surveillance of a person who has set routines, and who takes few precautions will take less time

Detecting surveillance requires a fairly constant state of alerts and therefore, must become a habit. A good sense of what normal, and what is unusual in your surroundings could be more important, than any other type of security precautions you may take. Above all, do not hesitate to

report any unusual event. There are three forms of surveillance foot, vehicular, and stationary. People who have well-established routines, permit surveillance to use methods that are much more difficult to detect: If for example, you leave the office at the same time each day, and travel by the most direct route to your home, or if you live in a remote area with few or no alternative routes to your home surveillants have no need to follow you all the way to your residence. Therefore, it is advisable to:

- 7 Vary your routes and times of travel
- 7 Be familiar with your route and have alternate routes
- 7 Check regularly for surveillance. Stationary surveillance is most commonly used by terrorists' organizations. Most attacks take place near the victims residence, because that part of the route is least easily varied. People are generally most vulnerable in the mornings when departing for work. because these times are more predictable than evening arrivals

Many surveillance teams use vans with windscreen in the sides or back, that permit observation from the interior of the van. Often, the van will have the name of a business or Utility Company to provide some pretext for being in the area. Where it is not possible to watch the residence unobserved, surveillants must come up with a plausible reason for being in the area. Women and children are often used to give an appearance of innocence. Try to check the street in front of your home from a window before you go out each day. If you suspect surveillance, report to the Police giving details of the vehicle including color and make.

- 7 Be alert to people disguised as public utility crews, road workers, vendors, etc. who might enter or station themselves near your home or office.

- 7 Always check your vehicle inside before entering

PERSONAL SAFETY TIPS

- . If you notice anything unusual, do not enter the vehicle.

- 7 Household staff and family members should be reminded to look for suspicious activities around your residence

- 7 Tell your household staff and family members to note descriptions and license numbers of suspicious vehicles Advise them to be alert for details.

MENTAL DISORDERS AND SAFETY

When dealing with people who show signs of mental disorders, ensuring personal safety should be paramount. Sometimes one may have to be willing to do certain things that seem uncomfortable, break some taboos, be firm, get out of dangerous situations, call for help or do whatever is necessary at that particular moment, to remove oneself from potential violence. abuse, and or threats.

A certain man narrated how he disguised as a madman in order to escape danger along a lonely rural road. A lady equally told the story of how she regained her freedom by begging a rapist gang in tears, not to rape her and be infected with HIV which she suffered from.

Conversation can occasionally lead to danger. It is easy to spend a great deal of time and energy in conversations that leave you feeling hurt and accomplish little or nothing. Some conversations may be urgent in nature or cannot be put off to another time. Note that in such circumstances,

less is more, and your words will have more impact if you don't repeat the same things over and over again. But just say what you have to say once. And then stop talking. Remember, a word is enough for the wise.

TIPS ON WHEN TO STOP A CONVERSATION TO AVOID GETTING HURT

8When your personal boundary is violated.

8When you are verbally insulted, mocked or invalidated

7When someone engages in thought-policing or tells you how you must think or feel.

7When you are engaged in a circular conversation.

7When your words start falling on deaf ears

7When the conversation is one way

7When you are angry and at a risk of taking the emotional elevator down to a less mature level.

7When you are tired and not able to engage in further constructive communication.

7When your attention is needed for a more responsible engagement.

WHEN TO LEAVE A PLACE TO AVOID BEING HURT

8When you feel unsafe or isolated.

8When you fear violence or abuse.

7When a person refuses to end the conversation or leave you alone

7When you are alone with an abusive or potentially dangerous person.

Do not hesitate to call for help or the Police when you are faced with situations such as:

7Someone threatens harm to him or herself.

7Someone threatens to hurt you.

7Someone threatens to hurt the children

7Someone performs any act of violence towards other people or property.

By applying these basic safety tips, and your own common sense, you can help protect yourself as much as possible. Above all, implicit trust in the Almighty Creator and godliness is good step towards a safe and Secure life.

The object of Defensive

TRAFFIC SAFETY(SAFETY DRIVING TECHNIQUE)

DEFENSIVE DRIVING CODE

The object of Defensive Driving is to drive without having a preventive accident.

Defensive Driving requires the knowledge and strict adherence to all traffic rules and regulation applicable to the area in which vehicle is being operated.

Defensive Driving requires a constant alertness for the illegal acts and driving errors of other drivers and the willingness to make timely adjustments in one`s own driving. So that these illegal acts and errors will not involve one in an accident.

Defensive Driving requires knowledge of and intelligent adjustments in one`s own driving to

the special hazards presented by abnormal, unusual or changing conditions in the mechanical functioning of one's vehicle, type of road surface, weather; degree of light, kind of traffic and one's physical condition and state of mind.

Defensive Driving requires a thorough knowledge of the rules of right-of-way and the willingness to yield the right-of-way to the other whenever necessary an accident.

In Defensive Driving requires an attitude of confidence that one can drive without ever having a preventable accident. Drivers are required by law to comply with road/traffic rules set forth in the Highway code.

THE DRIVER'S SIGNALS

An important rule of the road is that every driver must give signals to tell other drivers on the road, what he intends to do in guiding the vehicle he is driving. They must be given by hand or by mechanical/electrical devices which are maintained in good operating conditions. These signals are intended to alert other drivers or pedestrians in advance before you change speed, direction or stop in any direction. Give the signal for 100meters continuously before reaching the turning point. You are not the required to continue the signal actually making the turn. Be sure your electrical signal does not continue to flash after you have completed your turn.

When you plan a series of driving changes- such as stop for a stop sign, followed by a right turn you should always signal first for the action you intend to take. In the situation mentioned, you would signal first for a stop and come to complete stop; in this position you would then give the signal for your right turn. Common sense should guide you in giving and reading driver's signals.

At night when the arm to use electrical lamp signals. In bright sunlight it is often difficult to see the trafficator signals and it is better to use the hand and arm signal as well.

Give your own signals clearly and do what you have indicated by means of the signals. Read other driver's signal with the same care you give when making your own. Be alert to other drivers who may neglect to use signals.

STARTING, STANDING PARKING RULES

In starting from a proper parking place at the curb, walk around your vehicle and look for children, tool, materials, etc before you start up. Look for cars approaching from the rear before pulling away from the curb – slowly. Remember, you do not have the right-of-way.

In starting from a shop or a drive way be sure to watch for approaching persons and vehicles. Move continuously to join the traffic at a suitable speed.

BACKING

If your starting position requires backing, inspect to see what is behind your vehicle before you begin to move it. Then keep the situation well in view through your mirrors and windows. Look in the direction your vehicle is moving or using rear view camera on your dashboard which displays images from the vehicles on the center console screen, helping driver to park and maneuver more easily.

STOPPING AND PARKING

Always give a clear signal for slowing or stopping, Park your vehicle properly. Never leave your

vehicle until you have stop the engine, set the hand brake and remove the keys. Do not park in restricted or prohibited areas. These areas are designated by signs or by curbs that are painted yellow. When you have to stop on a high way, be sure to park with all four wheels off the pavement, if possible. Turn on your dome light and parking light at night. Do not stop, park or leave your vehicle in any of the following places.

1. Within an intersection
2. By fire hydrants
3. In front of a driveway
4. On a crosswalk or sidewalk.
5. Beside or opposite any street excavation or obstruction when such stopping would further obstruct traffic.
6. On the roadway side of any car parked at the curb or edge of the road.
7. Beside curb space marked as bus stop; either by sign or curb painting.
8. Beside curbs painted yellow, except to discharge passengers or cargo.
9. On hill, get curve on the highway. If you must park on a hill, get off the road, set the parking brake and transmission in 8park=, low or reverse gear, as appropriate. Block the the gear wheels of the heavily loaded trucks to prevents motion of the vehicle, if brake should fail.
10. On the vehicle on the wrong slide of a street or road, headed in the wrong direction

OTHER BASIC RULED OF THE BASIC OF THE DRIVE TO THE RIGHT

Always drive on the right hand side of the roadway except when passing cars going in the same direction. When turning left, when the right half of the roadway is closed or obstructed or on a one-way road.

SLOW VEHICLES

Any vehicle moving slower than normal flow of traffic should be driven as close to the right edge of the roadway as is safe.

DON`T WEAVE

Stay in one lane but if you have to change lanes look to be sure you can do so safely, then give the proper signal before changing.

FOLLOWING

Keep a safe distance from the vehicle ahead; for each 15 kph allow one car length. At speed over 50kph , increase your safe position to two car lengths for 15kph.

RIGHT OF WAY

The car approaching on your left has the right-way. He will probably insist on it so be prepared to stop. The man on your right does not have the right-of -way, but he may take it so again,be prepared to stop. Having the right-of-way will not save your life.

HOW TO MAKE TURNS

RULES FOR RIGHT AND LEFT TURNS

Keep your vehicle in the proper line of traffic for all types of turns. Keep to the far right side of the road for turns to the right.

Keep to the left side of your traffic lane as close to the center line as possible for a turn to the left. Signal for 100 meters before turning. You may need to signal for a longer distance if you must change lanes to make the turns properly.

Do not move to the right or the left on a roadway, unless you know such a movement can be made safely, and not until after you the proper signal. For safety's sake, avoid turning <wide= from one street to another. You may enter any lane that is safely available after first making sure that there are no conditions of danger present.

SPEED

Although the Nigerian Highway Code clearly defines speed limits, you should be guided by what is known as the <Basic Speed Rule=

It states, No person shall drive a vehicle upon a high way at speed greater than is reasonable or prudent having due regards for the traffic theon, and the surface and width of the highway, and in no event at which endangers the safety of persons or property=

This precept does not set an exact limit in miles or kilometers per hour at which you may drive. It considers all conditions that may affect your safety. These include the number and speed of other cars on the road; the present of pedestrians, the surface of other road whether smooth or rough, wet or dry, wide or narrow, how far ahead you can see in existing light conditions; and such other things as rain, fog, wind, dust and sand. For example, if you fail to notice that the car ahead of you has stopped and you strike it in the rear, you are, you are in violation of the Basic speed Rule=

WHEN YOU SHOULD SLOW DOWN

Regardless of speed limit signs you should always look for either cars or animals on the road and slow down if necessary under these conditions;

1. When you come to or across an intersection or railroad crossing.
2. When you come to or go around a curve.
3. When you come to the top or crest of a hill, keep well to the right
4. When driving upon any narrow or winding roadway.
5. When you approach any special conditions such as these:
 - a. Pedestrian,
 - b. A bicycle rider
 - c. Slow-moving vehicle
 - d. School zones and playgrounds
 - e. School guide safely patrol
 - f. Bad roads
 - g. Animals
 - h. When entering a town or village
 - i. When approaching the scene of an accident

You should slow down when there are children walking or playing in or near the street. Children often do not watch for cars so you must watch for them. You should always slow down after dark or at times when you cannot see clearly, as in dust, fog or rain. Even 40kilommeters an hour on the open road may be too fast if it violates the precept of the

basic speed rules.

PASSING AND CLEARANCE

When you overtake another vehicle and wish to pass it, pass to the left. There are also certain conditions under which you may pass to the, when such passing can be done safely. They are:

1. When the driver of the vehicle is making or about to make a left turn. In this case you may pass to the right with caution, because the driver ahead may turn right.
2. Upon a road where the pavement is wide enough and clearly marked for two or more lanes of traffic in your direction of travel. You may pass on either the left or the right in this case but only if it is done safely.

Before moving to the left or right on a highway, give a signal tell the vehicle ahead of you and the one behind, that you are changing position, don't make the change until you have looked in the rear view mirror and to the side you wish to move toward to be the lane is clear.

PASSING ON THE LEFT- AND RESTRICTION

- 1 .On the highway where there is a broken center line,

Passing is permitted. On the positions of the highway where there is a solid center line, or two lines with a solid line closest to your side of the road, passing is prohibited.

The solid line indicates some obstruction to vision(hill curve, intersection or railroad crossing) that makes passing at this point hazardous.

2. Make sure there is time to pass before changing lanes to make the pass. Make sure there is time to

pass before changing lanes to make the pass. Make sure you have time to get back on the right hand

side of the road to avoid getting dangerously close to any car coming from the opposite direction.

Allow 100 meters to make you pass.

3. In passing at a safe distance to the left , do not return to the right hand lane until you are sure that

you are safely out of the way of the vehicle behind you in the right lane. A safe rule is to wait until you

can see the car you have passed your rear view mirror, but do not turn sharply in front of it do not <cut

- in=

- 4.You must not pass or attempt to pass on hills, curve, hump of bridge or within 100meters of an intersection. Do not pass at any time when the road ahead is not clearly visible(87)

- 5.Except in a location where another vehicle may pass you on the right, when another vehicle comes

up behind you and sounds its horn, you must move over to your right and let it pass. You must never

speed up when another vehicle tries to pass you.

6. If you meet or overtake any animal or livestock, you must reduce your speed or stop as may be necessary or may be requested by any person in charge o the animal or livestock.

DRIVING AT NIGHT

Driving speed should be reduced for night driving; particularly so at dawn and dawn dusk. Too many

drivers try to drive just as fast at night as they do during the day. We are not as alert, nor do we see as

well when we are tired after a day`s work. Wide awake driving is necessary at all times and especially

at night , since we can see objects only a limited distance ahead. Driving at night with a sand pitted windshield is a safety hazard.

Don`t overdrive your headlights. You should hold to a speed that allows you to come to a complete stop when necessary in the distance you can clearly see ahead. After-dark driving requires different skills and extra care for safe travel. These are the things you should do:

1. When you meet a car at night, you should lower headlight beams(dim your lights)when within 500

meters of the approaching vehicle. Even if the other driver does not lower his beams., will be safer to keep your beams lowered and drive with due care.

2. Lower your headlight beams when following another vehicle within 200 meters

3. Lower your headlight beams when you are driving on well lighted streets.

4. Use your lower headlight beams when driving in fog or haze , and reduce your speed. Driving with

your headlight beams on in a fog is like shining your light on a mirror light is reflected back to your own eyes and blinds you.

5. Avoid looking directly into the light

6. Keep your headlights properly adjusted so that the lower beams are not aimed upward into an approaching driver`s eyes.

7. Ensure that your tail lights, stop lights and turn signals are working

8. Keep your headlights and windshield clean

9. Slow down when facing the glare from approaching headlights

10. Be sure that you can stop, when it is necessary, within the distance you can see clearly ahead and

watch carefully for pedestrians and animals along the road side.

CAUTION-CAR COONTROL

Keep your car under control and your brakes in good condition so that you can slow down or stop short of any possible danger or emergency. To do this, always allow yourself a SAFE DISTANCE between your car and another vehicle or person, so that you can make smooth stop, jerky stop often

cause dangerous skidding.

At high speeds, even on dry roads a car may go a long way before it will slide much further. That is why you must be very careful on slippery roads.

To avoid skidding on wet or slippery surfaces, drive at a slow speed. Allow more distance between your car and others. Apply your brakes or gas pedal slowly and easily so that you can start or stop

smoothly. Try to keep your wheels out of ruts. No matter what causes a skid, the method of control is

always the same. If your vehicle goes into a skid.

1. Turn your front wheels in the direction the rear end is skidding. That tends to straighten you out. Turning your wheels in the wrong direction will throw you into a dangerous spin.

2. Don't over steer. That could throw the rear end into a reverse skid in the other direction and you'll

be worse off than you were before.

3. Keep your foot on the accelerator to keep your rear wheels rolling, if there's nothing in front of you, feed gas gently to maintain traction.

4. Keep your foot off the brake. Jamming of the brake will make the skid worse. When you have the car under control, pump the brakes gently to stop.

If you turn off the pavement onto the sand to avoid an oncoming vehicle, do not attempt to return to

the pavement too quickly. By over correcting you're steering the front wheels tend to move sideways

until they dig-in the sand. This increases the chances of your vehicle overturning. Reduce your speed gradually and hold the steering firmly with both hands.

Turn slowly and carefully __ it's better to get stuck than to overturn your vehicle. Do not return to the

roadway until you are sure traffic is clear and you can do so safely. A complete stop is recommended.

SIGNS AND SIGNALS

For your guidance, a display of the traffic signs approved for use in Nigeria is included.

REMEMBER __ the stop means S-T-O-P, a complete stop. The yield sign means that you must yield right-of-way to vehicles on the street that you are about to cross or enter.

Nigerian traffic expect a straight ahead signal by hand when a driver is proceeding straight through at

an intersection under his control. This signal will assist the policeman in directing a smoother flow of safer traffic.

SAFETY EQUIPMENT OF YOUR VEHICLE

The Traffic Department and Division have many requests for the standards for efficiency of brakes, light and horns. We are not aware of any local laws giving a clean cut definition of all these standard

in some other areas.

BRAKES

Every passenger motor vehicle must have a service brake, which is usually a foot brake, capable of stopping it in 25 meters from a speed of 35 kilos per hour. An emergency brake (parking brake) is also required.

This is usually a hand brake, but in some newer vehicle models sometime it is applied by foot pressure separately from the service brake. For some late model cars, vigorous of the foot brake and

the hand brake is necessary to prevent the car from rolling back. The emergency brake must be able

to hold the vehicle on any grade upon which it is driven.

HEADLIGHTS

Your car must have at least two, headlights, one on each side of the front of the vehicle. The headlights must be strong enough to light the road so that you can see a person or vehicle 100 meters ahead on you, low beam, and 350 meters ahead on the high beam. They must not blind an

oncoming driver.

TAIL LIGHTS

Every vehicle must be EQUIPPED with at least one tail- light showing a red light for easy visibility a distance of 500meter to the rear. It is normally required that headlights(not parking light) and tail-lights be lighted when a vehicle is driven between 30 minutes after sunset and 30 minutes before sunrise, and at other times when needed.

STOPLIGHT

Every vehicle must be equipped with at least one rear stop light, which may be part of the tail-light or

a separate system. The stop light should display a red light when the service brake pedal is applied.

The red light must be strong enough to be seen 300meters to the car day or night.

TIRES

No tire should be used which has worn to the point that fabric is exposed or has sidewall cuts or breaks which have damaged the fabric. All regular tires must have a tread configuration on the part making contact with the road surface. Specific exception to this rule may be in cases of heavy trucks

and trailers

Some people think the chief function of tires is to provide a soft ride. They do far more than that, they

are an integral part of the braking system and the transmission of power from the motor to the road.

Under-inflation not only affects the safe performance of these systems, but can weaken the tires rapidly to the point when they fail and often result in dangerous blow outs .If you should have a blowout, positive smooth braking effort at intervals much more efficient and safer than quick, hard brake applications, Hold as straight a line as possible with your vehicle by means of a tight, firm grip

on the steering wheel; otherwise, the car will turn toward the side the blown out tire is on.

OTHER EQUIPMENT

A horn which can be heard 200 meters away should be on every vehicle.

Muffler systems should be gas-tight and should not release exhaust gases to the side of the vehicle between 2 feet and 11 feet above the ground.

Never close a fuel tank spout with such materials as a plug of wood, rags or other combustible items.

An inside rear view mirror should be attached so that you can see the road to the rear car for at least

200 meters. Left and right hand out side rear view mirrors will improve your visibility.

AUTOMOTIVE SEAT BELTS

It is the responsibility of the organization to which the vehicle is assigned to see that appropriate seat

belt installations are made if the vehicle is not already so equipped. It is the responsibility of each driver to use the seat belt. It is an additional responsibility of the driver to insist that his passengers also use seat belt as provided.

Owners of private vehicles are urged to install and use seat belts. We all know that seat belts do not

prevent accidents. We do not prevent accidents. We do know that the use of seat belt does minimize

the opportunity for, and severity of, injuries that might arise out of an accident.

WHEN ON THE SCENE OF AN ACCIDENT

1. When approaching the scene of an incident, slow down. Do not park your in a manner that will congest traffic.

2. Use available means to alert other traffic.

3. Assist the injured.

4. Dial emergency and briefly describe the accident, give the exact location and stay on the phone to

answer necessary questions.

5.

WHEN YOU ARE INVOLVED IN AN ACCIDENT

1. Assist injured persons.

2. Use available means to alert other traffic.

3. Do not accept responsibility or liability for the accident to anyone.

4. Do not make payment of money to anyone claiming that you are responsible or liable

5. Do not move the vehicle until permission is received from competent authority. (FRSC law permits vehicle to transport injured person to a medical facility if an ambulance is not readily available). The vehicle must then be return to the accident venue immediately.

SAFER DRIVING SKILLS

It takes three kinds of skills to drive successfully:

1. Mechanical operation of the vehicle.

2. Correct application of traffic laws.

3. How to see and what to look for.

Most of this guide deals with the first and second of these skills. The third skill, art of seeing and knowing what to look for, is as important as the first two but is usually overlooked or at least not given sufficient attention.

By understanding and practicing from the start the seeing methods used by the experts, most drivers can form seeing habits which are the safest approach to proper driving. These methods

emphasize allowance timing and spacing.

The secret is selective seeing. In a moving automobile, the driver has no time to see every detail along the road. This selective seeing is based on the following five seeing steps:

1. Aim High in Steering: Correct steering is simple ignore the fact that you sit left of the center in the car. Depend chiefly on your lower fringe vision for occasional glance well ahead at the center of your traffic lane. Seeing is usually of two kinds, sharp center eyesight which is a narrow three degree cone of vision and fringe vision which is the upper, lower and side parts your part of eye sight, your car then travels on center in its lane, and to the sides for traffic conditions.
2. Get the Big Picture: keep your central vision scanning a wide, deep roadway scene- foreground, background, side ground- for a block ahead in town; up to half a kilo away from towns.
3. Keep your eyes moving: Building a strong habit of forcing your eyes to move at least every two seconds in the driving. Check the rear view mirror each five second if any traffic is ahead or behind. Check the rear also before changing lanes and before slow down that could surprise a driver behind. The eye moving habit is restful for your eyes, keeps your mind straying too far from you driving and literally forces you to adjust speed to traffic. Weather and visibility conditions.
4. Make Sure They See You: When some one does create a conflict that threatens your safety, it means he either did not see your car or was counting on you to avoid the conflict. Be certain that the other person sees your car and shows clearly by his action that he will <stay put=.

Always tap your horn or flick your lights if in doubt. Do it early so you could stop if necessary.

5. Leave yourself an out: Recognize a basic rule of self preservation; protect yourself from being trapped by error of other people. Adjust your pace in traffic to keep a space cushion around your car. Whenever possible poise your foot lightly above the brake when in doubt. Allow extra space when the road slippery, when large vehicles block your view or when other cars drive too close behind you. With the habit of leaving yourself an out, you avoid impulsive moves and almost never need a hurried move. It`s the smart and easy way to drive.

A FINAL RULE

Never insist on taking the right-of-way even when you are sure it is yours. The right of-of-way should never be taken it must be given. This is a rule of common courtesy. which also requires that you yield the right-of-way to others when it eases a traffic situation to do so. Courtesy in yielding the right -of-way is an important factor in modern driving. Courtesy the roadway, and it saves lives.

OCCUPATIONAL HEALTH SAFETY PRACTICE INDUSTRIES

Safety is 25% Common sense, 80% Compliance, and the rest is goodliness – Safety spud.

Industrial safety can be defined as the ability to manage the risks inherent to operations, or related to the environment. Industrial safety is not a dislike of risk, rather it is a commitment to

clearly identify them in relation to production operations, assess them in terms of quality, and quantity and manage them. It is the policies and protections put in place, to ensure plant and factory worker protection from hazards that could cause injury.

INDUSTRIAL SAFETY TIPS

Total, a globally reputable company in her golden rules of occupational safety:(Thoroughness and vigilance) Summarized it in these words:

<Everyone must understand the basic rules of prevention and comply with them.=

These golden rules reinforce and complete our procedures for identifying dangers and controlling risks. The essential rules were established using records of accidents, observed in our different business. They must be followed to the letter.

Everyone should intervene as soon as a deviation is observed, and stop work if risk is not controlled. The twelve themes covered by these are:

High Risk Situations

Do not start-up or shut-down equipment or installations without using the appropriate written operating procedure. Report any situation where the risk level is temporally increased.

Identify and analyze induced risks implement compensatory measures.

Traffic: Machine/Vehicle/Cyclist/ Pedestrians

Do not exceed the speed limits. Traffic rules apply inside and outside sites.

Body Mechanics and Tools

Do not carry out work if you do not have the right tools the job and the environment: Adopting an inappropriate posture when handling objects or using tools can cause physical harm.

Protective Equipment

Do not access installations and perform work without wearing, general or task-specific personal Protective Equipment (PPE). Collective protection is preferred. Individual protection completes preventive measures already taken.

Work on Powered Systems

Do not perform work without checking that the power and product source supply has been rendered inoperative. Some works require the energy to be safely discharged or the equipment to be purged and vented. In Such cases, a lock-out system is set up to Isolate the energy or product according to a specific method.

Confined Spaces

Do not enter a confined space until isolation has been verified and the atmosphere checked. Do not enter a confined space(container, tanks, well, etc) without supervision, and only after the atmosphere and isolation have been checked.

Excavation Work

Do not perform excavation Work without valid work permit comprising a map of all underground hazards.

Manual or mechanical excavation work including dredging rivers and sea beds, can only start after a risk analysis has been done performed, to identify all hazards in the zone, and related precautions have to be taken.

Working at Height

Do not work at heights without a safety harness when there is no collective protective equipment. Work at height is performed on fixed or mobile platforms with a guardrail, designed for the task at hand. Work on rooftops (buildings, reservoirs) is performed only after the roof solidity has been checked and appropriate has been set. Ladders are a means access only, their use must remain exceptional.

Change Management

Do not make any technical or organization changes without prior authorization. Any change in technique or organization must be the subject of a risk analysis.

Simultaneous Operations or Co-activities

Do not perform simultaneous or co-activities without a prior visit. All work or operations inside a unit in operation can risk levels. Interferences due to simultaneous operations or joint-activities must be identified and controlled.

PREVENTING ELECTRICAL HAZARDS IN THE WORK PLACE

Good electrical safety habits can help protect you and your co-workers from injury. And you can play a role in recognizing and preventing workplace hazards. Keep in mind the following:

1. Electrocution is one of the top fire causes of workplace deaths.
2. Workers younger than 25 have the highest rate of death from electrical shock.
3. Many workplace electrocutions involve touching a power line with long or tall equipment.

STAYING ALERT

Many electrical injuries could be prevented if people were alert to hazards. Stay aware by keeping focused on your job and do not let emotions like anger and frustration get in the way.

CHECK CORDS AND CONNECTION

Before you start work, check electric cords for wears. If you are outside or in a wet location, be sure tools and extension cords are suitable for outdoor use and circuits are equipped with Ground Fault Circuit Interrupter (GFCI). The ground fault circuit interrupters (GFCI) is a fast-acting circuit-breaker designed to shut-off electric power in the event of a ground-fault, within as little as 1/40 of a second. It works by comparing the amount of current going to, and returning from equipment along the circuit conductors. It is also called Residual Current Device (RCD) and shuts off an unintended path, possibly through water or through a person. It is used to reduce the risk of electrical shocks. Also keep cords free of oil, heat and corrosion chemicals. Never yank. Kink or bend cords, and store them loosely coiled in a dry place.

POWER TOOLS

Never carry a tool by its cord. Be sure a tool is switched <off> before plugging or unplugging. This protects you and the next person who uses it. Watch out for energized areas when reaching into equipment. Shields, barriers, insulation and GFCI protects you, so do not modify them just to get a job done faster. Learn and follow your company lockout/tagout procedures. If in doubt ask a qualified electrical worker for help.

Dirt and dampness increase the risk of shock. Keep your tools, work areas and storage space clear and dry. When cleaning electrical equipment, be sure it is unplugged and follow the

manufacturer`s cleaning instructions.

USE SAFETY GEAR

Personal Protective Equipment(PPE) is your first line defense against shock and electrical burns. Keep boots, gloves and other gear in good condition, even a pinhole let electricity through. Wear non- conductive protection on your head, face, hand, and feet. Use insulated tools or handling equipment, such as no-conductive ropes and protective shields.

PREVENTING ELECTRICAL FIRE

Most electrical fires can be traced to overheated circuits or overloaded equipment. When abused, insulation may melt or burn, exposing live wires. Electrical fires can also occur when equipment is driven beyond capacity; when accumulated oil and dirt overheat a motor or when sparks ignite scraps, dirt, dust or flammable liquids. Observe the following rules;

- Use and maintain wiring, tools and equipment correctly. Keep everything oil and dust-free.
- Uncoil an extension cord fully before use. Find the amperage marked on it. Is it adequate ?
- Don not use equipment that gives off mild shock, unusual heat or odd smells. If in doubt, have it checked and repaired or replaced.
- Sweep up scraps and sawdust, and store flammable liquids in approved containers.
- Do not use electrical equipment when flammable gases, vapours, liquids, dust or fibers are present.

A FIRE SAFETY CHECKLIST

Visualize your plan of response in a fire, so you can move quickly if one occurs. Take into account;

- The nearest multi-purpose fire extinguisher and how to use it.
- The nearest emergency exit or fire escape.
- Your company procedures for notifying fire fighters and other emergency personal.
- Try to extinguish a fire yourself, only if you are trained and it is small or non-threatening. When in doubt, get out and take others with you.

INDUSTRIAL ACCIDENTS

An industrial accident is sudden and unanticipated event that can result in injuries, fatalities property damage and lost production time. Mishaps in the in the work place are fairly frequent, despite federal laws to protect workers safety. Ignoring safety procedures and using poor judgment are among factors that put workers at greater risk, resulting in serious job-related injury or illness. Industrial accident, work accident, work place accident, occupational accident or accident at work is a <discrete occurrence in the course of work is a <leading to physical or mental occupational injury.

According to the International Labor Organization (ILO). More than 337 million diseases in more than 337 million accidents happen on the job each year, resulting together with occupational disease in more than 2.3 million death annually. The phase <in the course of work= can include work-related accidents happening off the company`s premises, and can include accidents caused by third parties according to Eurostats. The definition of work accidents include occurring 8while engaged in an economic activity, or at work, or carrying on the business of the employer=. According to ILO, the phrase <Physical or mental harm= means

any injury, disease or death.

Occupational accidents differ from occupational disease, as accidents are unexpected and unplanned occurrences e.g(mine collapse,etc).while occupational disease, contracted as a result of an exposure over a period of time, to risk factors arising from work activity= e.g(miners lung)

Incidents that fall within the definition of occupational accidents include cases of acute poisoning, attacks by humans and animals, insect,slips and falls, on payments or staircases, traffic collisions, and accidents on board means of transportation in the course of work, accidents in airports stations and so on.

There is no consequences as to whether commuting accidents, that is, =accidents on the way to work and while returning home after work= should be considered to be work accident. The Europeans statistics on accidents at work(ESAW) methodology exclude them, The international Labour Organization (ILO) includes them in its conventions concerning health and Safety at work, although it list them as a separate category of accidents and some countries for example Greece do not distinguish them from other work accidents.

A fatal accident at work is defined as an accident which leads to death of a victim. The time within which the death may occur, varies among countries. In Netherland, an accident is registered as fatal if the victim dies during same day that the accident happened. In Germany, if death came within 30 days, while Belgium, France and Greece set no time limit. Where the accident involve multiple fatalities, they are often referred to as industrial, they are often referred to as industrial disasters.

COMMON CAUSES OF ACCIDENT

About 80 out of every 100 accidents are the fault of the person involved in the incident.

Unsafe acts cause four times time as many accidents and injuries conditions.

Accidents occur for many reasons. In most industries, people tend to look for things to blame= when an accident happened=; because it is easier than looking for= root causes=such as the underlisted:

Taking short-cuts: Everyday we, make decisions we hope wil make the job faster and more efficient. But do time-severs ever your own safety or that of other crew members? Short -cuts that reduce your safety on the job are not short-cuts, but an increased chance for injury.

Over-confidence: Confidence is a good thing; over-confident is too much of a good thing. <it will never happen to me= is an attitude that can lead to improper procedures, tool or methods in your work. Any of these can lead to an injury.

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OCCUPATIONAL HEALTH SAFETY PRACTICE

INDUSTRIES

Safety is 25% Common sense, 80% Compliance, and the rest is goodliness – Safety spud.

Industrial safety can be defined as the ability to manage the risks inherent to operations, or related to the environment. Industrial safety is not a dislike of risk, rather it is a commitment to clearly identify them in relation to production operations, assess them in terms of quality, and quantity and manage them. It is the policies and protections put in place, to ensure plant and factory worker protection from hazards that could cause injury.

INDUSTRIAL SAFETY TIPS

Total, a globally reputable company in her golden rules of occupational safety:(Thoroughness and vigilance) Summarized it in these words:

<Everyone must understand the basic rules of prevention and comply with them.=

These golden rules reinforce and complete our procedures for identifying dangers and controlling risks. The essential rules were established using records of accidents, observed in our different business. They must be followed to the letter.

Everyone should intervene as soon as a deviation is observed, and stop work if risk is not controlled. The twelve themes covered by these are:

High Risk Situations

Do not start-up or shut-down equipment or installations without using the appropriate written operating procedure. Report any situation where the risk level is temporally increased.

Identify and analyze induced risks implement compensatory measures.

Traffic: Machine/Vehicle/Cyclist/ Pedestrians

Do not exceed the speed limits. Traffic rules apply inside and outside sites.

Body Mechanics and Tools

Do not carry out work if you do not have the right tools the job and the environment: Adopting an inappropriate posture when handling objects or using tools can cause physical harm.

Protective Equipment

Do not access installations and perform work without wearing, general or task-specific personal Protective Equipment (PPE). Collective protection is preferred. Individual protection completes preventive measures already taken.

Work on Powered Systems

Do not perform work without checking that the power and product source supply has been rendered inoperative. Some works require the energy to be safely discharged or the equipment to be purged and vented. In Such cases, a lock-out system is set up to Isolate the energy or product according to a specific method.

Confined Spaces

Do not enter a confined space until isolation has been verified and the atmosphere checked. Do not enter a confined space (container, tanks, well, etc) without supervision, and only after the atmosphere and isolation have been checked.

Excavation Work

Do not perform excavation Work without valid work permit comprising a map of all underground hazards.

Manual or mechanical excavation work including dredging rivers and sea beds, can only start after a risk analysis has been done performed, to identify all hazards in the zone, and related precautions have to be taken.

Working at Height

Do not work at heights without a safety harness when there is no collective protective equipment. Work at height is performed on fixed or mobile platforms with a guardrail, designed for the task at hand. Work on rooftops (buildings, reservoirs) is performed only after the roof solidity has been checked and appropriate has been set. Ladders are a means access only, their use must remain exceptional.

Change Management

Do not make any technical or organization changes without prior authorization. Any change in technique or organization must be the subject of a risk analysis.

Simultaneous Operations or Co-activities

Do not perform simultaneous or co-activities without a prior visit.

All work or operations inside a unit in operation can risk levels.

Interferences due to simultaneous operations or joint-activities must be identified and controlled.

PREVENTING ELECTRICAL HAZARDS IN THE WORK PLACE

Good electrical safety habits can help protect you and your co-workers from injury. And you can play a role in recognizing and preventing workplace hazards. Keep in mind the following:

1. Electrocution is one of the top fire causes of workplace deaths.
2. Workers younger than 25 have the highest rate of death from electrical shock.
3. Many workplace electrocutions involve touching a power line with long or tall equipment.

STAYING ALERT

Many electrical injuries could be prevented if people were alert to hazards. Stay aware by

keeping focused on your job and do not let emotions like anger and frustration get in the way.

CHECK CORDS AND CONNECTION

Before you start work, check electric cords for wears. If you are outside or in a wet location, be sure tools and extension cords are suitable for outdoor use and circuits are equipped with Ground Fault Circuit Interrupter (GFCI). The ground f-fault circuit interrupters (GCFI) is a fast- acting circuit- breaker designed to shut-off electric power in the event of a ground-fault, within as little as 1/40 of a second. It works by comparing the amount of current going to, and returning from equipment along the circuit conductors. It is also called Residual Current Device (RCD) and shuts off an unintended path, possibly through water or through a person. It is used to reduce the risk of electrical shocks. Also keep cords free of oil, heat and corrosion chemicals. Never yank. Kink or bend cords, and store them loosely coiled in a dry place.

POWER TOOLS

Never carry a tool by its cord. Be sure a tool is switched <off= before plugging or unplugging. This protects you and the next person who uses it. Watch out for energized areas when reaching into equipment. Shields, barriers, insulation and GFCI protect you, so do not modify them just to get a job done faster. Learn and follow your company lockout/tag out procedures. If in doubt ask a qualified electrical worker for help.

Dirt and dampness increase the risk of shock. Keep your tools, work areas and storage space clear and dry. When cleaning electrical equipment, be sure it is unplugged and follow the manufacturer`s cleaning instructions.

USE SAFETY GEAR

Personal Protective Equipment (PPE) is your first line defense against shock and electrical burns. Keep boots, gloves and other gear in good condition, even a pinhole let electricity through. Wear non- conductive protection on your head, face, hand, and feet. Use insulated tools or handling equipment, such as no-conductive ropes and protective shields.

PREVENTING ELECTRICAL FIRE

Most electrical fires can be traced to overheated circuits or overloaded equipment. When abused, insulation may melt or burn, exposing live wires. Electrical fires can also occur when equipment is driven beyond capacity; when accumulated oil and dirt overheat a motor or when sparks ignite scraps, dirt, dust or flammable liquids. Observe the following rules;

- Use and maintain wiring, tools and equipment correctly. Keep everything oil and dust-free.
- Uncoil an extension cord fully before use. Find the amperage marked on it. Is it adequate?
- Don not use equipment that gives off mild shock, unusual heat or odd smells. If in doubt, have it checked and repaired or replaced.
- Sweep up scraps and sawdust, and store flammable liquids in approved containers.
- Do not use electrical equipment when flammable gases, vapours, liquids, dust or fibers are present.

A FIRE SAFETY CHECKLIST

Visualize your plan of response in a fire, so you can move quickly if one occurs. Take into account;

- The nearest multi-purpose fire extinguisher and how to use it.

- The nearest emergency exit or fire escape.
- Your company procedures for notifying fire fighters and other emergency personal.
- Try to extinguish a fire yourself, only if you are trained and it is small or non-threatening. When in doubt, get out and take others with you.

INDUSTRIAL ACCIDENTS

An industrial accident is sudden and unanticipated event that can result in injuries, fatalities property damage and lost production time. Mishaps in the in the work place are fairly frequent, despite federal laws to protect workers safety. Ignoring safety procedures and using poor judgment are among factors that put workers at greater risk, resulting in serious job-related injury or illness. Industrial accident, work accident, work place accident, occupational accident or accident at work is a <discrete occurrence in the course of work is a <leading to physical or mental occupational injury.

According to the International Labor Organization (ILO). More than 337 million diseases in more than 337 million accidents happen on the job each year, resulting together with occupational disease in more than 2.3 million death annually. The phase <in the course of work= can include work-related accidents happening off the company`s premises, and can include accidents caused by third parties according to Euro stats. The definition of work accidents include occurring 8while engaged in an economic activity, or at work, or carrying on the business of the employer=. According to ILO, the phrase <Physical or mental harm= means any injury, disease or death.

Occupational accidents differ from occupational disease, as accidents are unexpected and unplanned occurrences e.g(mine collapse,etc).while occupational disease, contracted as a result of an exposure over a period of time, to risk factors arising from work activity= e.g(miners lung)

Incidents that fall within the definition of occupational accidents include cases of acute poisoning, attacks by humans and animals, insect, slips and falls, on pavements or staircases, traffic collisions, and accidents on board means of transportation in the course of work, accidents in airports stations and so on.

There is no consequences as to whether commuting accidents, that is, <accidents on the way to work and while returning home after work= should be considered to be work accident. The Europeans statistics on accidents at work(ESAW) methodology exclude them, The international Labour Organization (ILO) includes them in its conventions concerning health and Safety at work, although it list them as a separate category of accidents and some countries for example Greece do not distinguish them from other work accidents.

A fatal accident at work is defined as an accident which leads to death of a victim. The time within which the death may occur, varies among countries. In Netherland, an accident is registered as fatal if the victim dies during same day that the accident happened. In Germany, if death came within 30 days, while Belgium, France and Greece set no time limit. Where the accidents involve multiple fatalities, they are often referred to as industrial, they are often referred to as industrial disasters.

COMMON CAUSES OF ACCIDENT

About 80 out of every 100 accidents are the fault of the person involved in the incident.

Unsafe acts cause four times time as many accidents and injuries conditions.

Accidents occur for many reasons. In most industries, people tend to look for things to blame= when an accident happened=; because it is easier than looking for= root causes= such as the under listed:

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Disaster

Management

INTRODUCTION

Disaster, means an event, which brings sudden great fortune bringing disruption to normal life and causing colossal loss of life and property and disruption of humans activities.

Disasters are situations of great calamity like earthquake, floods etc.

Disaster can also be defined as catastrophe, mishap, calamity or grave occurrence in any

area arising from natural or man made causes, or by accident or negligence which results in substantial loss of life or human suffering or damage to, and destruction of, property, or damage to degradation of environment, and is of such a nature or magnitude as to be beyond the coping capacity of the community of the affected areas.

Natural disasters are global phenomena, which may occur anywhere at any time. These adversely affect the lives of the people and cause considerable damage to property, environment and infrastructure worldwide. Although the basic cause of most of the natural disasters is also dependent on man made changes caused in the environment. However, hazards and disasters are untimely related and are synonymous to each other. Virtually hazard is a threat whereas disaster is an event.

□ India is one of the most disaster-prone countries of the world. The following data clearly show that India is a disaster-prone country:

- 65 per cent of the land area is vulnerable to earthquakes
- 12 per cent of the land area is vulnerable to cyclones
- 70 per cent of the land under cultivation is prone to drought.

Crisis is an event of acute danger, which can cause sudden disruption of work. The event may be caused due to natural cause, human error, equipment failure or sabotage by anti-social elements. The basic idea in crises (emergency situation) is to avoid wastage of crucial time. Terrorist threats and attacks, bomb explosion, strike, accidents, fire accidents, etc. are the examples of situations.

Disaster management is a comprehensive term which includes all aspects of preventive and protective measure, preparedness and relief operations for reducing the impact of disaster on humans.

Disaster management is a continuous and integrated process of planning, organizing, coordination and implementing measure, which are necessary or expedient for:

- (i) Prevention of danger or threat of any disaster
- (ii) Mitigation or reduction of risk of and disaster or its severity or consequences.
- (iii) Capacity building
- (iv) Preparedness to deal with any disaster
- (v) Prompt response to any threatening disaster situation or disaster
- (vi) Assessing the severity or magnitude of the effect of any disaster
- (vii) Evacuation, rescue and relief
- (viii) Rehabilitation and reconstruction

Poverty and lagging development amplify the adverse effects of natural disasters. Therefore developing countries (particularly their most densely populated regions) are

more vulnerable to natural disasters because of their limited capacity to prevent and absorb these effects. Poor communities get forced to live in crowded, makeshift houses and in areas vulnerable to bad weather or seismic activity having low quality of infrastructure.

Mitigation efforts lessen the impact disasters have on people and property. Mitigation measure include keeping homes away from flood plains, designing structures to withstand earthquakes. The vulnerability to natural hazards can be reduce by altering, averting avoiding and adapting to the hazard.

Beside the natural disasters, there are many man-made or technological oriented physical events, which result in a significant disruption in the normal working. The event which causes the sudden disruption of work, is known as crisis. The crisis management plan shall be able to respond quickly and effectively to such unexpected events and situation. The management includes the preventive measures need to be adopted to avoid any crisis and the ways and means to tackle a crisis, if it occurs in spite of preventive measures. Thus, crisis management helps in avoiding or minimizing wastage of crucial time and cost.

OBJECTIVES

The following are the major objectives of disaster/crisis management

1. To improve state of preparedness to meet any contingency,
2. To reduce response time in organizing the assistance,
3. To identify major resources, manpower and the equipment needed to make the disaster management plan operational.
4. Making optimum use of the combined resources,
5. To identify mitigating measures and take timely action,
6. To avoid panic situations and rumors at the time of contingency.

The objective of preparing disaster management plan is to enable various departments/agencies to plan for, quickly respond to and recover from unexpected events and situations. This plan shall provide necessary guidelines for assistance to various sections for ensuring of people, protection of property and

restoration of normal life including water and power supply, communication, etc.

CLASSIFICATION OF DISASTERS

The classification of disaster can be based on the origination agent, or the impact, or whether the disaster is preceded by a warning phase or not. The disasters may be broadly grouped into major or minor types depending upon their potential to cause damage to human life and property. The disasters are generally classification as major or minor.

NATURAL DISASTER

These can be further sub-classified as under as under:

1. Meteorological Disasters. These include:

(i)Cyclones ,(ii)Hurricanes,(iii)Floods,(iv)Droughts,(v)Hailstorm, cloudburst

2.Topographical Disasters. These include:

(i)Landslides,(ii)Avalanches(i.e, fall of mass of snow, rock and ice down a mountain)

3.Tectonic Disasters. These include:

(i)Earthquakes, (ii) Volcanoes

4.Infestic Disasters. These include:

(i)Epidemics,(ii)Pest attack, Locust invasion of crops

5.Chemical, Industrial and Nuclear Disasters

6.Accident related disasters. These include:

(i)Fires,(ii) Mine Flooding.(iii)Major building collapse(iv)Air, rail, road (v)stampede

The accidents may be either natural or man-made (intentionally like bomb blast) or due to mistake or carelessness, etc

Different natural calamities can be distinguished from each other in terms of their nature and extent of their impact.

(a) Calamities like earthquakes hailstorms, avalanches and landslides occurs quite suddenly but they are restricted in their impact in terms of time and space.

(b) Floods and Cyclones occur with some element of warning yet their occurrence is confined in duration.

(c) Drought spans over a much longer time-frame and its adverse impact on the economic activities and life of an area is of a more lasting nature.

Natural disasters may be broadly grouped into major and minor types depending upon their potential to cause damage. For example:

(i)Major disasters. Earthquake, flood, drought, cyclone.

(2) Minor disasters. Landslide, avalanche, tornadoes, hailstorm, heat wave, cold wave.

MAN-MADE DISASTERS

Man-made disasters are the results of follies and unwise acts of human beings. Following man-made factors increase vulnerability:

1. Density of population
2. Population living below poverty line(PBL)
3. Low percentage of literacy
4. Pattern of occupation and situation in employment.

About 70% people of our country reside in villages and the mode of their employment revolve around agriculture. The agriculture sector is always vulnerable to drought, floods and man-made problems. such as spills from oil and gas industrial activities By and large farmers lack knowledge to face disasters. Natural disasters with man-made factors of vulnerability make the overall situation highly volatile.

As compared to the developed countries, the third world countries suffer more relative loss to lives, livestock, and property. The third world countries have weak socio-economic infra-structure and do not have adequate prevention, preparedness and mitigation against disasters.

Other man-made disasters are:

(i)Communal riots,(ii)Ethnic conflict,(iii)Refuge situation,(iv)Terrorism,(v)stampede,(vi)Food poisoning,(vii)Alcohol/Liquor tragedies.

NATIONAL POLICY ON DISASTER MANAGENT

As discussed above disasters can strike our area at any time with or without any fore warning. Therefore disaster management policy is proactive and is comprehensive and includes the measures of preparedness, mitigation and prevention . In view of the changing dimension and probabilities of the natural disasters a new approach of strategic management is being followed in the national policy.

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1. SAFETY AND PERSONAL PROTECTION

1.1 WHY SAFETY?

It is a legal requirement; non-compliance attracts sanctions, punishment, etc.

It is economical; avoidable losses – injuries, asset damage/loss, fines, compensations, etc.

It is the right thing to do; it is morally expected. Duty of care

1.2 WHAT IS PERSONAL PROTECTIVE EQUIPMENT?

Personal protective equipment, commonly referred to as "PPE", is equipment worn to minimize exposure to serious workplace injuries and illnesses. These injuries and illnesses may result from contact with chemical, radiological, physical, electrical, mechanical, or other workplace hazards.

Personal protective equipment may include items such as gloves, safety glasses and shoes, earplugs or

muffs, hard hats, respirators, or coveralls, vests and full body suits.

PPE is used:

- as a way of controlling individual exposure to a hazard
- when there is occasional short exposure to a high risk
- when there is a residual risk of exposure remaining despite other controls

But:

☐ It is not a substitute for good working practices and personal hygiene standards

1.2.1 Types of PPE

☐ Protective clothing – Worn as a barrier against hazard that may cause injury to the body (coveralls or apron). Required:

- o In the field, laboratory, workshop, hospital, etc
- o When handling hazardous waste, chemicals or pesticides
- o When removing asbestos
- o In hot or cold environments

☐ Foot & Leg – can be boots or shoes. Protects against injuries from dropped objects, hot or cold liquids, stepping on something sharp

☐ Head – hard hat, helmet. Protect against injuries from dropped objects or striking something. Shelf life is about 5 years

☐ Eye & Face – Glasses, goggles, face shield, visor. Protect against injuries from flying objects, chemicals, welding flash, heat and glare

☐ Hand – hand gloves. Selection considerations include physical characteristics, permeability, allergies, maintenance and storage

☐ Hearing – Ear plugs and ear muffs (banded or helmet mounted). Should have ability to attenuate noise sufficiently

☐ Respiratory (RPE) – Air purifiers, Breathing apparatus. RPE is required when:

- o There is a risk of airborne infection
- o In operations generating dust or hazardous substances
- o Working in contaminated environments

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- o In oxygen-deficient environments
- o In emergency or rescue operations

1.2.2 What can be done to ensure proper use of personal protective equipment?

All personal protective equipment should be safely designed and constructed, and should be maintained in a clean and reliable fashion. It should fit comfortably, encouraging worker use. If the personal protective equipment does not fit properly, it can make the difference between being safely

covered or dangerously exposed. When engineering, work practice, and administrative controls are not

feasible or do not provide sufficient protection, employers must provide personal protective equipment

to their workers and ensure its proper use. Employers are also required to train each worker required to

use personal protective equipment to know:

- ☐ When it is necessary
- ☐ What kind is necessary:
- ☐ Exposure level

- Type of job
- Health implications
- How to properly put it on, adjust, wear and take it off
- The limitations of the equipment
- Proper care, maintenance, useful life, and disposal of the equipment

If PPE is to be used, a PPE programme should be implemented. This programme should address the hazards present; the selection, maintenance, and use of PPE; the training of employees on use, storage

and maintenance; and monitoring of the programme to ensure its ongoing effectiveness

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1.3 SAFE SYSTEM OF WORK

A safe system of work (SSW) is a formal procedure based on a systematic examination of work in order to identify the hazards and define safe methods of working which eliminates those hazards or minimize the risks associated with them to as low as reasonably practicable (ALARP). It is recorded in

form of procedures or work method statements.

2. BASIC FIRST AID

2.1 INTRODUCTION

At work, injuries and illnesses kill more than 2 million people in the world each year; i.e. 1 death every 15 seconds or 6000 deaths per day!

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Safe practices and healthy choices at work, home and play can prevent many injuries, diseases and death. Once injury or sudden illness has occurred, providing first aid can make a difference between

life and death; rapid versus prolonged recovery; and temporary versus permanent disability.

2.2 LEGAL ASPECTS

The <Good Samaritan= principle protects a first aid provider from being sued for wrong doing provided:

- o You are reasonably careful
- o You act in good faith (not for a reward)
- o You do not provide care beyond your skill level

You are also expected to remain with the injured until help arrives and you hand over to a more experienced person or professional unless it becomes dangerous for you to stay

Consent – means permission. To get consent, first introduce yourself and tell the victim your level of

competence or training, and ask if it is okay to help. Implied consent means that permission to perform

first aid care on an unresponsive victim is assumed. When caring for infants or elderly (with

dementia), consent shall be obtained from the parent/guardian or family member (as may be appropriate).

Care - Use appropriate PPE; it provides barrier between you and a victim's blood or body fluid. Disposable gloves and eye goggles (where there is risk of spatter) are most recommended. Be careful

to remove gloves, clothing, or any other contaminated stuff.

2.3 ROLES AND RESPONSIBILITIES

Roles - The basic role of a first aid provider is to recognize a medical emergency and make a decision

to help. In doing this, your personal safety is the highest priority, followed by safety of victim and that

of any by-standers

Responsibilities -

- o Maintain composure
- o Maintain personal health and safety
- o Maintain caring attitude
- o Maintain up-to-date knowledge and skills
- o Do not cause further harm

2.4 LEVELS OF CONSCIOUSNESS

Full consciousness - Able to speak and answer questions normally

Drowsiness - Easily roused, but lapses into unconsciousness

Stupor - Can be roused with difficulty. Aware of painful stimuli and when spoken to

Coma - Cannot be roused by stimuli

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2.5 EMERGENCY ACTION STEPS

These action steps are intended to help the first aid provider respond to an emergency and manage life-

threatening problems of the Airway, Breathing and Circulation in any victim of any age

Assess the Scene:

- ☐ If it is not safe, or becomes unsafe, GET OUT
- ☐ Observe universal precautions (use PPE)
- ☐ If victim is awake and talking, identify yourself and ask if it is ok to help
- ☐ If victim appears weak, seriously ill or unresponsive Call emergency service

Attend to the Victim:

A: Airway - Open airway

-If unresponsive tilt head, lift chin

B: Breathing - Check breathing

- Look, listen and feel for at least 5 secs (but not more than 10 secs)

- ☐ If unresponsive, perform CPR (Cardio-pulmonary resuscitation)
- ☐ If unresponsive, place in recovery position

C: Circulation - Look for and control severe bleeding with direct pressure

- Monitor tissue colour and temperature
- Help maintain normal body temperature
- Give emergency oxygen if available and if trained on it.

2.6 COMMON FIRST AID PROCEDURES

Severe bleeding

- Apply pressure to the wound
- Cover wound with clean pad and apply pressure
- Raise and support injured part
- Bandage wound and transport to hospital

Fracture

- Steady and support injured part
- Secure injured part to a sound part of the body
- Take or send casualty to a hospital

Muscle or Joint injury – sprain, strain

- Rest injured part

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- Apply ice-pack to injured area for 20 – 30 mins
- Apply elastic bandage to reduce swelling
- Raise and support injured limb

Burns

- Cool burn (apply cold water)
- Remove any constriction from the affected area
- Cover with clean piece of material
- Do not apply lotion, ointment or fat

Shock

- Make victim comfortable; loosen shirt
- Elevate feet

Choking

- Ask victim to cough
- Bend the victim forward
- Deliver 5 back blows
- Perform abdominal thrusts

Heart attack – persistent chest pain, spreading often to the jaw and down to the left arm. Faintness

- Call emergency service
- Make victim comfortable
- Check for medication
- Keep victim still
- Don't give stimulants

Heat exhaustion / Fainting

- ☐ Move victim to a cool place
- ☐ Lay victim down
- ☐ Elevate feet
- ☐ Loosen clothing
- ☐ Give fluids

2.7 THE FIRST AID KIT

The first aid kit contains materials required by the first aider in the management of first aid cases

- ☐ Contents are based on location and experience
- ☐ It must be easily accessible
- ☐ Contents must be labelled
- ☐ It must be kept clean and tidy
- ☐ It must be checked and replenished regularly

Common contents of the first aid kit include:

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- i. Paracetamol tabs
- ii. Hibitane
- iii. TBC
- iv. Cotton bandage
- v. Crepe bandage
- vi. Safety pins
- vii. Cotton wool and Gauze
- viii. Plaster strips
- ix. Scissors
- x. Record book

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3. HAZARDS

3.1 DEFINITIONS

A hazard is a situation that poses a level of threat to life, health, property, or environment. Most hazards are dormant or potential, with only a theoretical risk of harm; however, once a hazard becomes "active", it can create an emergency. A hazardous situation that has come to pass is called an incident.

3.1.1 Types of hazard

Hazards are generally labeled as one of four types:

- ☐ Physical hazards are conditions or situations that can cause the body physical harm or intense stress. Physical hazards can be both natural and human made elements. A physical hazard is a type of occupational hazard that involves environmental hazards that can cause harm with or without contact. Physical hazards include ergonomic and electrical hazards, radiation, heat

and cold stress, vibration hazards, and noise hazards. Engineering controls are often used to mitigate physical hazards e.g. use of PPE.

Physical hazards are a common source of injuries in many industries. They are perhaps unavoidable in certain industries, such as construction and mining, but over time people have developed safety methods and procedures to manage the risks of physical danger in the workplace.

□ Chemical hazards are substances that can cause harm or damage to the body, property or the environment. Chemical hazards can be natural or human made; examples include working with non-flammable chemicals, acids, toxins, etc.

□ Biological hazards are biological agents that can cause harm to the human body. These some biological agents can be viruses, parasites, bacteria, food, fungi, etc.

□ Psychological hazards are created during work related stress or a stressful environment. A person can be a hazard when he/she is affected psychological disturbance through -stress, shift patterns and also can be a hazard when a person is under the influence of alcohol, illness and lack of training.

Hazardous Area – A 3-dimensional space in which a flammable atmosphere may be expected to be present at such frequencies as to require special precautions for the construction, installation and use

of equipment or other sources of ignition.

Non Hazardous Area – An area where an explosive atmosphere is not expected to occur in quantities

such as to require special precautions for the construction, installation and use of electrical apparatus

and other sources of ignition.

3.2 HAZARDOUS AREA CLASSIFICATION (HAC)

Areas where gas (hydrocarbon) may collect or be present at any time are called hazardous areas. For

the purpose of assessing the degree of risk they are classified into three zones:

Zone 0 – An area in which an explosive gas/air mixture is continuously present, or present for long periods

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Zone 1 – An area in which an explosive gas/air mixture is likely to occur in normal operation

Zone 2 – An area in which an explosive gas/air mixture is not likely to occur in normal operation and,

if it occurs, it will exist only for a short time.

Area classification; who establishes it?

It's a team work, consisting of:

- Technical safety engineer (or HSSE professional)
- Process engineer
- Operations and maintenance
- Electrical engineer

□ Mechanical/Rotating equipment/HVAC engineer

3.3 HAZARDOUS AREA BASICS

Explosive gas mixture – a mixture of flammable gas/vapour and air, where after ignition, combustion

propagates throughout the unconsumed mixture (without additional supply of energy)

Lower explosive limit – Concentration of flammable gas or vapour in air, below which the gas atmosphere is not explosive

Upper explosive limit - Concentration of flammable gas or vapour in air, above which the gas atmosphere is not explosive

Flash point – the lowest temperature of a liquid at which sufficient vapour is given off to form an ignitable mixture with air near the surface of the liquid

Auto-ignition Temperature – the lowest temperature at which it will spontaneously ignite in a normal atmosphere without external source of ignition, such as flame or spark.

3.4 BENEFITS OF HAC

□ Basis for selection of protective equipment appropriate to defined areas to avoid ignition

□ Assists in the location of air inlet/outlet for ventilation and combustion systems

□ Helps determine location of flammable gas detectors

□ Assists in location of escape routes

□ Helps set the max allowable surface temperature of an equipment in an area

3.5 SAFETY IN CONSTRUCTION

□ Construction is a process that consists of the building or assembling of infrastructure

□ It involves clearing, excavation, grading, concreting, block laying, cementing, painting, roofing, operation of machines, plant and equipment, use of hand tools and other activities associated with buildings, structures, etc.

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□ Demolition is the tearing down of buildings and other structures either manually or mechanically.

□ Construction and demolition activities are some of the most dangerous land-based work

□ Construction related fatalities are in the region of 13 per 100,000 in Europe, as against 5 per 100,000 in the US. No clear statistics available for Nigeria but many reported cases of fatalities from collapsed buildings under construction and other related activities

Common hazards associated with construction and demolition are:

□ Motion; moving objects – earth moving equipment, rotating equipment, etc

□ Height; objects at height (dropped objects), lifting and hoisting, unstable work platform

□ Pressure; working with materials under pressure

□ Electricity; working with high voltage, power tools and equipment

□ Fire and Explosion; flammables

□ Toxic materials; dangerous materials

□ Thermal; extreme temperatures

☐ People; manual handling, exhaustion, noise, stress, security, violence, etc.

3.5.1 Construction HSSE Management System

☐ Employer is responsible by law to provide safe working environment for workers and third parties while workers are responsible for obeying safety rules and responsible for personal safety

☐ In providing safe working environment, employer must put a system in place for managing safety in construction and demolition operations

☐ Effective management of safety involves development and implementation of the HSSE management system:

☐ Leadership and commitment

☐ Policy and strategic objectives

☐ Organization, Responsibilities and Resources

☐ Risk management (risk identification, assessment, control and recovery measures)

☐ Planning and procedures

☐ Implementation and Monitoring

☐ Assurance

☐ Management review

Key steps in managing risks:

Risk identification – usually visual (walk-around, inspection of equipment and data sheets, etc)

Risk assessment – use risk assessment matrix and rank high, medium and low risks

Risk control – e.g. by elimination, substitution/replacement, technical controls, procedures, PPE, etc.

Recovery measures – Emergency response, Training/testing, Assurance (develop safe system of work,

performance monitoring, incident investigation, tool box meetings, consequence management, etc.)

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3.6 ELECTRICAL SAFETY

The basic rule of electrical safety is that no person should carry out work on any part of electrical equipment unless such parts are: -

☐ Properly identified as the equipment on which work is intended

☐ Made DEAD; switched off from the source of electrical power supply

☐ Isolated; action taken to prevent inadvertent reconnection and CAUTION notice attached

☐ Tested to prove DEAD condition

☐ Discharged, EARTHED and short-circuited; to provide back-up protection against danger from unforeseen conditions

3.6.1 Common Causes of Electrical Accidents

Accidents do not just occur; they are caused. Electrical accidents are caused by:

☐ Contact with live parts

☐ Faulty designs

- ☐ Improper installation of equipment
- ☐ Poor maintenance of equipment
- ☐ Improper use of equipment
- ☐ Poor grounding (earthing)

3.6.2 Electrical Equipment in Hazardous areas

Electrical equipment and instrumentation shall be installed in the least hazardous areas. Where they

have to be installed and used in hazardous areas, they need to be selected to be suitable for:

- ☐ The hazard category
- ☐ The temperature zone
- ☐ The zone of use

Any operational and environmental requirements become additional to above

3.6.2.1 Equipment classification

Two groups exist: -

- ☐ Group I- intended for use in underground mines, likely to become endangered by combustible dust
- ☐ Group II- intended for use in other places likely to become endangered by explosive atmospheres

Equipment group II is further divided into hazard categories 1, 2 and 3, corresponding to hazardous area zones 0, 1 and 2 described earlier in section 3.2; e.g. category 1 equipment are for use in areas

where explosive atmospheres are present continuously (i.e. in hazardous area zone 0).

3.6.2.2 Temperature classification

Equipment surface temperature must not be allowed to rise beyond the auto-ignition of gas or vapour

in that environment. Equipment labels will indicate corresponding temperature class, e.g. T1 for maximum surface temperature of 450C, T2 for 300C, T3 for 200C, etc.

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3.7 CHEMICAL/GAS HAZARDS

3.7.1 Chemicals and Handling

Over 100,000 chemicals are produced on an industrial scale. The chemicals are used in manufacturing,

medicine, agriculture, energy, home, etc.

The physical form of a chemical plays a big role in potential exposure:

- ☐ Gas – CO, H₂S
- ☐ Vapour – solvents
- ☐ Liquid – paints
- ☐ Mist – cutting fluids
- ☐ Solid – dust, pellets, fibres, cake

The nature/characteristics of the chemical also play an important role – organic/inorganic, volatility,

solubility, etc. These determine whether they will be absorbed, ingested or inhaled into the body.

Methods of exposure control are selected following a hierarchy of control:

- ☐ Elimination
- ☐ Substitution
- ☐ Isolation
- ☐ Engineering
- ☐ Procedural
- ☐ PPE

NOT all types of PPE protect against all types of chemical; careful accurate selection required, especially with gloves and RPE.

Treat all chemicals with respect. Read the label/MSDS/SHOC card and follow the instructions

Minimize exposure and avoid all body contact with chemicals

Clear spillages when they occur and dispose of waste chemicals responsibly

Know the emergency procedure

3.7.2 Types of Gas Hazard

Three main types of gas hazard: -

- ☐ Flammable – risk of fire and/or explosion (e.g. methane, butane propane)
- ☐ Toxic – risk of poisoning (e.g. carbon monoxide, hydrogen sulphide, chlorine)
- ☐ Asphyxiant – risk of suffocation (e.g. oxygen deficiency, nitrogen and carbon dioxide)

Flammable risk – considered in section 4

Toxic risk –

Some gases are poisonous and can be dangerous to life at very low concentrations

Some toxic gases have strong smells like rotten egg (H₂S)

Toxic gases can enter the body through -

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- ☐ Inhalation
- ☐ Absorption
- ☐ Ingestion
- ☐ Injection

3.7.3 Confined Space:

A space that -

- ☐ Has limited or restricted means of entry or exit
- ☐ Is not designed for continuous human occupancy

Examples include sewers, tunnels, storage tanks, silos, enclosure with only top or bottom access

Potential Hazards in Confined Space:

- ☐ Oxygen deficiency – <19.5% or >23.5% oxygen concentration
- ☐ Combustibles – methane, hydrogen, propane, gasoline fumes, etc.
- ☐ Toxic materials – welding fumes, CO, H₂S
- ☐ Electricity
- ☐ Mechanical hazards

3.7.4 Gas testing

The main purpose of carrying out gas test is:

- To test for flammable and toxic gases in the atmosphere and at what concentration they are present
- To ensure that no hazardous work condition exist from the presence of such gases
- To test for oxygen content in the environment

3.7.5 Safety equipment

- i. Gas detector
- ii. Respiratory protection for work
- iii. Respiratory protection for escape
- iv. Ventilation equipment
- v. Other equipment (PPE, communication equipment, etc.)

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4. FIRE AND EXPLOSION

4.1 THE PROCESS

Fire - A state of combustion in which fuels burn to produce heat, often flame and smoke.

Explosion - A release of mechanical, chemical, or nuclear energy in a sudden and often violent manner with the generation of high temperature accompanied with the release of gases.

4.1.1 The Fire Triangle:

- Oxygen, Fuel and Heat in certain proportions create a fire
- If any one of these three elements is removed, a fire cannot exist

4.1.2 The Explosion Mix

- Fuels - (Gas, Fuel vapour or Dust)
 - o Solids
 - o Liquids
 - o Chemicals
 - o Gases
- Air (Oxygen) - may be by planned or unplanned introduction
- An ignition source - which could be a minute spark from an instrument
 - o Open flames
 - o Sparks
- Chemical sources
- Electrical sources
- Natural sources
- Hot surfaces

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4.1.3 Ignition Heat Measurement

Flash point – the lowest temperature at which a substance gives off sufficient flammable vapour and

would ignite with the introduction of an ignition source but may not continue to burn. The lower the flash point the easier it is to ignite.

Auto-ignition temperature – the temperature at which a substance will automatically ignite by itself without the introduction of an ignition source.

Auto-ignition

Fuel Flash point
temperature

Ethanol (70%) 16.6 °C (61.9 °F) 363 °C (685 °F)

Gasoline (petrol) –43 °C (–45 °F) 280 °C (536 °F)

Diesel (2-D) >52 °C (126 °F) 256 °C (493 °F)

Jet fuel (A/A-1) >38 °C (100 °F) 210 °C (410 °F)

Kerosene >38–72 °C (100–162 °F) 220 °C (428 °F)

Vegetable oil (canola) 327 °C (621 °F)

Biodiesel >130 °C (266 °F)

*Diesel fuel flash points vary between 52 and 96 °C (126 and 205 °F).

*Both Jet A and A1 fuels have flash points between 38 and 66 °C

**Flash points >90°C are considered very low hazard

Explosive Atmosphere – An accumulation of gas, mist, dust or vapour, mixed with air with the potential to lead to a fire or explosion.

Explosive range – Managing gas and vapour concentration outside the explosive range is a very important consideration in operations activities in the oil and gas work environment

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In managing explosive gases, the focus is on LEL!

4.2 CAUSES OF FIRE AND EXPLOSION

- Carelessness (poor house-keeping, poor interface management, etc.)
- Negligence (smoking in prohibited areas)
- Non-compliance (sub-standard materials, non-intrinsically safe equipment)
- Ignorance
- Works of nature – extreme weather conditions; hurricanes, thunder and lightning strikes, etc.
- Poor/weak systems of work (poor planning, poor maintenance practice)
- Human error, Work overload, Sabotage, etc.

4.3 FIRE FIGHTING/CONTAINMENT

Fire containment derives on three key principles:

- Smothering – starving the fire of oxygen supply
- Cooling – reducing the heat
- Starvation – cutting of fuel supply

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4.3.1 Classification of Fires

4.3.2 Fire Protection Systems

Two sub-categories exist:

□ Passive protection -does not require any electrical, mechanical or manual (human) action to function. E.g.

- o Spacing and layout
- o Drainage/spill control systems
- o Fire proofing – insulators that protect substrate from damage from fire
- o Fire barriers (fire walls)

□ Active protection -

- o Water spray systems
- o Deluge systems
- o Drenching systems
- o Sprinkler systems
- o Fire cannons/hydrants
- o Fire extinguishers

4.3.2.1 Types of fire extinguishers

1. Water
2. Foam
3. Dry chemical powder

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4. Carbon dioxide

4.3.2.2 Use of fire extinguishers: PASS

- P – Pull (pull the pin)
- A – Aim
- S – Squeeze (squeeze the trigger)
- S – Sweep (spray side by side at the root of fire)

4.4 CONSEQUENCES OF FIRE

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- Disables operations
- Damage/loss of property
- Injury to personnel and/or loss of life
- Law suit
- Destruction of vital business and other information
- Loss of employment or livelihood
- Poor corporate image

- Effects on the environment

- Regulatory penalties

4.5 FIRE AND EXPLOSION PREVENTION

- Adoptable fire and explosion management process

- o Pre-job planning

- o Follow proper procedure

- o Toolbox meetings

- o Access control; Use of caution signs

- o Proper storage of chemicals

- o Fit-for-purpose design

- o Regular, timely and adequate maintenance

- o Use only intrinsically safe equipment (ATEX compliant – atmosphere explosibles)

- o Effective prevention mechanisms

- o Gas monitors

- o Emergency shutdown device

- o Depressurization systems

- o Flame, smoke and heat detectors

- Application of passive fire protection mechanisms

- o Area classification

- o Perimeter fencing

- o Plant/process area separation

- Provision of relevant fire extinguishment media

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INDUSTRIAL SAFETY ANALYSIS

GENERAL

Safety analysis is a necessity in today's litigious society. Companies that do not analyze may not prevent accidents which are preventable and may find their ability to defend themselves in a post accident lawsuit compromised. The standard to which the courts hold management is prudence, foresee ability, and reasonableness. Prudence requires that (1) production systems, products, and facilities be analyzed for safety, and (2) proper preventive measures follow the analysis. A prudent manager analyzes. Attempt must be made to foresee the foreseeable. This attempt must be reasonable. The key is reason. Analysis is simply the application of reason. There are two types of reason inductive and deductive. If a company can demonstrate that they made a reasonable attempt to foresee the foreseeable by applying system safety analysis inductive and deductive techniques and responding to the analyses with proper preventive actions, the company has a strong legal position. They can show they were reasonable and prudent in trying to foresee and avoid accidents, provided that they have carefully documented their efforts.

The philosophy developed by system safety engineers is that accident prevention can and must begin as soon as the idea for a new system or operation is conceived. Methods of analyses have been developed to this end which can be undertaken early. The analyses can indicate tentatively any hazards that might be present in any proposed operation, the types and degrees of accidents that might result from hazards, and the measures which can be taken to avoid or minimize accidents or their consequences. As more and more details become available after the initial analysis on the characteristics of the proposed system or operation, they can be expanded further and further to determine more intimately where there might be potential causes of accidents, their effects and safeguards needed. Almost 90 techniques have been developed, used, or proposed. This chapter includes a few important ones to indicate some of the methodologies now in wide use. This may make occupational safety engineers aware of some types of detailed analyses they might want to undertake.

Some of these techniques start at the component level and reason upward to find what "undesirable end events" might occur from that components fault or failure. These "bottom-up" techniques are based on inductive reasoning. Some techniques start with the undesired event which is to be avoided and reason down to identify which components of the system could contribute to the undesired event through their fault or failure. In systems where there is a critical potential for catastrophe, the prudent manager must make every effort to foresee and avoid the catastrophe. Everything is not foreseeable, but the manager must make a thorough effort to foresee when there is a potential for a catastrophe. This means the manager will use both inductive and deductive system safety analysis, if feasible. Preferably, the inductive analysis will be conducted blind to the deductive analysis, and vice versa. The results of the two analysis will then be compared. Hopefully, the undesirable end events identified in the inductive analysis will match those the deductive analysis started with; the component faults and failures identified in

the deductive analysis will match those components the inductive analysis started with. Where the inductive and deductive techniques do not match, the deficient analysis is corrected. This double-blind cross-check is a strong indicator of a reasonable effort to foresee the foreseeable, and when accompanied by prudent corrective actions, places management in a very defensible position.

PRELIMINARY HAZARDS ANALYSIS

The first method generally used in the systematic process of determining hazard causes, effects, and deterrents is the preliminary hazards analysis. As its name implies, this is an initial study from which analysis efforts can be expanded further. It is fairly broad in scope, investigates what hazards might be present, whether they can be eliminated entirely, or if not, the best way to control them. If the hazard cannot be eliminated, the analyst determines whether there are standards or methods by which the hazard could, should, or must be controlled. A review is made of the functions to be performed and whether the environments in which they must be performed will have any adverse effects on personnel, equipment, facilities, or operations. According to Greek mythology, Daedalus wanted to escape from Crete where he and his son Icarus were being held captive by the king because of his skill as an artificer. Daedalus made wings of feathers, flax, and beeswax with which he and Icarus could fly to Greece. Before they flew off, Daedalus told Icarus:

My boy, take care

To wing your course

Along the middle air: If low, the surges

If high, the sun The melting wax consumes.

Daedalus's instruction to his son can be taken to be an elementary preliminary hazard analysis. However, Daedalus, not being an experienced safety engineer, did not go far modern Safety Analysis

Safety engineers are smarter than designers. We safety engineers can figure out potential ways of getting killed which Daedalus didn't even consider, and we can even indicate safety measures by which he could have prevented Icarus's accident (he flew too close to the sun, the wax melted off his wings and he fell to his death). Modern designers, on the other hand are not nearly as smart as Dardalus, they have never been able to figure out how to make flyable wings out of feathers, flax, and beeswax.] Each product or operation have a limited number of hazards, which can be determined as soon as a few facts are known. For example, the proposed product is to use electrical power. The hazards, which can potentially be present when electrical power is in me, could include any or all of the following electrical shock, burns from hot equipment, fire due to arcing, sparking, or very hot surfaces; inadvertent starts of equipment, failure of equipment to operate at a critical time so that an accident will occur, radiation effects, and electrical explosions. Numerous conditions can lead to each of these, but these are the basic hazards. If the electrical product uses small batteries such as for transistorized equipment, most of the hazards mentioned will not present. The analysis can therefore be limited to those which could possibly exist, no

matter how improbable Progress of the design can be monitored thereafter to determine whether the hazard can be eliminated; and if it cannot be, how best it could be controlled, and if it is controlled, whether or not the control appears to be adequate and how the adequacy of the control can be verified. At this point, very rough estimates might be made of the probability of an accident due to that hazard, and of the severity of its probable effects.

A practice has been initiated in which the preliminary hazards analysis list, which is prepared in tabular form, is broken down into individual items. Each item is recorded (Fig. 15-2) and then tracked through development or modification of a system, product, or operation to ensure that adequate consideration is given to its hazard elimination control. (To assist in the tracking, visualizations, such as in the fault-tree method described later, can be used.) When all necessary actions are completed to eliminate or acceptably control the hazard, the item is signed off, since no further action is required. (An item should never be signed off with a comment that an action will be taken, is scheduled to take place, or a similar entry Signoff is considered accomplished only after the proposed action has been completed and shown to be adequate.) As design or planning progresses, studies are made of the hardware and facilities through reviews of assemblies of their major components, their proposed interrelationships and interfaces with each other and with personnel, environments that could affect them, and the effects they could generate on personnel, other components and assemblies, and the environment.

Some information does not lend itself well to an analysis presented solely in a tabular form. A narrative format can then be added to include addition data. For example an analysis presented in columnar form might indicate that a hazard was a accidental fire. To have a fire requires a fuel, oxidizer, ignitable mixture, and source of ignition. Detailed information on each item could then be listed separately, with details on the many facts, oxidizers, and sources of modern Safety Analysis.

A practice has been initiated in which the preliminary hazards analysis list, which is prepared in tabular form, is broken down into individual items. Each item is recorded and then tracked through development or modification of a system, product, or operation to ensure that adequate consideration is given to its hazard elimination control. (To assist in the tracking, visualizations, such as in the fault-tree method described later, can be used.) When all necessary actions are completed to eliminate or acceptably control the hazard, the item is signed off, since no further action is required. (An item should never be signed off with a comment that an action will be taken, is scheduled to take place, or a similar entry Signoff is considered accomplished only after the proposed action has been completed and shown to be adequate.) As design or planning progresses, studies are made of the hardware and facilities through reviews of assemblies of their major components, their proposed interrelationships and interfaces with each other and with personnel, environments that could affect them, and the effects they could generate on personnel, other components and assemblies, and the environment.

FAILURE MODES AND EFFECTS ANALYSIS

One of the methods used to accomplish safety analyses has been derived from reliability engineering Failure Modes and Effects Analysis (FMEA). FMEA is an inductive technique. It is

used to determine how long a piece of complex equipment will operate satisfactorily and what the effects of any failure of individual components might be. The method is intended for analyses of proposed equipment and systems. An occasion might arise in which it would be highly desirable for an occupational safety engineer to determine whether or not the manufacturer of industrial equipment and system to be purchased and installed had had analysis made, and to understand how failure might occur, modes and frequencies of failures, and the necessity for proper and timely maintenance and replacements.

In this method of analysis, the constituent major assemblies of the product to be analyzed are listed. Each assembly broken down into subassemblies and their components. Each component is studied to determine how it could malfunction and cause downstream effects. Effect might result on other components, and then higher-level subassemblies, assemblies, and the entire product or system. Failure rates for each item are determined and listed. The calculations are used to determine how long a piece of hardware is expected to operate between failures, and the overall probability that it will operate for a specific length of time. It is the best and principal means of determining where components and designs must be improved to increase the operational life of a product. It is best used to analyze how often and when parts must be replaced if a failure, possibly affecting safety, must be avoided.

Until the use of Boolean mathematics, described in the next paragraph, FMEA calculations were often erroneous. Also, because many component failures would have to effect on safety, that aspect of an FMEA does not involve accident possibilities. Also failure modes and effects analysis is limited to determination of all causes and effects, hazardous or not. Further, the FMEA does very little to analyze problem which could arise from operator errors (unless the system analysis includes the human as a component), or hazardous characteristics of equipment created by bad design of adverse environments (unless the scope of the analysis includes these elements). The FMEA is excellent for determining optimum points for improving and controlling product quality. Another method, fault-tree analysis has been found more effective for safety purposes.

FAULT-TREE ANALYSIS (FTA)

In 1959 the Air Force became concerned with the potentially catastrophic events that could occur with the Minuteman missile then being developed by Boeing. The Air Force contracted with Bell Laboratories to develop a method of analysis by which probabilities of occurrence of events with which they were concerned could be computed. The Air Force wanted to know the possibilities and probabilities that a missile could be launched or a warhead activated inadvertently, and the chance of either of these being done intentionally by an unauthorized person in an act of sabotage. Two years later Bell Laboratories completed the project. The new method, fault-tree analysis involved Boolean logic in ways similar to those being used increasingly in electronics industries. It is a deductive technique. Safety engineers of the Boeing Company adopted the proposed method and became its foremost proponents. They added to the symbols that Bell Laboratories had proposed. They organized the method so it could be computerized to permit calculating the probabilities of the problem with which the analyst was concerned. Although still used to determine probabilities of failures in complex systems or

operations, fault tree analysis is being used far more frequently in logically analyze the possibilities of potential accidents due to hazards, and the quantitative safety level,

At first the fact that Boolean logic (and arithmetic), with which few engineers were familiar at the time was its basis gave many people the impression the method could be used only by mathematicians (Some of the first Boeing employees involved in making those early safety analyses were mathematicians) It turned out that Boolean logic was fairly simple to understand. One of the fundamental principles is that any statement, condition, act, situation, or process could be described as being in either one of two states Something could be true or false, on or off, up or down-but not both, it would go or not go; or it could be fully open or not fully open (closed or partially open). There were no middle intermediate positions.

FAULT-TREE SYMBOLS

Primary Event Symbols

Basic Event, such as a component failure Circle.

-Conditioning Event, a condition which must be satisfied before the event above the gate can occur- Ellipse

- Undeveloped Event, an event not fully developed because of a lack of information or significance - Diamond

Normal Event, expected to occur during system operation-House-shape

Intermediate and Top Event Symbol

-Output Event, those that should be developed or analyzed further to determine how they can occur (the top output event is called the end event; others are called intermediate events) - Rectangle for gate): Any event leading into it from underneath will cause the event leading out of it at the top to occur.

Limitations of Fault-Tree Analyses One of the very few limitations of fault-tree analysis is that the tree indicates only the events included which will contribute to the occurrence of the top event.

SAFETY ANALYSIS METHODS MANDATED FOR PROCESS SAFETY MANAGEMENT

Safety analysis techniques are now mandated by regulations for some industries. This program requirement is examined below, and the techniques it suggests that are not already covered in this chapter are described and explained.

In the mid-1990s, Industries with processes that involved "highly hazardous chemicals" (HHC) were placed under requirements by OSHA (29 CFR 1910.119) to conduct process hazard analysis and have a process safety management program. The program is triggered by above-threshold quantities of any of 136 different, listed chemicals. The purpose of this standard is to prevent or minimize the consequences of a catastrophic release of toxic, reactive, flammable, or explosive highly hazardous chemicals from their use, storage, manufacturing, handling or on-site movement by industry.

The importance of this standard is that it requires safety analysis and names certain analytic

techniques to use or their equivalent. The employer shall one or more of the following methodologies that are appropriate to determine and evaluate the hazards of the process being analyzed... What-if, Checklist, What-if/Checklist. Hazard and Operability Study (HAZOP), Failure Mode and Effects Analysis (FMEA), Fault Tree Analysis, or an appropriate equivalent methodology" [29 CFR 1925.04(012))

What-if. In "What-if?" analysis, an analysis team reviews the covered process from beginning to end. At each step in the process they think of questions that begin with "What-if... They address procedural, hardware and software errors and produce a tabular listing of these questions and their answers. This constitutes a set of potential accident scenarios, their consequences, and remedial recommendations Report form can be narrative, but use of a matrix table often provides clarity

Checklist. The "Checklist" method applies a list of specific items, usually from industry standards, consensus codes, industry guidelines, etc. The list is prepared by an experienced safety engineer, familiar with the covered process and the Limitations of Fault-Tree Analyses

Process Safety Management program

Process Safety Information

Requires compilation of written process safety information, including hazard information on HHCS and information on technology and equipment at the covered process

Employee Involvement

Requires a written plan of action for involving employees in the conduct and development of process hazard analyses and process safety management and for providing employees access to the information required by the standard.

Process Hazard Analysis

Specifies that process hazard analysis be conducted and updated and revalidated at least every five years. Must be retained for the life of the process

Operating procedure

Requires these be in writing and include clear instructions for safely conducting activities of the covered process Steps for each operating phase, operating limits, safety and health considerations and safety systems and their functions must be readily accessible to involved workers. They must reflect current operating practice, and must implement safe work practices for special circumstances such as lockout/tag out and confined-space entry.

Training

Operators of the covered process must be trained in the overview of the process and in the operating procedures. Training must emphasize specific safety and health hazards, emergency operations, and safe work practices. Refresher training is required at least every three years.

Contractors

Contract employers are required to train their employees to safely perform their jobs at the process site. Must document that employees received and understood training and sure that contract employees know about the potential process hazards and the worksite employer's

emergency action plan. Must assure that contract employees follow safety rules of the facility
Must advise the worksite employer of hazards that contract work itself poses or hazards identified by contract employees.

Prestart up Safety Review

A safety review is required for new facilities and significantly modified work sites to confirm that the construction and equipment of a process are to design specs. Must assure that adequate safety, operating, maintenance, and emergency procedures are in place and that process operator training has been completed For new facilities process hazard analysis must be performed and its outstanding section items resolved and safety implemented before startup. Modified facilities must meet management of change requirements

Mechanical integrity

Written procedures must be established and maintained for the ongoing integrity of process equipment, particularly those components which are part of a covered process.

Hot Work

Hot work permits must be issued for hot work operations conducted in or near a covered process

Management of Change

Written procedures must be established and implemented to manage changes to facilities that affect a covered process Employees must be informed of changes

Incident Investigation

Require s employers to investigate as soon as possible, but no later than 48 hours after incidents which did result or could reasonably have resulted in catastrophic releases of covered chemicals. An investigation team which includes at least one person knowledgeable in the process involved and others with knowledge and experience in incident investigation and analysis must develop a written report on the incident. Reports must be retained for five years

Emergency Planning and Response Compliance Audits

An emergency action plan must be developed and implemented. It must include procedures for handling small releases

Employers must certify that they have evaluated compliance with process safety

Requirements at fast every three years. Prompt response to audit findings and documentation that deficiencies are corrected are required. The two most recent audit reports must be retained.

Trade Secret

Requires that certain information be available to employees Employers may enter into confidentiality agreement with employees to prevent disclosure of trade secrets questionnaire be applied to the covered process or system. Those parts of the design which are not adequate should be revealed when the checklist is applied. Then corrective measures are identified. What-it/Checklist. The What-if/Checklist Analysis uses the list of What-d questions and supplements it with a checklist. Therefore, it is a combination of the two methods, as the name suggests.

Hazard and Operability Study (HAZOP), HAZOP addresses each element in a process. The

elements must first be identified. The expected way each element operates is determined. Deviations from the intended or expected operations and parameters are identified using guide words. The consequences from the deviating element are estimated. Causes of the deviations, currently planned (or used) hazard controls, and inadequate controls are identified. The study is conducted by a team made up of five or six persons usually including a team leader familiar with HAZOP safety and health expert, a manager/supervisor, a technical person, a senior process operator, a maintenance person, a chemist, etc. The quality of the results will depend upon the quality of the team selected. The team will start by collecting process information from piping and instrumentation diagrams (P&ID), written operating instructions, material safety data sheets, plant models, Safety Analysis Reports (SARs), etc. They then review each element step by step through each line or process. At each element they apply key words a guide words to tease out the potential deviations from normal expectations of a design parameter.

The hazard and operability consequences, if any, for each deviation are determined and documented. The actions or changes in plant methods which will prevent the deviation or reduce the consequences are identified. If the cost of the changes can be justified, the team then must agree on the change and who will be responsible for the action. The team then must follow up to make sure that the action has been taken. If the cost is not considered to be justifiable, the team must agree to accept the risk. One of more reports of the evaluation are made, usually including tabular representation of the results. Team decision making is at the heart of this methodology. This technique as well as the others in this chapter are supported by software. The reader is advised to "surf the net" for the latest. Arthur D. Little Company is one source.

HAZOP GUIDE WORDS

No/Not Complete negation of design intention Quantitative increase or decrease

No part of the design intention happens e.g. "No flow", No pressure" etc

More, less Quantitative increase or decrease

Refers to quantities properties and duration of design parameters, such as flow rate, temperature ,etc

As well as Something else occurs in addition to design intention

Design intention are achieved, plus additional things occur.

Part of Only part of design is achieved

Partial addition or removal of material occurs. Activities are incomplete.

Reverse The opposite of the design intention

Reverse flow of material, electrical current.voltage polarity, Backward installation (e.g. check valve), opposite chemical reaction (eg.decomposition).

Other than Complete substitution

Original design intention is not accomplished, a different activity occurs-eg.leak.Rupture

Job Safety Analysis (JSA). The purpose of this technique is to uncover and correct hazards which are intrinsic to or inherent in the workplace. It should be done with a team that includes the

worker, supervisor, safety engineer, and management. Its success depends upon the rigor this team exercises during five steps: (1) Select a job, (2) Break the job down into steps (3) Identify the hazards and determine the necessary controls of the hazards, (4) Apply the controls to the hazards, and (5) Evaluate the controls

Eliminating, controlling, and minimizing hazards are becoming more and more necessary for work plants to comply with the standards of regulatory agencies. Formerly, chemical plants were built in which hazardous environments existed and where operators were safeguarded, theoretically, by use of protective equipment, such as gas masks, whose need was established after operations had begun. Some government agencies, such as OSHA, have adopted policies of "deterrence" and will not countenance the existence of hazardous conditions. Intrinsically safe environments are being required in which operators can work without the need for protective equipment, which is often inadequate, and which workers find burdensome and inconvenient and will not wear. To ensure that regulatory agencies do not require extensive and expensive retrofits of corrective equipment, it is highly advisable that analyses be made early to recognize potential problems and their solutions. The billions required to revise and correct designs and procedures at Three Mile Island and other nuclear power plants are an example of need for proper and adequate analysis and planning for safety purposes.

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PEDESTRIAN SAFETY (PERSONAL SAFETY TIPS)

The Nigeria Police once wrote:

"You have the primary responsibility to prevent crime against yourself, loved ones and your properties. Thinking that crime can only happen to someone else and doing nothing to prepare yourself, or take precautionary preventive measures, makes you a perfect target for criminals. The ultimate goal of crime prevention is to reduce the risk of being a victim. In order to accomplish this effectively, it is important to remove opportunities for a criminal to take advantage of you or your property.

You are attempting to prevent either victimization or criminalization by presenting an unattractive target to the criminal. This effort. removing opportunities is often referred to as

target-hardening. Target -hardening can be as complex as installing a high tech alarm system, or taking a self defense course, to something as simple as locking your door. Making yourself less of a target, is simpler than you might think it further gave the following information on safety at home, ATM Security tips

Make Sure They See You: When someone does create a . conflict that threatens your safety, it means he either did not see your car or was counting on you to avoid the conflict. Be certain that the other person sees your car and shows clearly by his actions that he will "stay put. Always tap your horn or flick your lights if in doubt. Do it early so you could stop if necessary.

Never insist on taking the right-of-way, even when you are sure it is yours. The right-of-way should never be taken - it must be given. This is a rule of common courtesy, which also requires that you yield the right-of-way to others when it cases a traffic situation to do so. Courtesy in yielding the right-of-way is an important factor in modern driving Courtesy smooths the roadway, and it saves lives .Never drive without a Valid Driver's License in your immediate possession.

AFINAL RULE

No driver will give perfect attention to his driving, but habits based on observance of correct seeing in every day driving will prevent you from becoming involved in many critical situations.

AFINAL REMINDER

Keep Your Eyes Moving Build a strong habit of forcing your eyes to move at least every two seconds in driving. Check the rear view mirror each five seconds if any traffic is ahead or behind. Check the rear also before changing lanes and before any slowdown that could surprise a driver behind. The eye moving habit is restful for your eyes, keeps your mind straying too far from your driving and literally forces you to adjust speed to traffic. weather and visibility conditions.

SAFETY AT HOME

- ✓ Make sure there are good locks on exterior doors and windows.
- ✓ Lock up when you go out, are asleep, or alone, even if it is only for a few minutes. It takes a thief ten seconds or less to enter an open room and steal property.
- ✓ Get to know your neighbors and their daily routines,
- ✓ Keep a list of emergency numbers especially of the Police Station close to you.
- ✓ Make sure your home is fenced and if you can afford it.
- ✓ Trim flowers that hide doorways or windows.
- ✓ Set timers on lights and radios to make your home appear lived-in, when you are away.
- ✓ Have a neighbor regularly check on your home when you are away, and report any suspicious activities. Remember that one good turn deserves another
- ✓ Mark your valuables with a number or identification unique to you, with an engraving pen and keep a list of them in a safe place .

Replace solid walls in front yard with open fencing to eliminate hiding places and make climbing more difficult

- Install a wide-angle peephole in your front door. so you am look out to an on-coming visitor without being seen yourself

Do not give out information about your neighbors.

Don't make it easy on burglars

✓Keep valuables away from windows

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ATM SECURITY TIPS

✓Be alert and conscious of your surroundings when using ATM.

Don't use an ATM machine located in a dark or obscure location.

✓Never give your card or Personal Identification Number (PIN) to anyone, for any reason

Be very conscious of persons hanging around the ATM machine Do not write your PIN on the card or anything that is kept with the card

Do not insert your card until asked to do so by the display screen

Never use an ATM with a blank screen, and if the ATM is obscured from view or properly lit, leave immediately and

find another ATM

✓ Stand close to the ATM and use your body and hand as shield to make sure nobody sees you keying in your PIN Never accept help from strangers when using an ATM. You should be wary of strangers asking for help. The elderly, or physically challenged persons should take along with them trusted persons, to assist them..

✓ Criminals work in teams, for example, one to distract you while the other steals your card or money. If your card is retained or swallowed by the ATM, it is advisable to contact your bank immediately Sign up for SMS alert, so that you are alerted whenever any withdrawal is made in your account

VEHICLE SECURITY TIPS

Most car crime is preventable. Police are making a concerted effort to tackle car crime, and target offenders: but motorists need to help themselves avoid becoming victims, by removing the opportunity for crime in the first place

Don't leave anything on display: it is a prime invitation to passersby. Even an old coat or plastic bag can tempt a thief. If items have to be left in the car put them out of sight at the start of journey.

Get a music system that is removable, or has a front that can be taken off, and remove ipads and mp3 players.

Driving documents and personal correspondence could help a thief to sell your vehicle or provide a cover story, if stopped by the police. Do not keep these in your vehicle Some car thieves break into houses to steal car keys.

Always keep the keys in a safe place. Do not leave them in the hallway or kitchen, as these are the obvious places.

A-Z PERSONAL SAFETY TIPS

Personal Safety is the act of ensuring that one is free from harm or danger, and or the ability to take steps towards preventing such harm

In other words, it is a list of actions that are designed to keep situations from occurring or escalating, and to make sure that physical, emotional and verbal abuse is avoided, stopped at the first moment, if it at all begins to happen. It borders on ideas such as when to stop a conversation, when to leave the room, and or call for help.

The following tips can help secure you in your daily living, while hopefully inspiring more awareness, when you are out and about

- Always be alert and aware of your surroundings, and of the people around you
- Make casual eye contact with people when walking
- Do not be easily distracted or take your eyes off someone you feel looks suspicious in a possibly dangerous situations.

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- Always carry ID Card and only the much cash need
- Whenever possible buddy up to walk out for dark parking
- Informants trusted family or friends if you are traveling your itinery dates and location if necessary
- Stay away from the bush or trees when walking of running Always be prepared to run away from an attacker and scream of confront where possible.
- Turn and go in the opposite direction of the car if a stranger approaches you, offering a ride. If possible rite down the license plate number and description of the vehicle.
- .. Do not respond to passing comments or conversation from strangers on the street or elsewhere.
- Walk confidently at a steady pace, and have your keys ready in your hand
- Avoid being on a cell phone with an ipod in your other ear unaware of your surroundings. This makes you an easy target.
- Avoid dangerous places at night time. Visit them during the time hours.
- Stop and look around if you feel unsafe, entering an area Trust your instincts
- Avoid isolated bus stops otherwise, continually look all around. Be Alart
- Avoid suspicious passenger cars/buses, one chance or all male etc.
- Don't stay in the same spot at bus stops for long. It makes you an easy target.

If you are carrying money, don't open your purse or wallet while boarding a vehicle Have your fare ready.

- Keep flashy jewelry or other valuables of easy prey out of sight
- If you must be in an isolated vehicle, sit close to the driver
- Check your purse or wallet, if someone is jostling, crowding or pushing you
- Never leave your purse, backpack or briefcase in plain view Lock it up when you leave your desk or office
- Keep the office door locked if you work alone or before and after normal business hours
- Try to find another worker or a security guard to walk out with you, if you work late
- Do not get in the elevator with another person if you do not feel comfortable with that person, take the next one. If you have to get in, stand next to the control panel, so that if you are attacked you can press the alarm and the control buttons as hard as possible
- Be alert for pick pockets on crowded elevators, and other busy places
- Be aware of escape routes for emergencies and post phone numbers of police, security agencies and fire departments in your Cell phones, so that you do not have to fumble pushing many numbers.

Young women should avoid provocative clothing, and carelessly immoral gesture.

TRAVELLING, SAFETY TIPS

- ✓ Notify your Resident Security Officer (RSC) or Personal Security Officer (PSO) your departure and return dates but don't otherwise publicize your travel or vacation plans.
- ✓ Check plane, tram and bus times before you travel,
- ✓ Sit near other people, aisles or doors Learn the location of emergency alarms and exits
- ✓ Stay awake and alert when using public transportation
- ✓ Conceal your passport, money, or credit cards when traveling
- ✓ Keep other valuables out of sight, and luggage where possible close at hand if carrying a handbag, keep it in front of you, closed with the fastening towards your body. Keep wallet in your front pant's pocket and not back
- ✓ Let go if your bag is snatched. Do not run after, especially in unfamiliar surroundings
- ✓ Do some research on the area you are visiting, especially at first time. Talk to your security officer or host regarding travel advisories and warnings.
- ✓ When traveling, dress casually dress down where appropriate. Be aware of local customs.
- ✓ Do not wear excess jewelry Reduce wallet and purse contents, particularly cards denoting affiliations, memberships and accounts, etc
- ✓ At airports proceed through security checks, and go to the boarding area, as quick as possible. These areas are usually the most secure in the airport. Be very careful anytime you make a telephone conversation.

LETTER AND PARCEL BOMBS SAFETY TIPS

Letter and parcel bombs generally are "victim activated" incaring that a victim or intended target must activate the device by opening it. They do not normally contain timing devices.

Bombs can range from the size of a cigarette packet to a large parcel. Letter and package bombs have been disguised as letters, books, candy and figurines. Delivery methods have included mail systems, personal delivery, or placement at the recipient's site. A letter or parcel bomb might have some of the following indication.

- ✓ Suspicious origin especially if the postmark or name of sender is unusual, unknown, or no further address is given.
- ✓ Excessive inadequate postage:
- ✓ Off-balance or lopsided letter or package.
- ✓ Unusual weight for the size of the letter or package. Letters also may be unusually thick. Stiffness or spring of contents when checking do not bend excessively.
- Protruding wires or components, unusual grease or oil stains on the envelope.
- ✓ Strange smell, particularly almond or other suspicious odours
- ✓ handwriting of sender is not familiar, or indicates foreign style not normally received by recipient. Common words or names are misspelled
- ✓ Rub-on or black lettering
- ✓ Restrictive markings such as confidential "personal". or an horrific title appended to the name of the addressee
- ✓ Wrapping that could be a provision for an arming safety wire.
- ✓ Rattling inside the envelope or package, possibly loose components of a device.
- ✓ Visual distractions for example currency or pornography etc

WARNING

If you identify a letter or package as suspicious, don't let anyone near it. Notify your local Bomb Disposal Unit or the nearest Police Station immediately. Leave the letter or package in open area such as a court yard, where it is easily accessible to Bomb Squad Personnel. Never submerge it in water or burn.

CAR JACKING SAFETY TIPS

- ✓ When in your car, always keep the doors locked. Anytime you drive through areas containing stoplights, stop-signs or anything that significantly reduces vehicular speed, keep your windows up, especially in narrow ways and isolated places
- ✓ Leave ample maneuvering space between your vehicle and the one in front of you.. If you are approached by suspicious persons while you are stopped do not roll down windscreen, drive away quickly

If you are being followed or harassed by another driver, try to find the nearest police station, hotel or other public facility. Once you find a place of safety, don't worry about using a legal parking space, park as close as you can, and get inside fast, especially if the threat is on your life.

- ✓ If another driver tries to force you to pull over, or to cut you off, keep driving and try to get away. Try to note the license plate number of the car, and a description of the driver. If this effort places you in danger, don't do it. The information is not as important as your safety

If you are being trailed, never lead the person back to your home or stop and get out. Drive to the nearest police station or public facility

✓ If you are traveling alone and a car "bumps" into you, don't stop to exchange accident information. Go to the nearest service station, or other public place to call the police if necessary.

✓ When you park, look for a spot that offers good lighting and is close to a location where there are a lot of people. Lock valuables in the trunk and lock all doors

✓ Be wary of unsolicited assistance.

✓ Be extra careful while shopping and loading your car trunk.

SURVEILLANCE

The purpose of surveillance is to identify a potential target, based on the security precautions that an individual takes, and the most suitable time, location, and method of attack. Surveillance may last for days or weeks. Naturally the surveillance of a person who has set routines, and who takes few precautions will take less time

Detecting surveillance requires a fairly constant state of alerts and therefore, must become a habit. A good sense of what normal, and what is unusual in your surroundings could be more important, than any other type of security precautions you may take. Above all, do not hesitate to report any unusual event. There are three forms of surveillance foot, vehicular, and stationary

People who have well-established routines, permit surveillance to use methods that are much more difficult to detect: If for example, you leave the office at the same time each day, and travel by the most direct route to your home, or if you live in a remote area with few or no alternative routes to your home surveillants have no need to follow you all the way to your residence

Therefore, it is advisable to:

✓ Vary your routes and times of travel ✓ Be familiar with your route and have alternate routes

✓ Check regularly for surveillance. Stationary surveillance is most commonly used by terrorists' organizations. Most attacks take place near the victims residence, because that part of the route is least easily varied. People are generally most vulnerable in the mornings when departing for work. because these times are more predictable than evening arrivals

Many surveillance teams use vans with windscreen in the sides or back, that permit observation from the interior of the van. Often, the van will have the name of a business or Utility Company to provide some pretext for being in the area. Where it is not possible to watch the residence unobserved, surveillants must come up with a plausible reason for being in the area. Women and children are often used to give an appearance of innocence. Try to check the street in front of your home from a window before you go out each day. If you suspect surveillance, report to the Police giving details of the vehicle including color and make.

✓ Be alert to people disguised as public utility crews, road workers, vendors, etc. who might enter or station themselves near your home or office.

✓ Always check your vehicle inside before entering

PERSONAL SAFETY TIPS

. If you notice anything unusual, do not enter the vehicle.

✓ Household staff and family members should be reminded to look for suspicious activities around your residence

✓ Tell your household staff and family members to note descriptions and license numbers of suspicious vehicles Advise them to be alert for details.

MENTAL DISORDERS AND SAFETY

When dealing with people who show signs of mental disorders, ensuring personal safety should be paramount. Sometimes one may have to be willing to do certain things that seem uncomfortable, break some taboos, be firm, get out of dangerous situations, call for help or do whatever is necessary at that particular moment, to remove oneself from potential violence. abuse, and or threats.

A certain man narrated how he disguised as a madman in order to escape danger along a lonely rural road. A lady equally told the story of how she regained her freedom by begging a rapist gang in tears, not to rape her and be infected with HIV which she suffered from.

Conversation can occasionally lead to danger. It is easy to spend a great deal of time and energy in conversations that leave you feeling hurt and accomplish little or nothing. Some conversations may be urgent in nature or cannot be put off to another time. Note that in such circumstances, less is more, and your words will have more impact if you don't repeat the same things over and again. But just say what you have to say once. And then stop talking. Remember, a word is a enough for the wise.

TIPS ON WHEN TO STOP A CONVERSATION TO AVOID GETTING HURT

✓When your personal boundary is violated.

✓When you are verbally insulted, mocked or invalidated

✓When someone engages in thought-policing or tells you how you must think or feel.

✓When you are engaged in a circular conversation.

✓When your words start falling on deaf ears

✓When the conversation is one way

✓When you are angry and at a risk of taking the emotional elevator down to a less mature level.

✓When you are tired and not able to engage in further constructive communication.

✓When your attention is needed for a more responsible engagement.

WHEN TO LEAVE A PLACE TO AVOID BEING HURT

✓When you feel unsafe or isolated.

✓When you fear violence or abuse.

✓When a person refuses to end the conversation or leave you alone

✓When you are alone with an abusive or potentially dangerous person.

Do not hesitate to call for help or the Police when you are faced with situations such as:

✓Someone threatens harm to him or herself.

- ✓ Someone threatens to hurt you.
- ✓ Someone threatens to hurt the children
- ✓ Someone performs any act of violence towards other people or property.

By applying these basic safety tips, and your own common sense, you can help protect yourself as much as possible. Above all, implicit trust in the Almighty Creator and godliness is good step towards a safe and Secure life.

The object of Defensive

TRAFFIC SAFETY(SAFETY DRIVING TECHNIQUE)

DEFENSIVE DRIVING CODE

The object of Defensive Driving is to drive without having a preventive accident.

Defensive Driving requires the knowledge and strict adherence to all traffic rules and regulation applicable to the area in which vehicle is being operated.

Defensive Driving requires a constant alertness for the illegal acts and driving errors of other drivers and the willingness to make timely adjustments in one`s own driving. So that these illegal acts and errors will not involve one in an accident.

Defensive Driving requires knowledge of an d intelligent adjustments in one`s own driving to the special hazards presented by abnormal, unusual or changing conditions in the mechanical functioning of one`s vehicle, type of road surface, weather; degree of light, kind of traffic and one`s physical condition and state of mind.

Defensive Driving requires a thorough knowledge of the rules of right - of way and the willingness to yield the right-of-way to the other whenever necessary an accident.

In Defensive Driving requires an attitude of confidence that one can drive without ever having a preventable accident. Drivers are required by law to comply with road/traffic rules set forth in the Highway code.

THE DRIVER`S SIGNALS

An important rule of the road is that every driver must give signals to tell other drivers on the road, what he intends to do in guiding the vehicle he is driving. They must be given by hand or by mechanical/electrical devices which are maintained in good operating conditions. These signals are intended to alert other drivers or pedestrians in advance before you change speed, direction or stop in any direction. Give the signal for 100meters continuously before reaching the turning point. You are not the required to continue the signal actually making the turn. Be sure your electrical signal does not continue to flash after you have completed your turn.

When you plan a series of driving changes- such as stop for a stop sign, followed by a right turn you should always signal first for the action you intend to take. In the situation mentioned, you would signal first for a stop and come to complete stop; in this position you would then give the signal for your right turn. Common sense should guide you in giving and reading driver`s signals.

At night when the arm to use electrical lamp signals. In bright sunlight it is often difficult to see the trafficator signals and it is better to use the hand and arm signal as well.

Give your own signals clearly and do what you have indicated by means of the signals. Read other driver's signal with the same care you give when making your own. Be alert to other drivers who may neglect to use signals.

STARTING, STANDING PARKING RULES

In starting from a proper parking place at the curb, walk around your vehicle and look for children, tool, materials, etc before you start up. Look for cars approaching from the rear before pulling away from the curb – slowly. Remember, you do not have the right-of-way.

In starting from a shop or a drive way be sure to watch for approaching persons and vehicles. Move continuously to join the traffic at a suitable speed.

BACKING

If your starting position requires backing, inspect to see what is behind your vehicle before you begin to move it. Then keep the situation well in view through your mirrors and windows. Look in the direction your vehicle is moving or using rear view camera on your dashboard which displays images from the vehicles on the center console screen, helping driver to park and maneuver more easily.

STOPPING AND PARKING

Always give a clear signal for slowing or stopping, Park your vehicle properly. Never leave your vehicle until you have stop the engine, set the hand brake and remove the keys. Do not park in restricted or prohibited areas. These areas are designated by signs or by curbs that are painted yellow. When you have to stop on a high way, be sure to park with all four wheels off the pavement, if possible. Turn on your dome light and parking light at night. Do not stop, park or leave your vehicle in any of the following places.

1. Within an intersection
2. By fire hydrants
3. In front of a driveway
4. On a crosswalk or sidewalk.
5. Beside or opposite any street excavation or obstruction when such stopping would further obstruct traffic.
6. On the roadway side of any car parked at the curb or edge of the road.
7. Beside curb space marked as bus stop; either by sign or curb painting.
8. Beside curbs painted yellow, except to discharge passengers or cargo.
9. On hill, get curve on the highway. If you must park on a hill, get off the road, set the parking brake and transmission in 'park', low or reverse gear, as appropriate. Block the the gear wheels of the heavily loaded trucks to prevents motion of the vehicle, if brake should fail.
10. On the vehicle on the wrong side of a street or road, headed in the wrong direction

OTHER BASIC RULED OF THE BASIC OF THE DRIVE TO THE RIGHT

Always drive on the right hand side of the roadway except when passing cars going in the same direction. When turning left, when the right half of the roadway is closed or obstructed or on a one-way road.

SLOW VEHICLES

Any vehicle moving slower than normal flow of traffic should be driven as close to the right edge of the roadway as is safe.

DON`T WEAVE

Stay in one lane but if you have to change lanes look to be sure you can do so safely, then give the proper signal before changing.

FOLLOWING

Keep a safe distance from the vehicle ahead; for each 15 kph allow one car length. At speed over 50kph , increase your safe position to two car lengths for 15kph.

RIGHT OF WAY

The car approaching on your left has the right-way. He will probably insist on it so be prepared to stop. The man on your right does not have the right-of -way, but he may take it so again,be prepared to stop. Having the right-of-way will not save your life.

HOW TO MAKE TURNS

RULES FOR RIGHT AND LEFT TURNS

Keep your vehicle in the proper line of traffic for all types of turns. Keep to the far right side of the road for turns to the right.

Keep to the left side of your traffic lane as close to the center line as possible for a turn to the left.Signal for 100 meters before turning. You may need to signal for a longer distance if you must change lanes to make the turns properly.

Do not move to the right or the left on a roadway, unless you know such a movement can be made safely , and not until after you the proper signal. For safety`s sake, avoid turning “wide” from one street to another. You may enter any lane that is safely available after first making sure that there are no conditions of danger present.

SPEED

Although the Nigerian Highway Code clearly defines speed limits, you should be guided by what is known as the “Basic Speed Rule”

It states, No person shall drive a vehicle upon a high way at speed greater than is reasonable or prudent having due regards for the traffic theon, and the surface and width of the highway, and in no event at which endangers the safety of persons or property”

This precept does not set an exact limit in miles or kilometers per hour at which you may drive. It considers all conditions that may affect your safety. These include the number and speed of other cars on the road; the present of pedestrians, the surface of other road whether smooth or rough, wet or dry, wide or narrow, how far ahead you can see in existing light conditions; and such other things as rain , fog, wind, dust and sand. For example, if you fail to notice that the car ahead of you has stopped and you strike it in the rear, you are, you are in violation of the Basic speed Rule”

WHEN YOU SHOULD SLOW DOWN

Regardless of speed limit signs you should always look for either cars or animals on the road and slow down if necessary under these conditions;

1. When you come to or across an intersection or railroad crossing.
2. When you come to or go around a curve.
3. When you come to the top or crest of a hill, keep well to the right
4. When driving upon any narrow or winding roadway.
5. When you approach any special conditions such as these:
 - a. Pedestrian,
 - b. A bicycle rider
 - c. Slow-moving vehicle
 - d. School zones and playgrounds
 - e. School guide safely patrol
 - f. Bad roads
 - g. Animals
 - h. When entering a town or village
 - i. When approaching the scene of an accident

You should slow down when there are children walking or playing in or near the street.

Children often do not watch for cars so you must watch for them. You should always slow down after dark or at times when you cannot see clearly, as in dust, fog or rain. Even 40kilommeters an hour on the open road may be too fast if it violates the precept of the basic speed rules.

PASSING AND CLEARANCE

When you overtake another vehicle and wish to pass it, pass to the left. There are also certain conditions under which you may pass to the, when such passing can be done safely. They are:

1. When the driver of the vehicle is making or about to make a left turn. In this case you may pass to the right with caution, because the driver ahead may turn right.
2. Upon a road where the pavement is wide enough and clearly marked for two or more lanes of traffic in your direction of travel. You may pass on either the left or the right in this case but only if it is done safely.

Before moving to the left or right on a highway, give a signal tell the vehicle ahead of you and the one behind, that you are changing position, don`t make the change until you have looked in the rear view mirror and to the side you wish to move toward to be the lane is clear.

PASSING ON THE LEFT- AND RESTRICTION

- 1 .On the highway where there is a broken center line,

Passing is permitted. On the positions of the highway where there is a solid center line, or two lines with a solid line closest to your side of the road, passing is prohibited.

The solid line indicates some obstruction to vision(hill curve, intersection or railroad crossing) that makes passing at this point hazardous.

2. Make sure there is time to pass before changing lanes to make the pass. Make sure there is time to

pass before changing lanes to make the pass. Make sure you have time to get back on the right hand

side of the road to avoid getting dangerously close to any car coming from the opposite direction. Allow 100 meters to make you pass.

3. In passing at a safe distance to the left , do not return to the right hand lane until you are sure that

you are safely out of the way of the vehicle behind you in the right lane. A safe rule is to wait until you

can see the car you have passed your rear view mirror, but do not turn sharply in front of it do not "cut

– in"

4.You must not pass or attempt to pass on hills, curve, hump of bridge or within 100meters of an intersection. Do not pass at any time when the road ahead is not clearly visible(87)

5.Except in a location where another vehicle may pass you on the right, when another vehicle comes

up behind you and sounds its horn, you must move over to your right and let it pass. You must never

speed up when another vehicle tries to pass you.

6. If you meet or overtake any animal or livestock, you must reduce your speed or stop as may be necessary or may be requested by any person in charge o the animal or livestock.

DRIVING AT NIGHT

Driving speed should be reduced for night driving; particularly so at dawn and dawn dusk. Too many

drivers try to drive just as fast at night as they do during the day. We are not as alert, nor do we see as

well when we are tired after a day`s work. Wide awake driving is necessary at all times and especially

at night , since we can see objects only a limited distance ahead. Driving at night with a sand pitted windshield is a safety hazard.

Don`t overdrive your headlights. You should hold to a speed that allows you to come to a complete stop when necessary in the distance you can clearly see ahead. After-dark driving requires different skills and extra care for safe travel. These are the things you should do:

1. When you meet a car at night, you should lower headlight beams(dim your lights)when within 500

meters of the approaching vehicle. Even if the other driver does not lower his beams., will be safer to keep your beams lowered and drive with due care.

2. Lower your headlight beams when following another vehicle within 200 meters

3. Lower your headlight beams when you are driving on well lighted streets.

4. Use your lower headlight beams when driving in fog or haze , and reduce your speed. Driving with

your headlight beams on in a fog is like shining your light on a mirror light is reflected back to your own eyes and blinds you.

5. Avoid looking directly into the light

6. Keep your headlights properly adjusted so that the lower beams are not aimed upward into an approaching driver's eyes.
7. Ensure that your tail lights, stop lights and turn signals are working
8. Keep your headlights and windshield clean
9. Slow down when facing the glare from approaching headlights
10. Be sure that you can stop, when it is necessary, within the distance you can see clearly ahead and watch carefully for pedestrians and animals along the road side.

CAUTION-CAR COONTROL

Keep your car under control and your brakes in good condition so that you can slow down or stop short of any possible danger or emergency. To do this, always allow yourself a SAFE DISTANCE between your car and another vehicle or person, so that you can make smooth stop, jerky stop often cause dangerous skidding.

At high speeds, even on dry roads a car may go a long way before it will slide much further. That is why you must be very careful on slippery roads.

To avoid skidding on wet or slippery surfaces, drive at a slow speed. Allow more distance between your car and others. Apply your brakes or gas pedal slowly and easily so that you can start or stop smoothly. Try to keep your wheels out of ruts. No matter what causes a skid, the method of control is always the same. If your vehicle goes into a skid.

1. Turn your front wheels in the direction the rear end is skidding. That tends to straighten you out. Turning your wheels in the wrong direction will throw you into a dangerous spin.

2. Don't over steer. That could throw the rear end into a reverse skid in the other direction and you'll be worse off than you were before.

3. Keep your foot on the accelerator to keep your rear wheels rolling, if there's nothing in front of you, feed gas gently to maintain traction.

4. Keep your foot off the brake. Jamming of the brake will make the skid worse. When you have the car under control, pump the brakes gently to stop.

If you turn off the pavement onto the sand to avoid an oncoming vehicle, do not attempt to return to

the pavement too quickly. By over correcting you're steering the front wheels tend to move sideways

until they "dig-in" the sand. This increases the chances of your vehicle overturning. Reduce your speed gradually and hold the steering firmly with both hands.

Turn slowly and carefully __ it's better to get stuck than to overturn your vehicle. Do not return to the

roadway until you are sure traffic is clear and you can do so safely. A complete stop is recommended.

SIGNS AND SIGNALS

For your guidance, a display of the traffic signs approved for use in Nigeria is included.

REMEMBER __ the stop means S-T-O-P, a complete stop. The yield sign means that you must yield right-of-way to vehicles on the street that you are about to cross or enter.

Nigerian traffic expect a straight ahead signal by hand when a driver is proceeding straight through at

an intersection under his control. This signal will assist the policeman in directing a smoother flow of safer traffic.

SAFETY EQUIPMENT OF YOUR VEHICLE

The Traffic Department and Division have many requests for the standards for efficiency of brakes, light and horns. We are not aware of any local laws giving a clean cut definition of all these standard

in some other areas.

BRAKES

Every passenger motor vehicle must have a service brake, which is usually a foot brake, capable of stopping it in 25 meters from a speed of 35 kilos per hour. An emergency brake (parking brake) is also required.

This is usually a hand brake, but in some newer vehicle models sometime it is applied by foot pressure separately from the service brake. For some late model cars, vigorous of the foot brake and

the hand brake is necessary to prevent the car from rolling back. The emergency brake must be able

to hold the vehicle on any grade upon which it is driven.

HEADLIGHTS

Your car must have at least two, headlights, one on each side of the front of the vehicle. The headlights must be strong enough to light the road so that you can see a person a person or vehicle 100 meters ahead on you, low beam, and 350 meters ahead on the high beam. They must not blind an

oncoming driver.

TAIL LIGHTS

Every vehicle must be EQUIPPED with at least one tail- light showing a red light for easy visibility a distance of 500meter to the rear. It is normally required that headlights(not parking light) and tail-lights be lighted when a vehicle is driven between 30 minutes after sunset and 30 minutes before sunrise, and at other times when needed.

STOPLIGHT

Every vehicle must be equipped with at least one rear stop light, which may be part of the tail-light or

a separate system. The stop light should display a red light when the service brake pedal is applied. The red light must be strong enough to be seen 300meters to the car day or night.

TIRES

No tire should be used which has worn to the point that fabric is exposed or has sidewall cuts or breaks which have damaged the fabric. All regular tires must have a tread configuration on the part making contact with the road surface. Specific exception to this rule may be in cases of heavy trucks

and trailers

Some people think the chief function of tires is to provide a soft ride. They do far more than that, they

are an integral part of the braking system and the transmission of power from the motor to the road.

Under-inflation not only affects the safe performance of these systems, but can weaken the tires rapidly to the point when they fail and often result in dangerous blow outs .If you should have a blowout, positive smooth braking effort at intervals much more efficient and safer than quick, hard brake applications, Hold as straight a line as possible with your vehicle by means of a tight, firm grip

on the steering wheel; otherwise, the car will turn toward the side the blown out tire is on.

OTHER EQUIPMENT

A horn which can be heard 200 meters away should be on every vehicle.

Muffler systems should be gas-tight and should not release exhaust gases to the side of the vehicle between 2 feet and 11 feet above the ground.

Never close a fuel tank spout with such materials as a plug of wood, rags or other combustible items.

An inside rear view mirror should be attached so that you can see the road to the rear car for at least

200 meters. Left and right hand out side rear view mirrors will improve your visibility.

AUTOMOTIVE SEAT BELTS

It is the responsibility of the organization to which the vehicle is assigned to see that appropriate seat

belt installations are made if the vehicle is not already so equipped. It is the responsibility of each driver to use the seat belt. It is an additional responsibility of the driver to insist that his passengers also use seat belt as provided.

Owners of privates vehicles are urged to install and use seat belts. we all know that seat belts do not

prevent accidents. We do not prevent accidents. we do know that the use of seat belt does minimize

the opportunity for, and severity of,injuriesthat might arise out of an accident.

WHEN ON THE SCENCE OF AN ACCIDENT

1. When approaching the scene of an incident, slow down. Do not park your in a manner that will congest traffic.

2. Use available means to alert other traffic.

3. Assist the injured.

4. Dial emergency and briefly describe the accident, give the exact location and stay on the phone to

answer necessary questions.

5.

WHEN YOU ARE INVOLVED IN AN ACCIDENT

1. Assist injured persons.

2. Use available means to alert other traffic.
3. Do not accept responsibility or liability for the accident to anyone.
4. Do not make payment of money to anyone claiming that you are responsible or liable
5. Do not move the vehicle until permission is received from competent authority. (FRSC law permits vehicle to transport injured person to a medical facility if an ambulance is not readily available). The vehicle must then be return to the accident venue immediately.

SAFER DRIVING SKILLS

It takes three kinds of skills to drive successfully:

1. Mechanical operation of the vehicle.
2. Correct application of traffic laws.
3. How to see and what to look for.

Most of this guide deals with the first and second of these skills. The third skill, art of seeing and knowing what to look for, is as important as the first two but is usually overlooked or at least not given sufficient attention.

By understanding and practicing from the start the seeing methods used by the experts, most drivers can form seeing habits which are the safest approach to proper driving. These methods emphasize allowance timing and spacing.

The secret is selective seeing. In a moving automobile, the driver has no time to see every detail along the road. This selective seeing is based on the following five seeing steps:

1. Aim High in Steering: Correct steering is simple ignore the fact that you sit left of the center in the car. Depend chiefly on your lower fringe vision for occasional glance well ahead at the center of your traffic lane. Seeing is usually of two kinds, sharp center eyesight which is a narrow three degree cone of vision and fringe vision which is the upper, lower and side parts your part of eye sight, your car then travels on center in its lane, and to the sides for traffic conditions.
2. Get the Big Picture: keep your central vision scanning a wide, deep roadway scene- foreground, background, side ground- for a block ahead in town; up to half a kilo away from towns.
3. Keep your eyes moving: Building a strong habit of forcing your eyes to move at least every two seconds in the driving. Check the rear view mirror each five second if any traffic is ahead or behind. Check the rear also before changing lanes and before slow down that could surprise a driver behind. The eye moving habit is restful for your eyes, keeps your mind straying too far from you driving and literally forces you to adjust speed to traffic. Weather and visibility conditions.
4. Make Sure They See You: When some one does create a conflict that threatens your safety, it means he either did not see your car or was counting on you to avoid the conflict. Be certain that the other person sees your car and shows clearly by his action that he will “stay put”.

Always tap your horn or flick your lights if in doubt. Do it early so you could stop if necessary.

5. Leave yourself an out: Recognize a basic rule of self preservation; protect yourself from

being trapped by error of other people. Adjust your pace in traffic to keep a space cushion around your car. Whenever possible poise your foot lightly above the brake when in doubt. Allow extra space when the road is slippery, when large vehicles block your view or when other cars drive too close behind you. With the habit of leaving yourself an out, you avoid impulsive moves and almost never need a hurried move. It's the smart and easy way to drive.

A FINAL RULE

Never insist on taking the right-of-way even when you are sure it is yours. The right-of-way should never be taken; it must be given. This is a rule of common courtesy, which also requires that you yield the right-of-way to others when it eases a traffic situation to do so. Courtesy in yielding the right-of-way is an important factor in modern driving. Courtesy the roadway, and it saves lives.

OCCUPATIONAL HEALTH SAFETY PRACTICE INDUSTRIES

Safety is 25% Common sense, 80% Compliance, and the rest is goodliness – Safety spud.

Industrial safety can be defined as the ability to manage the risks inherent to operations, or related to the environment. Industrial safety is not a dislike of risk, rather it is a commitment to clearly identify them in relation to production operations, assess them in terms of quality, and quantity and manage them. It is the policies and protections put in place, to ensure plant and factory worker protection from hazards that could cause injury.

INDUSTRIAL SAFETY TIPS

Total, a globally reputable company in her golden rules of occupational safety: (Thoroughness and vigilance) Summarized it in these words:

“Everyone must understand the basic rules of prevention and comply with them.”

These golden rules reinforce and complete our procedures for identifying dangers and controlling risks. The essential rules were established using records of accidents, observed in our different business. They must be followed to the letter.

Everyone should intervene as soon as a deviation is observed, and stop work if risk is not controlled. The twelve themes covered by these are:

High Risk Situations

Do not start-up or shut-down equipment or installations without using the appropriate written operating procedure. Report any situation where the risk level is temporally increased.

Identify and analyze induced risks implement compensatory measures.

Traffic: Machine/Vehicle/Cyclist/ Pedestrians

Do not exceed the speed limits. Traffic rules apply inside and outside sites.

Body Mechanics and Tools

Do not carry out work if you do not have the right tools the job and the environment: Adopting an inappropriate posture when handling objects or using tools can cause physical harm.

Protective Equipment

Do not access installations and perform work without wearing, general or task-specific personal Protective Equipment (PPE). Collective protection is preferred. Individual protection

completes preventive measures already taken.

Work on Powered Systems

Do not perform work without checking that the power and product source supply has been rendered inoperative. Some works require the energy to be safely discharged or the equipment to be purged and vented. In Such cases, a lock-out system is set up to Isolate the energy or product according to a specific method.

Confined Spaces

Do not enter a confined space until isolation has been verified and the atmosphere checked. Do not enter a confined space(container, tanks, well, etc) without supervision, and only after the atmosphere and isolation have been checked.

Excavation Work

Do not perform excavation Work without valid work permit comprising a map of all underground hazards.

Manual or mechanical excavation work including dredging rivers and sea beds, can only start after a risk analysis has been done performed, to identify all hazards in the zone, and related precautions have to be taken.

Working at Height

Do not work at heights without a safety harness when there is no collective protective equipment. Work at height is performed on fixed or mobile platforms with a guardrail, designed for the task at hand. Work on rooftops(buildings, reservoirs) is performed only after the roof solidity has been checked and appropriate has been set. Ladders are a means access only, their use must remain exceptional.

Change Management

Do not make any technical or organization changes without prior authorization. Any change in technique or organization must be the subject of a risk analysis.

Simultaneous Operations or Co-activities

Do not perform simultaneous or co-activities without a prior visit.

All work or operations inside a unit in operation can risk levels.

Interferences due to simultaneous operations or joint-activities must be identified and controlled.

PREVENTING ELECTRICAL HAZARDS IN THE WORK PLACE

Good electrical safety habits can help protect you and your co-workers from injury. And you can play a role in recognizing and preventing workplace hazards. Keep in mind the following:

1. Electrocution is one of the top fire causes of workplace deaths.
2. Workers younger than 25 have the highest rate of death from electrical shock.
- 3.Many workplace electrocutions involve touching a power line with long or tall equipment.

STAYING ALERT

Many electrical injuries could be prevented if people where alert to hazards. Stay aware by keeping focused on your job and do not let emotions like anger and frustration get in the way.

CHECK CORDS AND CONNECTION

Before you start work, check electric cords for wears. If you are outside or in a wet location, be

sure tools and extension cords are suitable for outdoor use and circuits are equipped with Ground Fault Circuit Interrupter (GFCI). The ground f-fault circuit interrupters (GCFI) is a fast- acting circuit- breaker designed to shut-off electric power in the event of a ground-fault, within as little as 1/40 of a second. It works by comparing the amount of current going to, and returning from equipment along the circuit conductors. It is also called Residual Current Device (RCD) and shuts off an unintended path, possibly through water or through a person. It is used to reduce the risk of electrical shocks. Also keep cords free of oil, heat and corrosion chemicals. Never yank. Kink or bend cords, and store them loosely coiled in a dry place.

POWER TOOLS

Never carry a tool by its cord. Be sure a tool is switched “off” before plugging or unplugging. This protects you and the next person who uses it. Watch out for energized areas when reaching into equipment. Shields, barriers, insulation and GFCI protects you, so do not modify them just to get a job done faster. Learn and follow your company lockout/tagout procedures. If in doubt ask a qualified electrical worker for help.

Dirt and dampness increase the risk of shock. Keep your tools, work areas and storage space clear and dry. When cleaning electrical equipment, be sure it is unplugged and follow the manufacturer`s cleaning instructions.

USE SAFETY GEAR

Personal Protective Equipment(PPE) is your first line defense against shock and electrical burns. Keep boots, gloves and other gear in good condition, even a pinhole let electricity through. Wear non- conductive protection on your head, face, hand, and feet. Use insulated tools or handling equipment, such as no-conductive ropes and protective shields.

PREVENTING ELECTRICAL FIRE

Most electrical fires can be traced to overheated circuits or overloaded equipment. When abused, insulation may melt or burn, exposing live wires. Electrical fires can also occur when equipment is driven beyond capacity; when accumulated oil and dirt overheat a motor or when sparks ignite scraps, dirt, dust or flammable liquids. Observe the following rules;

- Use and maintain wiring, tools and equipment correctly. Keep everything oil and dust-free.
- Uncoil an extension cord fully before use. Find the amperage marked on it. Is it adequate ?
- Don not use equipment that gives off mild shock, unusual heat or odd smells. If in doubt, have it checked and repaired or replaced.
- Sweep up scraps and sawdust, and store flammable liquids in approved containers.
- Do not use electrical equipment when flammable gases, vapours, liquids, dust or fibers are present.

A FIRE SAFETY CHECKLIST

Visualize your plan of response in a fire, so you can move quickly if one occurs. Take into account;

- The nearest multi-purpose fire extinguisher and how to use it.
- The nearest emergency exit or fire escape.
- Your company procedures for notifying fire fighters and other emergency personal.
- Try to extinguish a fire yourself, only if you are trained and it is small or non-threatening. When

in doubt, get out and take others with you.

INDUSTRIAL ACCIDENTS

An industrial accident is sudden and unanticipated event that can result in injuries, fatalities property damage and lost production time. Mishaps in the in the work place are fairly frequent, despite federal laws to protect workers safety. Ignoring safety procedures and using poor judgment are among factors that put workers at greater risk, resulting in serious job-related injury or illness. Industrial accident, work accident, work place accident, occupational accident or accident at work is a “discrete occurrence in the course of work is a “leading to physical or mental occupational injury.

According to the International Labor Organization (ILO). More than 337 million diseases in more than 337 million accidents happen on the job each year, resulting together with occupational disease in more than 2.3 million death annually. The phrase “in the course of work” can include work-related accidents happening off the company`s premises, and can include accidents caused by third parties according to Eurostats. The definition of work accidents include occurring ‘while engaged in an economic activity, or at work, or carrying on the business of the employer’. According to ILO, the phrase “Physical or mental harm” means any injury, disease or death.

Occupational accidents differ from occupational disease, as accidents are unexpected and unplanned occurrences e.g(mine collapse,etc).while occupational disease, contracted as a result of an exposure over a period of time, to risk factors arising from work activity” e.g(miners lung)

Incidents that fall within the definition of occupational accidents include cases of acute poisoning, attacks by humans and animals, insect,slips and falls, on payments or staircases, traffic collisions, and accidents on board means of transportation in the course of work, accidents in airports stations and so on.

There is no consequences as to whether commuting accidents, that is, “accidents on the way to work and while returning home after work” should be considered to be work accident. The Europeans statistics on accidents at work(ESAW) methodology exclude them, The international Labour Organization (ILO) includes them in its conventions concerning health and Safety at work, although it list them as a separate category of accidents and some countries for example Greece do not distinguish them from other work accidents.

A fatal accident at work is defined as an accident which leads to death of a victim. The time within which the death may occur, varies among countries. In Netherland, an accident is registered as fatal if the victim dies during same day that the accident happened. In Germany, if death came within 30 days, while Belgium, France and Greece set no time limit. Where the accident involve multiple fatalities, they are often referred to as industrial, they are often referred to as industrial disasters.

COMMON CAUSES OF ACCIDENT

About 80 out of every 100 accidents are the fault of the person involved in the incident. Unsafe acts cause four times time as many accidents and injuries conditions.

Accidents occur for many reasons. In most industries, people tend to look for things to blame” when an accident happened”; because it is easier than looking for” root causes”such as the underlisted:

Taking short-cuts: Everyday we, make decisions we hope will make the job faster and more efficient. But do time-severs ever your own safety or that of other crew members? Short -cuts that reduce your safety on the job are not short-cuts, but an increased chance for injury.

Over-confidence: Confidence is a good thing; over-confident is too much of a good thing. “it will never happen to me” is an attitude that can lead to improper procedures, tool or methods in your work. Any of these can lead to an injury.

Incomplete Instructions: To do the job safely and right the first time, you need complete information. Don` t be shy about asking for explanations about work procedures and safety precautions.

Poor House Keeping: When clients, managers or safety professionals walk through your work site, house keeping s an accurate indication of every ones attitude quality, production and safety. Poor house keeping creates hazards of all types. A well- maintained area sets a standard for others to follow. Good house keeping involves both pride and safety.

Ignoring Safety Procedures: Purposely failing to observe safety procedures can endanger you and your coworkers. You are being paid to follow the company safety policies: not to make your own rules. Being casual about safety can lead to a casualty.

Mental Distractions: Having a bad day at home and worrying about it at work is hazardous combination. Dropping your “mental” guard can pull your focus away from work procedures. You can also be distracted when answering phones not relating to your work., or when a friend comes by to talk while you are busy with work. Do not become a statistic because you took your eyes or focus off the machines “just a minute”.

Failure to plan: There is a lot of talk today about job hazard Analysis. JHA`S are an effective way to figure out the smartest ways to work safely and effectively .Being hasty in starting a task. Or in not thinking through the process can put you in a harm`s way. Instead plan your work, and work your plan.

Remember. Mark Twain said

“It is better to careful many times than to die once”

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- Use and maintain wiring, tools and equipment correctly. Keep everything oil and dust-free.
- Uncoil an extension cord fully before use. Find the amperage marked on it. Is it adequate?
- Don not use equipment that gives off mild shock, unusual heat or odd smells. If in doubt, have it checked and repaired or replaced.
- Sweep up scraps and sawdust, and store flammable liquids in approved containers.
- Do not use electrical equipment when flammable gases, vapours, liquids, dust or fibers are present.

A FIRE SAFETY CHECKLIST

Visualize your plan of response in a fire, so you can move quickly if one occurs. Take into account;

- The nearest multi-purpose fire extinguisher and how to use it.
- The nearest emergency exit or fire escape.
- Your company procedures for notifying fire fighters and other emergency personal.
- Try to extinguish a fire yourself, only if you are trained and it is small or non-threatening. When in doubt, get out and take others with you.

INDUSTRIAL ACCIDENTS

An industrial accident is sudden and unanticipated event that can result in injuries, fatalities property damage and lost production time. Mishaps in the in the work place are fairly frequent, despite federal laws to protect workers safety. Ignoring safety procedures and using poor judgment are among factors that put workers at greater risk, resulting in serious job-related injury or illness. Industrial accident, work accident, work place accident, occupational accident or accident at work is a “discrete occurrence in the course of work is a “leading to physical or mental occupational injury.

According to the International Labor Organization (ILO). More than 337 million diseases in more than 337 million accidents happen on the job each year, resulting together with occupational disease in more than 2.3 million death annually. The phrase “in the course of work” can include work-related accidents happening off the company`s premises, and can include accidents caused by third parties according to Euro stats. The definition of work accidents include occurring ‘while engaged in an economic activity, or at work, or carrying on the business of the employer’. According to ILO, the phrase “Physical or mental harm” means any injury, disease or death.

Occupational accidents differ from occupational disease, as accidents are unexpected and unplanned occurrences e.g(mine collapse,etc).while occupational disease, contracted as a result of an exposure over a period of time, to risk factors arising from work activity”

e.g(miners lung)

Incidents that fall within the definition of occupational accidents include cases of acute poisoning, attacks by humans and animals, insect, slips and falls, on pavements or staircases, traffic collisions, and accidents on board means of transportation in the course of work, accidents in airports stations and so on.

There is no consequences as to whether commuting accidents, that is, “accidents on the way to work and while returning home after work” should be considered to be work accident. The Europeans statistics on accidents at work(ESAW) methodology exclude them, The international Labour Organization (ILO) includes them in its conventions concerning health and Safety at work, although it list them as a separate category of accidents and some countries for example Greece do not distinguish them from other work accidents.

A fatal accident at work is defined as an accident which leads to death of a victim. The time within which the death may occur, varies among countries. In Netherland, an accident is registered as fatal if the victim dies during same day that the accident happened. In Germany, if death came within 30 days, while Belgium, France and Greece set no time limit. Where the accidents involve multiple fatalities, they are often referred to as industrial, they are often referred to as industrial disasters.

COMMON CAUSES OF ACCIDENT

About 80 out of every 100 accidents are the fault of the person involved in the incident.

Unsafe acts cause four times time as many accidents and injuries conditions.

Accidents occur for many reasons. In most industries, people tend to look for things to blame” when an accident happened”; because it is easier than looking for” root causes” such as the under listed:

Taking short-cuts: Every day we, make decisions we hope will make the job faster and more efficient. But do time-severs ever your own safety or that of other crew members? Short -cuts that reduce your safety on the job are not short-cuts, but an increased chance for injury.

Over-confidence: Confidence is a good thing; over-confident is too much of a good thing. “it will never happen to me” is an attitude that can lead to improper procedures, tool or methods in your work. Any of these can lead to an injury.

Incomplete Instructions: To do the job safely and right the first time, you need complete information. Don` t be shy about asking for explanations about work procedures and safety precautions.

Poor House Keeping: When clients, managers or safety professionals walk through your work site, housekeeping an accurate indication of every ones attitude quality, production and safety. Poor housekeeping creates hazards of all types. A well- maintained area sets a standard for others to follow. Good housekeeping involves both pride and safety.

Ignoring Safety Procedures: Purposely failing to observe safety procedures can endanger you and your coworkers. You are being paid to follow the company safety policies: not to make your own rules. Being casual about safety can lead to a casualty.

Mental Distractions: Having a bad day at home and worrying about it at work is hazardous combination. Dropping your “mental” guard can pull your focus away from work procedures.

You can also be distracted when answering phones not relating to your work., or when a friend comes by to talk while you are busy with work. Do not become a statistic because you took your eyes or focus off the machines “just a minute”.

Failure to plan: There is a lot of talk today about job hazard Analysis. JHA`S are an effective way to figure out the smartest ways to work safely and effectively .Being hasty in starting a task. Or in not thinking through the process can put you in a harm`s way. Instead plan your work, and work your plan.

Remember. Mark Twain said

“It is better to be careful many times than to die once”

Disaster

Management

INTRODUCTION

Disaster, means an event, which brings sudden great fortune bringing disruption to normal life and causing colossal loss of life and property and disruption of humans activities. Disasters are situations of great calamity like earthquake, floods etc.

Disaster can also be defined as catastrophe, mishap, calamity or grave occurrence in any area arising from natural or man made causes, or by accident or negligence which results in substantial loss of life or human suffering or damage to, and destruction of, property, or damage to degradation of environment, and is of such a nature or magnitude as to be beyond the coping capacity of the community of the affected areas.

Natural disasters are global phenomena, which may occur anywhere at any time. These adversely affect the lives of the people and cause considerable damage to property, environment and infrastructure worldwide. Although the basic cause of most of the natural disasters is also dependent on man made changes caused in the environment. However, hazards and disasters are untimely related and are synonymous to each other. Virtually hazard is a threat whereas disaster is an event.

□ India is one of the most disaster-prone countries of the world. The following data clearly show that India is a disaster-prone country:

- 65 per cent of the land area is vulnerable to earthquakes
- 12 per cent of the land area is vulnerable to cyclones
- 70 per cent of the land under cultivation is prone to drought.

Crisis is an event of acute danger, which can cause sudden disruption of work.

The event may be caused due to natural cause, human error, equipment failure or sabotage by anti-social elements. The basic idea in crises(emergency situation) is to avoid wastage of crucial time. Terrorist threats and attacks, bomb explosion, strike, accidents, fire accidents, etc. are the examples of situations.

Disaster management Is a comprehensive term which includes all aspects of preventive and protective measure, preparedness and relief operations for reducing the impact of disaster humans.

Disaster management is a continuous and integrated process of planning, organizing, coordination and implementing measure, which are necessary or expedient for:

- (i) Prevention of danger or threat of any disaster
- (ii) Mitigation or reduction of risk of and disaster or its severity or consequences.
- (iii) Capacity building
- (iv) Preparedness to deal with any disaster
- (v) Prompt response to any threatening disaster situation or disaster
- (vi) Assessing the severity or magnitude of the effect of any disaster
- (vii) Evacuation, rescue and relief
- (viii) Rehabilitation and reconstruction

Poverty and lagging development amplify the adverse effects of natural disasters. Therefore developing countries (particularly their most densely populated regions) are more vulnerable to natural disasters because of their limited capacity to prevent and absorb these effects. Poor communities get forced to live in crowded, makeshift houses and in areas vulnerable to bad weather or seismic activity having low quality of infrastructure.

Mitigation efforts lessen the impact disasters have on people and property. Mitigation measures include keeping homes away from flood plains, designing structures to withstand earthquakes. The vulnerability to natural hazards can be reduced by altering, averting, avoiding and adapting to the hazard.

Beside the natural disasters, there are many man-made or technological oriented physical events, which result in a significant disruption in the normal working. The event which causes the sudden disruption of work, is known as crisis. The crisis management plan shall be able to respond quickly and effectively to such unexpected events and situation. The management includes the preventive measures need to be adopted to avoid any crisis and the ways and means to tackle a crisis, if it occurs in spite of preventive measures. Thus, crisis management helps in avoiding or minimizing wastage of crucial time and cost.

OBJECTIVES

The following are the major objectives of disaster/crisis

management

1. To improve state of preparedness to meet any contingency,
2. To reduce response time in organizing the assistance,
3. To identify major resources, manpower and the equipment needed to make the disaster management plan operational.
4. Making optimum use of the combined resources,
5. To identify mitigating measures and take timely action,
6. To avoid panic situations and rumors at the time of contingency.

The objective of preparing disaster management plan is to enable various departments/agencies to plan for, quickly respond to and recover from unexpected events and situations. This plan shall provide necessary guidelines for assistance to various sections for ensuring of people, protection of property and restoration of normal life including water and power supply, communication, etc.

CLASSIFICATION OF DISASTERS

The classification of disaster can be based on the origination agent, or the impact, or whether the disaster is preceded by a warning phase or not. The disasters may be broadly grouped into major or minor types depending upon their potential to cause damage to human life and property. The disasters are generally classification as major or minor.

NATURAL DISASTER

These can be further sub-classified as under as under:

1. Meteorological Disasters. These include:
 - (i)Cyclones ,(ii)Hurricanes,(iii)Floods,(iv)Droughts,(v)Hailstorm, cloudburst
- 2.Topographical Disasters. These include:
 - (i)Landslides,(ii)Avalanches(i.e, fall of mass of snow, rock and ice down a mountain)
- 3.Tectonic Disasters. These include:
 - (i)Earthquakes, (ii) Volcanoes
- 4.Infestic Disasters. These include:
 - (i)Epidemics,(ii)Pest attack, Locust invasion of crops
- 5.Chemical, Industrial and Nuclear Disasters
- 6.Accident related disasters. These include:
 - (i)Fires,(ii) Mine Flooding.(iii)Major building collapse(iv)Air, rail, road
 - (v)stampede

The accidents may be either natural or man-made (intentionally like bomb blast) or due to mistake or carelessness, etc

Different natural calamities can be distinguished from each other in terms of their nature and extent of their impact.

(a) Calamities like earthquakes hailstorms, avalanches and landslides occurs quite suddenly but they are restricted in their impact in terms of time and space.

(b) Floods and Cyclones occur with some element of warning yet their occurrence is confined in duration.

(c) Drought spans over a much longer time-frame and its adverse impact on the economic activities and life of an area is of a more lasting nature.

Natural disasters may be broadly grouped into major and minor types depending upon their potential to cause damage. For example:

(i) Major disasters. Earthquake, flood, drought, cyclone.

(2) Minor disasters. Landslide, avalanche, tornadoes, hailstorm, heat wave, cold wave.

MAN-MADE DISASTERS

Man-made disasters are the results of follies and unwise acts of human beings. Following man-made factors increase vulnerability:

1. Density of population
2. Population living below poverty line(PBL)
3. Low percentage of literacy
4. Pattern of occupation and situation in employment.

About 70% people of our country reside in villages and the mode of their employment revolve around agriculture. The agriculture sector is always vulnerable to drought, floods and man-made problems. such as spills from oil and gas industrial activities By and large farmers lack knowledge to face disasters. Natural disasters with man-made factors of vulnerability make the overall situation highly volatile.

As compared to the developed countries, the third world countries suffer more relative loss to lives, livestock, and property. The third world countries have weak socio-economic infra-structure and do not have adequate prevention, preparedness and mitigation against disasters.

Other man-made disasters are:

(i) Communal riots, (ii) Ethnic conflict, (iii) Refugee situation, (iv) Terrorism, (v) stampede, (vi) Food poisoning, (vii) Alcohol/Liquor tragedies.

NATIONAL POLICY ON DISASTER MANAGENT

As discussed above disasters can strike our area at any time with or without any fore warning. Therefore disaster management policy is proactive and is comprehensive and includes the measures of preparedness, mitigation and prevention . In view of the changing dimension and probabilities of the natural disasters a new approach of strategic management is being followed in the national policy.

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Shell 12 Life Saving Rules.

By GudIsaac123 (self media writer) | 1 year ago

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Shell's 12 Life-Saving Rules.

The following are Shell's 12 Life Saving Rules and how they should be followed by employees and contractors. The bottom line is, if you choose to break the rules, you choose not to work for Shell.

Contents

- 1 Work with a valid Work Permit when required.
- 2 Conduct Gas Tests when required.
- 3 Verify isolation before work begins and use the specified life protecting equipment.
- 4 Obtain authorization before entering a confined space.
- 5 Obtain authorization before overriding or disabling safety critical equipment.
- 6 Protect Yourself against a fall when working at height.
- 7 Do not walk under a suspended load.
- 8 Do not smoke outside of designated smoking areas.
- 9 No alcohol or drugs while working or driving.
- 10 While driving, do not use your phone and do not exceed speed limits.
- 11 Wear your seat belt.
- 12 Follow prescribed Journey Management Plan.
- 13 External links

Work with a valid Work Permit when required.

This rule is associated with the US Safe System of Work and the Global Permit to Work system. It doesn't apply to permits obtained through oversight agencies, such as encroachment permits, drilling permits, grading permits, building permits, etc. Those permits need to be obtained where required, and not having those permits can result in fines imposed by the oversight agencies, but this Life Saving Rule is specific to the Safe System of Work and the Global Permit to Work System. A Work Permit describes what you must do to stay safe.

You should:

Understand the Work Permit and follow it.

Confirm that the Opera

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permit

APP is valid. About

Confirm with the Supervisor or the Person in Charge of the work that it is safe to start work.

If you are the Supervisor or the Person in Charge of the work you should:

Confirm if a Work Permit is required for this work.

Confirm that the workplace

HEADLINES LATEST has POLITICS

been inspected before work

SPORTS starts.

ENTERTAINMENT BUSINESS EDUCATION HEALTH TECHNOLOGY

Explain how the Work Permit is signed.

Confirm the Work Permit is signed.

Confirm that it is safe to start work.

Get a new Work Permit when the work or the situation changes.

Confirm that the work is completed.

Conduct Gas Tests when required.

The air should be tested to prevent explosions and/or make sure the workers and people on-site can breathe the air safely.

You should:

Confirm with the Supervisor or the Person in Charge of the work that the air is tested.

Confirm with the Supervisor or the Person in Charge of the work that it is safe to start work.

Stop work if you smell gas.

If you are a Gas Tester you should:

Understand which tests the Work Permit requires and how often.

Use certified equipment for the tests.

If you are the Supervisor or the Person in Charge of the work you should:

Confirm that gas testing is carried out as per Work Permit.

Request more gas tests if necessary to keep the workers safe.

Confirm that it is safe to start work.

Verify isolation before work begins and use the specified life protecting equipment.

Isolation separates the worker and keeps them safe from the many dangers that can exist at any given job site. There is always a risk around any electricity, pressure, toxic materials, poisonous gas, chemicals, hot liquids, or radiation. Specified life-protecting equipment, such as breathing apparatus, electrical arc flash protection, or chemical resistant suits help to protect from danger.

You should:

Understand the isolations that protect you from danger.

Confirm with the Supervisor or the Person in Charge of the work that isolations are in place.

Confirm with the Supervisor or the Person in Charge of the work that it is safe to start work.

If you are the Supervisor or the Person in Charge of the work you should:

Confirm isolation is in place, for example, lock switches, separate pipes with spades, or lock access doors.

Confirm no stored energy or other dangers remain.

Confirm that it is safe to start work.

Obtain authorization before entering a confined space.

A confined space, such as a vessel, tank or pipe, can contain explosive gas, poisonous air or other dangers such as lack of oxygen, things that can fall on you or you can fall from. Authorised access keeps everyone safe.

You should:

Confirm with the Supervisor or the Person in Charge of the work that it is safe to start work.

Confirm with the Attendant that you can enter a confined space.

Follow the requirements of the Work Permit.

If you are the Attendant you should:

Approve and control access to a confined space.

Have means of communication with people in the confined space.

If you are the Supervisor or the Person in Charge of the work you should:

Confirm that the requirements of the Work Permit are in place.

Confirm that a qualified Attendant is always present when people are in a confined space.

Confirm that gas testing is carried out as per Work Permit.

Confirm that it is safe to start work.

Obtain authorization before overriding or disabling safety critical equipment.

Download

Obviously, Opera News

safety-critical APP

equipment must work correctly to keep workers safe. Some xamples of About

safety-critical equipment include isolation devices/emergency shut down valves, lock out/tag out devices, trip systems, relief valves, fire and gas alarm systems, certain level controls, alarms,

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You should:

Obtain authorisation from the Supervisor or the Person in Charge before overriding or disabling safety-critical equipment.

If you are the Supervisor or the Person in Charge you should:

Point our the safety-critical equipment in your work place.

Confirm the authorisation comes from the right level.

Protect Yourself against a fall when working at height.

Always use fall protection equipment when working outside a protective environment where you can fall over 1.8 metres (6 feet) to keep you safe. A protective environment includes approved scaffolds, stairs with handrails and man lifts.

You should:

Have authorization to work at height outside a protective environment.

Be aware of what fall protection equipment to use and how to use it.

Check equipment before using it.

Always tie off when at height outside of a protective environment.

If you are the Supervisor or the Person in Charge of the work you should:

Confirm that it is safe to start work at height.

Do not walk under a suspended load.

Working or walking immediately under a suspended load is unsafe as the load can fall on you. A suspended load is an object that is temporarily lifted and hangs above the ground. (Rig floors are excluded from this rule).

You should:

Never cross a barrier controlling an area with a suspended load without authorization.

Follow the instructions of the Flagman or the Person in Charge of the lift.

If you are the Person in Charge of the lift you should:

Mark the unsafe area and put barriers in place.

Ensure that nobody walks under a suspended load.

Do not smoke outside of designated smoking areas.

Smoking or use of matches or cigarette lighters could set on fire flammable materials. Designated smoking areas, such as a smoking hut or a smoking room, will keep you safe from causing fire and explosion.

You should:

Know where the designated smoking areas are.

Intervene if you see someone smoking outside a designated area.

If you are the Supervisor or Person in Charge you should:

Inform people about designated smoking areas.

Ensure that designated smoking areas are clearly marked.

No alcohol or drugs while working or driving.

Using alcohol, illegal drugs and misusing legal drugs or other substances will reduce your ability to do your job safely.

You should:

Always inform the Supervisor or the Person in Charge if you are taking medicine that may have an affect on your performance.

If in doubt always check with the Supervisor or the Person in Charge who may seek medical advice.

Not use, keep, sell or distribute illegal drugs.

Intervene if you see a case of alcohol or drugs abuse.

If you are the Supervisor or Person in Charge you should:

Only assign work to people who are fit to work.

While driving,

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Opera News not use your phone and do not exceed About speed limits.

Speeding or using your phone while driving increases the risk of losing control of your vehicle.

HEADLINES

If you are a Driver LATEST

you should: POLITICS SPORTS ENTERTAINMENT BUSINESS EDUCATION HEALTH TECHNOLOGY

Not use a mobile phone or pager, send or read a text message, or use a hands-free mobile phone device.

Stay at or below the maximum allowable speed for the road your are driving on as indicated by road signs or Journey Management instructions.

Stay at or below the maximum allowable speed limit for the vehicle you are driving.

Adjust your speed to the prevailing conditions.

If you are a Passenger you should:

Intervene if a Driver is using a phone in a moving vehicle.

Intervene if a Driver is exceeding the maximum allowable speed.

Wear your seat belt.

This rule is one of the most often violated, as people often don't think that driving even a short distance requires a seatbelt. A seatbelt protects you from injury in the event of an incident while driving and keeps you safe. No matter how small of a distance or how slow the speed, safety belts should always ben used, whether in (rental) cars, taxis, (mini) buses, trucks, cranes, or forklift trucks, and the situation involves persons in moving vehicles when engaged on Shell business. Putting on a seatbelt should be a well formed habit. The only exceptions include vehicles where only lap seatbelts are available or in public transport where seat belts are not available.

You (Drivers and Passengers) should:

Always use a 3-point seatbelt (please note exceptions above).

Check that your seatbelt works properly.

Keep your seatbelt properly fastened while in a moving vehicle.

Check that everyone in the vehicle is wearing a seatbelt properly before starting to drive.

Intervene when your fellow passengers are not wearing seatbelts properly.

Follow prescribed Journey Management Plan.

A Journey Management Plan is a plan for you as a Driver that will help you to travel and arrive safely.

Journey Management Plans are required for any travel time that is over 4.5 hours.

If you are a Driver you should:

Confirm if a Journey Management Plan is required before starting the journey.

Discuss the Journey Management Plan with the authorised person.

Understand the Journey Management Plan before starting the journey.

Comply with the duty, driving and rest hours specified in the Journey Management Plan.

Follow the route specified in the Journey Management Plan.

Always tell the authorised person immediately if changes occur.

If you are the Supervisor or Person in Charge you should routinely:

Check that the Journey Management Plan is in place and being followed.

Check that the Driver understands and complies with the Journey Management Plan.

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1. SAFETY AND PERSONAL PROTECTION

1.1 WHY SAFETY?

It is a legal requirement; non-compliance attracts sanctions, punishment, etc.

It is economical; avoidable losses – injuries, asset damage/loss, fines, compensations, etc.

It is the right thing to do; it is morally expected. Duty of care

1.2 WHAT IS PERSONAL PROTECTIVE EQUIPMENT?

Personal protective equipment, commonly referred to as "PPE", is equipment worn to minimize exposure to serious workplace injuries and illnesses. These injuries and illnesses may result from contact with chemical, radiological, physical, electrical, mechanical, or other workplace hazards.

Personal protective equipment may include items such as gloves, safety glasses and shoes, earplugs or

muffs, hard hats, respirators, or coveralls, vests and full body suits.

PPE is used:

- as a way of controlling individual exposure to a hazard
- when there is occasional short exposure to a high risk
- when there is a residual risk of exposure remaining despite other controls

But:

- It is not a substitute for good working practices and personal hygiene standards

1.2.1 Types of PPE

□ Protective clothing – Worn as a barrier against hazard that may cause injury to the body (coveralls or apron). Required:

- o In the field, laboratory, workshop, hospital, etc
- o When handling hazardous waste, chemicals or pesticides
- o When removing asbestos
- o In hot or cold environments

□ Foot & Leg – can be boots or shoes. Protects against injuries from dropped objects, hot or cold liquids, stepping on something sharp

□ Head – hard hat, helmet. Protect against injuries from dropped objects or striking something.

Shelf life is about 5 years

□ Eye & Face – Glasses, goggles, face shield, visor. Protect against injuries from flying objects, chemicals, welding flash, heat and glare

- Hand – hand gloves. Selection considerations include physical characteristics, permeability, allergies, maintenance and storage
- Hearing – Ear plugs and ear muffs (banded or helmet mounted). Should have ability to attenuate noise sufficiently
- Respiratory (RPE) – Air purifiers, Breathing apparatus. RPE is required when:
 - o There is a risk of airborne infection
 - o In operations generating dust or hazardous substances
 - o Working in contaminated environments

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- o In oxygen-deficient environments
- o In emergency or rescue operations

1.2.2 What can be done to ensure proper use of personal protective equipment?

All personal protective equipment should be safely designed and constructed, and should be maintained in a clean and reliable fashion. It should fit comfortably, encouraging worker use. If the personal protective equipment does not fit properly, it can make the difference between being safely

covered or dangerously exposed. When engineering, work practice, and administrative controls are not

feasible or do not provide sufficient protection, employers must provide personal protective equipment

to their workers and ensure its proper use. Employers are also required to train each worker required to

use personal protective equipment to know:

- When it is necessary
- What kind is necessary:
- Exposure level
- Type of job
- Health implications
- How to properly put it on, adjust, wear and take it off
- The limitations of the equipment
- Proper care, maintenance, useful life, and disposal of the equipment

If PPE is to be used, a PPE programme should be implemented. This programme should address the hazards present; the selection, maintenance, and use of PPE; the training of employees on use, storage

and maintenance; and monitoring of the programme to ensure its ongoing effectiveness

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1.3 SAFE SYSTEM OF WORK

A safe system of work (SSW) is a formal procedure based on a systematic examination of work in

other to identify the hazards and define safe methods of working which eliminates those hazards or minimize the risks associated with them to as low as reasonably practicable (ALARP). It is recorded in form of procedures or work method statements.

2. BASIC FIRST AID

2.1 INTRODUCTION

At work, injuries and illnesses kill more than 2 million people in the world each year; i.e. 1 death every 15 seconds or 6000 deaths per day!

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Safe practices and healthy choices at work, home and play can prevent many injuries, diseases and death. Once injury or sudden illness has occurred, providing first aid can make a difference between life and death; rapid versus prolonged recovery; and temporary versus permanent disability.

2.2 LEGAL ASPECTS

The “Good Samaritan” principle protects a first aid provider from being sued for wrong doing provided:

- o You are reasonably careful
- o You act in good faith (not for a reward)
- o You do not provide care beyond your skill level

You are also expected to remain with the injured until help arrives and you hand over to a more experienced person or professional unless it becomes dangerous for you to stay

Consent – means permission. To get consent, first introduce yourself and tell the victim your level of

competence or training, and ask if it is okay to help. Implied consent means that permission to perform

first aid care on an unresponsive victim is assumed. When caring for infants or elderly (with dementia), consent shall be obtained from the parent/guardian or family member (as may be appropriate).

Care - Use appropriate PPE; it provides barrier between you and a victim’s blood or body fluid. Disposable gloves and eye goggles (where there is risk of spatter) are most recommended. Be careful

to remove gloves, clothing, or any other contaminated stuff.

2.3 ROLES AND RESPONSIBILITIES

Roles – The basic role of a first aid provider is to recognize a medical emergency and make a decision

to help. In doing this, your personal safety is the highest priority, followed by safety of victim and that

of any by-standers

Responsibilities –

- o Maintain composure

- o Maintain personal health and safety
- o Maintain caring attitude
- o Maintain up-to-date knowledge and skills
- o Do not cause further harm

2.4 LEVELS OF CONSCIOUSNESS

Full consciousness – Able to speak and answer questions normally

Drowsiness – Easily roused, but lapses into unconsciousness

Stupor – Can be roused with difficulty. Aware of painful stimuli and when spoken to

Coma – Cannot be roused by stimuli

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2.5 EMERGENCY ACTION STEPS

These action steps are intended to help the first aid provider respond to an emergency and manage life-

threatening problems of the Airway, Breathing and Circulation in any victim of any age

Assess the Scene:

- ☐ If it is not safe, or becomes unsafe, GET OUT
- ☐ Observe universal precautions (use PPE)
- ☐ If victim is awake and talking, identify yourself and ask if it is ok to help
- ☐ If victim appears weak, seriously ill or unresponsive Call emergency service

Attend to the Victim:

A: Airway - Open airway

-If unresponsive tilt head, lift chin

B: Breathing - Check breathing

- Look, listen and feel for at least 5 secs (but not more than 10 secs)

- ☐ If unresponsive, perform CPR (Cardio-pulmonary resuscitation)
- ☐ If unresponsive, place in recovery position

C: Circulation - Look for and control severe bleeding with direct pressure

- Monitor tissue colour and temperature
- Help maintain normal body temperature
- Give emergency oxygen if available and if trained on it.

2.6 COMMON FIRST AID PROCEDURES

Severe bleeding

- ☐ Apply pressure to the wound
- ☐ Cover wound with clean pad and apply pressure
- ☐ Raise and support injured part
- ☐ Bandage wound and transport to hospital

Fracture

- ☐ Steady and support injured part
- ☐ Secure injured part to a sound part of the body

☐ Take or send casualty to a hospital

Muscle or Joint injury – sprain, strain

☐ Rest injured part

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☐ Apply ice-pack to injured area for 20 – 30 mins

☐ Apply elastic bandage to reduce swelling

☐ Raise and support injured limb

Burns

☐ Cool burn (apply cold water)

☐ Remove any constriction from the affected area

☐ Cover with clean piece of material

☐ Do not apply lotion, ointment or fat

Shock

☐ Make victim comfortable; loosen shirt

☐ Elevate feet

Choking

☐ Ask victim to cough

☐ Bend the victim forward

☐ Deliver 5 back blows

☐ Perform abdominal thrusts

Heart attack – persistent chest pain, spreading often to the jaw and down to the left arm. Faintness

☐ Call emergency service

☐ Make victim comfortable

☐ Check for medication

☐ Keep victim still

☐ Don't give stimulants

Heat exhaustion / Fainting

☐ Move victim to a cool place

☐ Lay victim down

☐ Elevate feet

☐ Loosen clothing

☐ Give fluids

2.7 THE FIRST AID KIT

The first aid kit contains materials required by the first aider in the management of first aid cases

☐ Contents are based on location and experience

☐ It must be easily accessible

☐ Contents must be labelled

☐ It must be kept clean and tidy

☐ It must be checked and replenished regularly

Common contents of the first aid kit include:

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- i. Paracetamol tabs
- ii. Hibitane
- iii. TBC
- iv. Cotton bandage
- v. Crepe bandage
- vi. Safety pins
- vii. Cotton wool and Gauze
- viii. Plaster strips
- ix. Scissors
- x. Record book

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3. HAZARDS

3.1 DEFINITIONS

A hazard is a situation that poses a level of threat to life, health, property, or environment. Most hazards are dormant or potential, with only a theoretical risk of harm; however, once a hazard becomes "active", it can create an emergency. A hazardous situation that has come to pass is called an incident.

3.1.1 Types of hazard

Hazards are generally labeled as one of four types:

□ Physical hazards are conditions or situations that can cause the body physical harm or intense stress. Physical hazards can be both natural and human made elements. A physical hazard is a type of occupational hazard that involves environmental hazards that can cause harm with or without contact. Physical hazards include ergonomic and electrical hazards, radiation, heat and cold stress, vibration hazards, and noise hazards. Engineering controls are often used to mitigate physical hazards e.g. use of PPE.

Physical hazards are a common source of injuries in many industries. They are perhaps unavoidable in certain industries, such as construction and mining, but over time people have developed safety methods and procedures to manage the risks of physical danger in the workplace.

□ Chemical hazards are substances that can cause harm or damage to the body, property or the environment. Chemical hazards can be natural or human made; examples include working with non-flammable chemicals, acids, toxins, etc.

□ Biological hazards are biological agents that can cause harm to the human body. These some biological agents can be viruses, parasites, bacteria, food, fungi, etc.

□ Psychological hazards are created during work related stress or a stressful environment. A person can be a hazard when he/she is affected psychological disturbance through -stress, shift

patterns and also can be a hazard when a person is under the influence of alcohol, illness and lack of training.

Hazardous Area – A 3-dimensional space in which a flammable atmosphere may be expected to be present at such frequencies as to require special precautions for the construction, installation and use

of equipment or other sources of ignition.

Non Hazardous Area – An area where an explosive atmosphere is not expected to occur in quantities

such as to require special precautions for the construction, installation and use of electrical apparatus

and other sources of ignition.

3.2 HAZARDOUS AREA CLASSIFICATION (HAC)

Areas where gas (hydrocarbon) may collect or be present at any time are called hazardous areas. For

the purpose of assessing the degree of risk they are classified into three zones:

Zone 0 – An area in which an explosive gas/air mixture is continuously present, or present for long periods

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Zone 1 – An area in which an explosive gas/air mixture is likely to occur in normal operation

Zone 2 – An area in which an explosive gas/air mixture is not likely to occur in normal operation and,

if it occurs, it will exist only for a short time.

Area classification; who establishes it?

It's a team work, consisting of:

- ☐ Technical safety engineer (or HSSE professional)
- ☐ Process engineer
- ☐ Operations and maintenance
- ☐ Electrical engineer
- ☐ Mechanical/Rotating equipment/HVAC engineer

3.3 HAZARDOUS AREA BASICS

Explosive gas mixture – a mixture of flammable gas/vapour and air, where after ignition, combustion

propagates throughout the unconsumed mixture (without additional supply of energy)

Lower explosive limit – Concentration of flammable gas or vapour in air, below which the gas atmosphere is not explosive

Upper explosive limit - Concentration of flammable gas or vapour in air, above which the gas atmosphere is not explosive

Flash point – the lowest temperature of a liquid at which sufficient vapour is given off to form an ignitable mixture with air near the surface of the liquid

Auto-ignition Temperature – the lowest temperature at which it will spontaneously ignite in a normal atmosphere without external source of ignition, such as flame or spark.

3.4 BENEFITS OF HAC

- Basis for selection of protective equipment appropriate to defined areas to avoid ignition
- Assists in the location of air inlet/outlet for ventilation and combustion systems
- Helps determine location of flammable gas detectors
- Assists in location of escape routes
- Helps set the max allowable surface temperature of an equipment in an area

3.5 SAFETY IN CONSTRUCTION

- Construction is a process that consists of the building or assembling of infrastructure
- It involves clearing, excavation, grading, concreting, block laying, cementing, painting, roofing, operation of machines, plant and equipment, use of hand tools and other activities associated with buildings, structures, etc.

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- Demolition is the tearing down of buildings and other structures either manually or mechanically.
- Construction and demolition activities are some of the most dangerous land-based work
- Construction related fatalities are in the region of 13 per 100,000 in Europe, as against 5 per 100,000 in the US. No clear statistics available for Nigeria but many reported cases of fatalities from collapsed buildings under construction and other related activities

Common hazards associated with construction and demolition are:

- Motion; moving objects – earth moving equipment, rotating equipment, etc
- Height; objects at height (dropped objects), lifting and hoisting, unstable work platform
- Pressure; working with materials under pressure
- Electricity; working with high voltage, power tools and equipment
- Fire and Explosion; flammables
- Toxic materials; dangerous materials
- Thermal; extreme temperatures
- People; manual handling, exhaustion, noise, stress, security, violence, etc.

3.5.1 Construction HSSE Management System

- Employer is responsible by law to provide safe working environment for workers and third parties while workers are responsible for obeying safety rules and responsible for personal safety
- In providing safe working environment, employer must put a system in place for managing safety in construction and demolition operations
- Effective management of safety involves development and implementation of the HSSE management system:
 - Leadership and commitment
 - Policy and strategic objectives
 - Organization, Responsibilities and Resources
 - Risk management (risk identification, assessment, control and recovery measures)

- Planning and procedures
- Implementation and Monitoring
- Assurance
- Management review

Key steps in managing risks:

Risk identification – usually visual (walk-around, inspection of equipment and data sheets, etc)

Risk assessment – use risk assessment matrix and rank high, medium and low risks

Risk control – e.g. by elimination, substitution/replacement, technical controls, procedures, PPE, etc.

Recovery measures – Emergency response, Training/testing, Assurance (develop safe system of work,

performance monitoring, incident investigation, tool box meetings, consequence management, etc.)

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3.6 ELECTRICAL SAFETY

The basic rule of electrical safety is that no person should carry out work on any part of electrical equipment unless such parts are: -

- Properly identified as the equipment on which work is intended
- Made DEAD; switched off from the source of electrical power supply
- Isolated; action taken to prevent inadvertent reconnection and CAUTION notice attached
- Tested to prove DEAD condition
- Discharged, EARTHED and short-circuited; to provide back-up protection against danger from unforeseen conditions

3.6.1 Common Causes of Electrical Accidents

Accidents do not just occur; they are caused. Electrical accidents are caused by:

- Contact with live parts
- Faulty designs
- Improper installation of equipment
- Poor maintenance of equipment
- Improper use of equipment
- Poor grounding (earthing)

3.6.2 Electrical Equipment in Hazardous areas

Electrical equipment and instrumentation shall be installed in the least hazardous areas. Where they

have to be installed and used in hazardous areas, they need to be selected to be suitable for:

- The hazard category
- The temperature zone
- The zone of use

Any operational and environmental requirements become additional to above

3.6.2.1 Equipment classification

Two groups exist: -

- Group I- intended for use in underground mines, likely to become endangered by combustible dust
- Group II- intended for use in other places likely to become endangered by explosive atmospheres

Equipment group II is further divided into hazard categories 1, 2 and 3, corresponding to hazardous area zones 0, 1 and 2 described earlier in section 3.2; e.g. category 1 equipment are for use in areas

where explosive atmospheres are present continuously (i.e. in hazardous area zone 0).

3.6.2.2 Temperature classification

Equipment surface temperature must not be allowed to rise beyond the auto-ignition of gas or vapour

in that environment. Equipment labels will indicate corresponding temperature class, e.g. T1 for maximum surface temperature of 450C, T2 for 300C, T3 for 200C, etc.

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3.7 CHEMICAL/GAS HAZARDS

3.7.1 Chemicals and Handling

Over 100,000 chemicals are produced on an industrial scale. The chemicals are used in manufacturing,

medicine, agriculture, energy, home, etc.

The physical form of a chemical plays a big role in potential exposure:

- Gas – CO, H₂S
- Vapour – solvents
- Liquid – paints
- Mist – cutting fluids
- Solid – dust, pellets, fibres, cake

The nature/characteristics of the chemical also play an important role – organic/inorganic, volatility, solubility, etc. These determine whether they will be absorbed, ingested or inhaled into the body.

Methods of exposure control are selected following a hierarchy of control:

- Elimination
- Substitution
- Isolation
- Engineering
- Procedural
- PPE

NOT all types of PPE protect against all types of chemical; careful accurate selection required, especially with gloves and RPE.

Treat all chemicals with respect. Read the label/MSDS/SHOC card and follow the instructions

Minimize exposure and avoid all body contact with chemicals

Clear spillages when they occur and dispose of waste chemicals responsibly

Know the emergency procedure

3.7.2 Types of Gas Hazard

Three main types of gas hazard: –

- ☐ Flammable – risk of fire and/or explosion (e.g. methane, butane propane)
- ☐ Toxic – risk of poisoning (e.g. carbon monoxide, hydrogen sulphide, chlorine)
- ☐ Asphyxiant – risk of suffocation (e.g. oxygen deficiency, nitrogen and carbon dioxide)

Flammable risk – considered in section 4

Toxic risk –

Some gases are poisonous and can be dangerous to life at very low concentrations

Some toxic gases have strong smells like rotten egg (H₂S)

Toxic gases can enter the body through –

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- ☐ Inhalation
- ☐ Absorption
- ☐ Ingestion
- ☐ Injection

3.7.3 Confined Space:

A space that –

- ☐ Has limited or restricted means of entry or exit
- ☐ Is not designed for continuous human occupancy

Examples include sewers, tunnels, storage tanks, silos, enclosure with only top or bottom access

Potential Hazards in Confined Space:

- ☐ Oxygen deficiency – <19.5% or >23.5% oxygen concentration
- ☐ Combustibles – methane, hydrogen, propane, gasoline fumes, etc.
- ☐ Toxic materials – welding fumes, CO, H₂S
- ☐ Electricity
- ☐ Mechanical hazards

3.7.4 Gas testing

The main purpose of carrying out gas test is:

- ☐ To test for flammable and toxic gases in the atmosphere and at what concentration they are present
- ☐ To ensure that no hazardous work condition exist from the presence of such gases
- ☐ To test for oxygen content in the environment

3.7.5 Safety equipment

- i. Gas detector
- ii. Respiratory protection for work
- iii. Respiratory protection for escape
- iv. Ventilation equipment
- v. Other equipment (PPE, communication equipment, etc.)

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4. FIRE AND EXPLOSION

4.1 THE PROCESS

Fire - A state of combustion in which fuels burn to produce heat, often flame and smoke.

Explosion - A release of mechanical, chemical, or nuclear energy in a sudden and often violent manner with the generation of high temperature accompanied with the release of gases.

4.1.1 The Fire Triangle:

- Oxygen, Fuel and Heat in certain proportions create a fire
- If any one of these three elements is removed, a fire cannot exist

4.1.2 The Explosion Mix

- Fuels - (Gas, Fuel vapour or Dust)
 - o Solids
 - o Liquids
 - o Chemicals
 - o Gases
- Air (Oxygen) - may be by planned or unplanned introduction
- An ignition source - which could be a minute spark from an instrument
 - o Open flames
 - o Sparks
- Chemical sources
- Electrical sources
- Natural sources
- Hot surfaces

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4.1.3 Ignition Heat Measurement

Flash point - the lowest temperature at which a substance gives off sufficient flammable vapour and

would ignite with the introduction of an ignition source but may not continue to burn. The lower the flash point the easier it is to ignite.

Auto-ignition temperature - the temperature at which a substance will automatically ignite by itself without the introduction of an ignition source.

Auto-ignition

Fuel Flash point

temperature

Ethanol (70%) 16.6 °C (61.9 °F) 363 °C (685 °F)

Gasoline (petrol) -43 °C (-45 °F) 280 °C (536 °F)

Diesel (2-D) >52 °C (126 °F) 256 °C (493 °F)

Jet fuel (A/A-1) >38 °C (100 °F) 210 °C (410 °F)

Kerosene >38–72 °C (100–162 °F) 220 °C (428 °F)

Vegetable oil (canola) 327 °C (621 °F)

Biodiesel >130 °C (266 °F)

*Diesel fuel flash points vary between 52 and 96 °C (126 and 205 °F).

*Both Jet A and A1 fuels have flash points between 38 and 66 °C

**Flash points >90°C are considered very low hazard

Explosive Atmosphere – An accumulation of gas, mist, dust or vapour, mixed with air with the potential to lead to a fire or explosion.

Explosive range – Managing gas and vapour concentration outside the explosive range is a very important consideration in operations activities in the oil and gas work environment

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In managing explosive gases, the focus is on LEL!

4.2 CAUSES OF FIRE AND EXPLOSION

- Carelessness (poor house-keeping, poor interface management, etc.)
- Negligence (smoking in prohibited areas)
- Non-compliance (sub-standard materials, non-intrinsically safe equipment)
- Ignorance
- Works of nature – extreme weather conditions; hurricanes, thunder and lightning strikes, etc.
- Poor/weak systems of work (poor planning, poor maintenance practice)
- Human error, Work overload, Sabotage, etc.

4.3 FIRE FIGHTING/CONTAINMENT

Fire containment derives on three key principles:

- Smothering – starving the fire of oxygen supply
- Cooling – reducing the heat
- Starvation – cutting of fuel supply

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4.3.1 Classification of Fires

4.3.2 Fire Protection Systems

Two sub-categories exist:

- Passive protection -does not require any electrical, mechanical or manual (human) action to function. E.g.
 - o Spacing and layout
 - o Drainage/spill control systems
 - o Fire proofing – insulators that protect substrate from damage from fire
 - o Fire barriers (fire walls)
- Active protection -

- o Water spray systems
- o Deluge systems
- o Drenching systems
- o Sprinkler systems
- o Fire cannons/hydrants
- o Fire extinguishers

4.3.2.1 Types of fire extinguishers

1. Water
 2. Foam
 3. Dry chemical powder
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4. Carbon dioxide

4.3.2.2 Use of fire extinguishers: PASS

- P – Pull (pull the pin)
- A – Aim
- S – Squeeze (squeeze the trigger)
- S – Sweep (spray side by side at the root of fire)

4.4 CONSEQUENCES OF FIRE

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- Disables operations
- Damage/loss of property
- Injury to personnel and/or loss of life
- Law suit
- Destruction of vital business and other information
- Loss of employment or livelihood
- Poor corporate image
- Effects on the environment
- Regulatory penalties

4.5 FIRE AND EXPLOSION PREVENTION

- Adoptable fire and explosion management process
- o Pre-job planning
- o Follow proper procedure
- o Toolbox meetings
- o Access control; Use of caution signs
- o Proper storage of chemicals
- o Fit-for-purpose design
- o Regular, timely and adequate maintenance
- o Use only intrinsically safe equipment (ATEX compliant – atmosphere explosibles)

- o Effective prevention mechanisms
- o Gas monitors
- o Emergency shutdown device
- o Depressurization systems
- o Flame, smoke and heat detectors
- Application of passive fire protection mechanisms
- o Area classification
- o Perimeter fencing
- o Plant/process area separation
- Provision of relevant fire extinguishment media

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